Probability and Statistics

I try to follow the chain from probability models through sampling and inference, rather than memorizing disconnected formulas.

Probability theory content (Intro)

Probability Space

⟵ **Proof idea** Condition explicitly; it often simplifies the calculation.

randomness

random event: a subset A of a sample space.

directly observable event (closed under the intersection) $\rightarrow$ algebraic extension (finite union, complement) $\rightarrow$ continuously extension

e.g. Borel set (analogy to rational number) to the measurable set (analogy to the real number). The approximate of Borel set to the measurable set is related to outer regularity (inner regularity).

Random variable

Given $(\Omega, \mathcal{F}, \mathbb{P})$, an $\mathcal{F}$-measurable mapping $X : \Omega \rightarrow \mathbb{R}$ is called a random variable (r.v.).

(it is a function defined on the sample space)

Here, measurability means $\{\omega \in \Omega : X(\omega) \leq a\} \in \mathcal{F}$ for every $a \in \mathbb{R}$.

The probability is defined on the subset of the sample space. When we calculate the probability of the random variable given a certain value, we are calculating the probability of this subset of the sample space and here the subset is mapped from the random variable to the value observed.

- Discrete r.v.
- continuous r.v.
- remark (not be silly): do not calculate the expectation of a discrete r.v. by calculate its integral!

$P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\})$ –remark: the two p on two sides' meanings are not the same.

distributions —see prof. 张颢– 现代数字信号处理 bilibili

Prof.Zhou's notes about Stochastic Processes derived distributions

⟵ **My thought** From a probabilistic model to a usable numerical quantity: what is observed, what is random?

周老师的笔记的逻辑性真的很舒服，那些分布的关系还有从抛硬币伯努利到 counting process 的 binomial，还有 waiting time 的 geometric or negative binomial，然后再通过对 geometric 除以期望转换单位，取极限得到 exponential（这个思维和昨天上课的那个讲的换个视角看世界 □ 取极限使得 expand knowledge boundary 真的一模一样），然后因为相互独立的 geometric 想加得到的是 negative binomial，所以对 negative binomial 也有个极限，恰好就是 gamma 从离散到连续：几何分布 → 指数分布；负二项分布 → 伽马分布的逻辑就串起来了（当时 113 学的时候真没有这种 sense 感觉老师很重要）

poisson 就是由二项分布 n 趋于无穷，p 趋于 0 的出来的，然后 poisson 和第 k 个事件发生的连续等待时间有个 duality，duality 是只是用在证明第 k 个事件发生的连续等待时间是 gamma (exponential) 分布的，因为 notes

上写的是根据 duality 对从 n=k 到无穷的 poisson 相加相当于是 waiting time 的 cdf，然后求导得到 gamma，说明 waiting time 就是服从 gamma

sufficient statistics

Fisher information matrix

Rao boundary

Simple Random Sampling

We can get information about the population by taking a sample from it. The sample should be representative of the population.

A simple random sample of size n is a sample selected from a population in such a way that every possible sample of size n has the same chance of being selected and without replacement.

Unbiased estimator

⟵ **Motivation** Inference uses data to distinguish plausible parameters; assumptions still matter.

An estimator is a statistic used to estimate a population parameter. An estimator is unbiased if the expected value of the estimator equals the population parameter.

$E(\hat{\theta}) = \theta$

where $\hat{\theta}$ is the estimator and θ is the population parameter.

eg.

$$\mathbb{E}[\hat{p}] = \mathbb{E}\left[\frac{X_1 + X_2 + \cdots + X_n}{n}\right] = \frac{1}{n}\left(\mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n]\right) = p$$

Variance

$$\mathrm{Var}[X] = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2$$

$$\begin{aligned}
\mathbb{E}\left[\hat{p}^2\right] &= \mathbb{E}\left[\left(\frac{X_1 + X_2 + \cdots + X_n}{n}\right)^2\right] \\
&= \frac{1}{n^2}\mathbb{E}\left[X_1^2 + \cdots + X_n^2 + 2\left(X_1X_2 + X_1X_3 + \cdots + X_{n-1}X_n\right)\right] \\
&= \frac{1}{n^2}\left(n\mathbb{E}\left[X_1^2\right] + 2\binom{n}{2}\mathbb{E}\left[X_1X_2\right]\right) \\
&= \frac{1}{n}\mathbb{E}\left[X_1^2\right] + \frac{n-1}{n}\mathbb{E}\left[X_1X_2\right]
\end{aligned}$$

$$\mathbb{E}\left[\hat{p}^2\right] = \frac{1}{n}\mathbb{E}\left[X_1^2\right] + \frac{n-1}{n}\mathbb{E}\left[X_1X_2\right]$$

Since X_1 is 0 or $1, X_1 = X_1^2$. Then $\mathbb{E}\left[X_1^2\right] = \mathbb{E}\left[X_1\right] = p$.

Notice: X_1 and X_2 are not independent.

$$\mathbb{E}\left[X_1X_2\right] = \mathbb{P}\left[X_1 = 1, X_2 = 1\right] = \mathbb{P}\left[X_1 = 1\right]\mathbb{P}\left[X_2 = 1 \mid X_1 = 1\right]$$

$$\mathbb{P}\left[X_1 = 1\right] = p, \quad \mathbb{P}\left[X_2 = 1 \mid X_1 = 1\right] = \frac{Np - 1}{N - 1}$$

$$
\begin{aligned}
\operatorname{Var}[\hat{p}] &= \mathbb{E}\left[\hat{p}^2\right] - (\mathbb{E}[\hat{p}])^2 \\
&= \frac{1}{n}p + \frac{n-1}{n}p\left(\frac{Np-1}{N-1}\right) - p^2 \\
&= \left(\frac{1}{n} - \frac{n-1}{n}\frac{1}{N-1}\right)p + \left(\frac{n-1}{n}\frac{N}{N-1} - 1\right)p^2 \\
&= \frac{N-n}{n(N-1)}p + \frac{n-N}{n(N-1)}p^2 \\
&= \frac{p(1-p)}{n}\frac{N-n}{N-1} = \frac{p(1-p)}{n}\left(1 - \frac{n-1}{N-1}\right)
\end{aligned}
$$

When N is much bigger than n, it is $\frac{p(1-p)}{n}$, which is like we sample n things with replacement (independently).

from Coin tossing process

counting

the importance of the Unit Standard

⟵ **Proof idea**
Identify the exact statement before proving it.

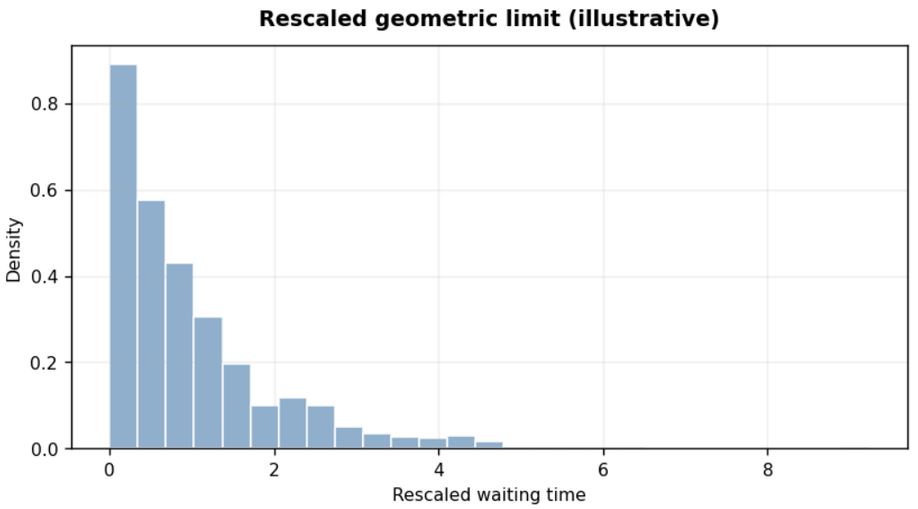


Figure 1: Output from code example 1. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: rgeom, hist.

Sampling distribution

The sampling distribution of a statistic is the probability distribution of that statistic based on a random sample.

Sample mean of i.i.d. normals

Sample mean follows a normal distribution with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

Simulation

Below codes explains the central limit theorem (CLT) and the sampling distribution of the sample proportion.

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

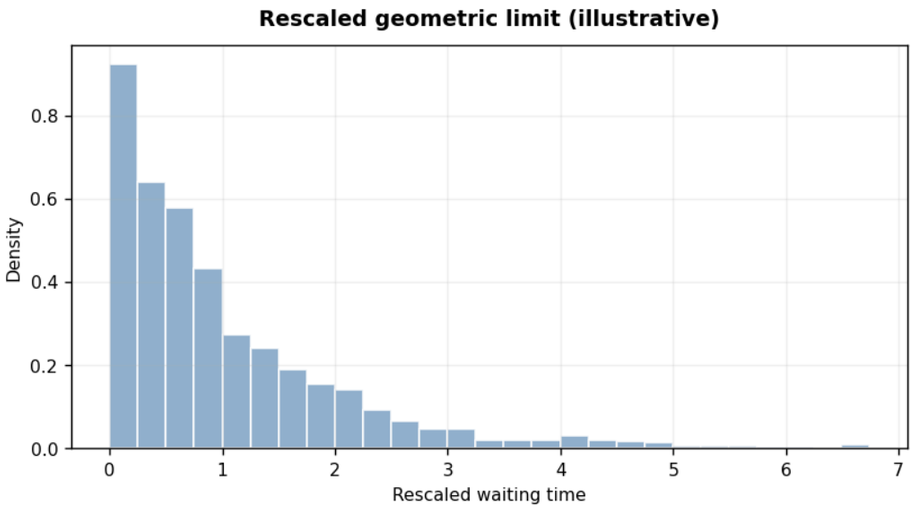


Figure 2: Output from code example 2. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: rgeom, ggplot, geom_histogram, density.

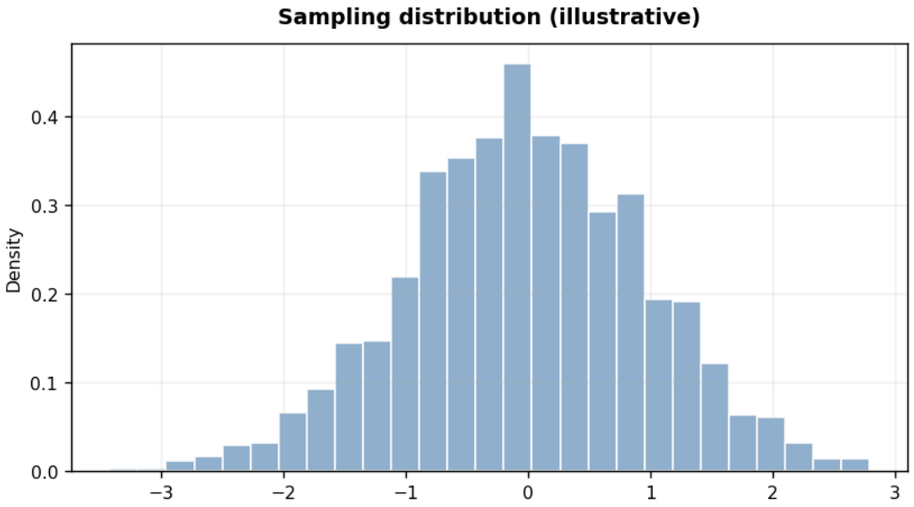


Figure 3: Output from code example 3. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: ggplot, geom_histogram.

simple simulation

The relative frequency will converge to the theoretical value (0.6) as the number of trials increases.

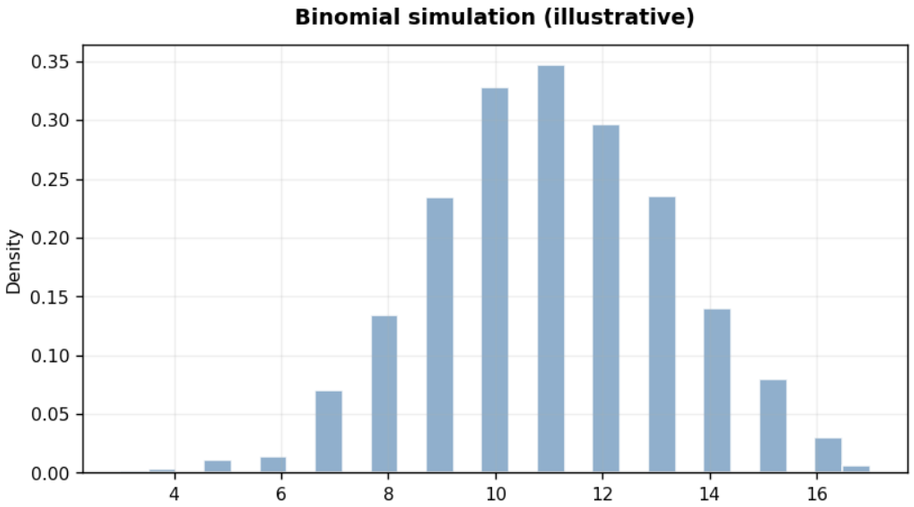


Figure 4: Output from code example 4. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: rbinom, plot, abline.

Moment generating functions

Moment generating function (MGF) and characteristic function are powerful functions that describe the underlying features of a random variable.

Definition

$M_x(t) = \mathbb{E}(e^{tx})$

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Theorems about MGF

1. If X and Y are random variables with the same MGF, which is finite on $[-t_0, t_0]$ for some $t_0 > 0$, then X and Y have the same distribution.

MGFs can be used as a tool to determine if two random variables have the identical CDF.

2. Let $X_1, \cdots, X_n$ be independent random variables, with MGFs $M_{X_1}, \cdots, M_{X_n}$. Then the MGF of their sum is given by
$$M_{X_1+\cdots+X_n}(t) = M_{X_1}(t) \cdots M_{X_n}(t)$$

Example of Gamma and Exponential MGF A gamma distribution with shape $r = 1$ is an exponential distribution. If $X \sim \text{Gamma}(r_X, \lambda)$ and $Y \sim \text{Gamma}(r_Y, \lambda)$ are independent, then we have $X + Y \sim \text{Gamma}(r_X + r_Y, \lambda)$. As a special case, if $X_1, X_2, \cdots, X_n$ are i.i.d. with $\text{Exp}(\lambda)$ distribution, then $X_1 + X_2 + \cdots + X_n$ has $\text{Gamma}(n, \lambda)$ distribution.

Suppose $X \sim \text{Exp}(\lambda)$, for $\lambda > 0$. Then

$$M_X(t) = \mathbb{E}\left[e^{tX}\right] = \int_0^\infty e^{tx}\lambda e^{-\lambda x} dx = \lambda \int_0^\infty e^{(t-\lambda)x} dx$$

Similar to Gamma MGF, the integral of Exponential MGF converges only for $t < \lambda$. For $t < \lambda$, we can integrate:

$$M_X(t) = \lambda \left[\frac{e^{(t-\lambda)x}}{t-\lambda}\right]_0^\infty = \frac{\lambda}{\lambda - t}$$

Since the Gamma MGF is $M_X(t) = \frac{\lambda^r}{(\lambda - t)^r}$ for any $t < \lambda$ For shape $r = 1$, Exponential MGF = Gamma MGF.

Coin tossing process

3 different ways to calculate the expectation of Y=$X_1/X2$ where X_1 and X_2 are r.v. —eg of example 2.1.9 in the textbook

⟵ **Proof idea** Identify the exact statement before proving it.

double integral

mathematical thinking

probability thinking —-conditional expectation

transformation (remark:) —I like this way best

defination of expectation

Distribution Transformation

Universality of the Uniform—From uniform you can get everything

Let u~unif(0,1), F be a CDF(assume F is strictly increased, continuous). Then there comes a theorem:

$$x = F^{-1}(u)$$

Then

$$X \sim F$$

(Proof:

$$P(X \leq x) = P(F^{-1}(u) \leq x) = P(F(F^{-1}(u)) \leq F(x)) = P(u \leq F(x)) = F(x)$$

)

You can convert from the random uniforms to whatever you want to simulate. One example is the simulation of Logistic distribution: F(X)=$e^x/(1 + e^x)$

u~unif(0,1),

$$X = F^{-1}(u) = log(u/1 - u)$$

and X~F

Another example is that we could try to use uniform to simulate normal distribution:

The Box-Muller transform generates pairs of independent standard normally distributed (zero mean, unit variance) random numbers, given a source of uniformly distributed random numbers.

Given two independent random variables U_1 and U_2 that are uniformly distributed on the interval (0, 1), we can generate two independent standard normal random variables Z_0 and Z_1 using the following formulas:

$$Z_0 = \sqrt{-2 \ln U_1} \cos(2\pi U_2)$$

$$Z_1 = \sqrt{-2 \ln U_1} \sin(2\pi U_2)$$

Inversely, the example could be: let Z_0 and Z_1 be standard normal random variables with the following values: Z_0 = 0.5 and Z_1 = -1.0.

1. **Compute CDF values**:

   For standard normal distribution, the CDF

   $$\Phi(z)$$

   is given by:

$$\Phi(z) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{z}{\sqrt{2}}\right)\right]$$

(ps:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

)

- For $Z_0 = 0.5$:

$$\Phi(0.5) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{0.5}{\sqrt{2}}\right)\right]$$

- For $Z_1 = -1.0$:

$$\Phi(-1.0) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{-1.0}{\sqrt{2}}\right)\right]$$

2. **Compute Uniform values**:
   Since the CDF values $\Phi(Z_0)$ and $\Phi(Z_1)$ are in the range [0, 1], we can directly use them as uniform random variables U_0 and U_1.
   - $U_0 = \Phi(0.5)$
   - $U_1 = \Phi(-1.0)$
3. **Result**:
   - Using the error function values:
     - $\Phi(0.5) \approx 0.6915$
     - $\Phi(-1.0) \approx 0.1587$

   Thus, the corresponding uniform distribution values are:
   - $U_0 \approx 0.6915$
   - $U_1 \approx 0.1587$

Another example is that we can create a function that has good quality as F in the theorem, for example,

$$u = F(x) = 1 - e^{-x}(x > 0)$$

then we can simulate X~F:X=-ln(1-u)~F

Also, if

$$X \sim F$$

Then,

$$F(x) \sim unif(0,1)$$

e.g. let X~F, if

$$F(x_0) = 1/3$$

then

$$P(F(X) \leq 1/3) = P(X \leq x_0) = F(x_0) = 1/3$$

which follows the uniform distribution(0,1) because 1/3 generates 1/3. (Uniform distribution: Probability(CDF) is proportional to length)

Normal Distribution

From Standard Normal Distribution to Normal Distribution $f(z) = ce^{-z^2/2}$ is a function with good qualities such as symmetric……..

⟵ **My thought** From a probabilistic model to a usable numerical quantity: what is observed, what is random?

To generate normalization constant c using CDF=1, instead of using impossible integral methods to compute it we should try to find the area under it. In that case, we transfer the integral shape to the double integral's multiplication then we get c:

$$\int_{-\infty}^{\infty} \exp(-z^2/2)dz = \int_{-\infty}^{\infty} \exp(-z^2/2)dz = \int_{-\infty}^{\infty} \exp(-x^2/2)dx \int_{-\infty}^{\infty} \exp(-y^2/2)dy$$

$$= \int_0^{\infty} \exp(-r^2/2)rdr \int_0^{2\pi} d\theta \ = \sqrt{2}\pi$$

So

$$c = 1/\sqrt{2}\pi$$

…….

From…..

111

Exponential distribution the only one parameter is the rate parameter. The probability density function (pdf) of an exponential distribution with rate parameter (□> 0) is given by:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0, \\ 0 & \text{for } x < 0. \end{cases}$$

The cumulative distribution function (cdf) of an exponential distribution with rate parameter (□> 0) is given by:

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & \text{for } x \geq 0, \\ 0 & \text{for } x < 0. \end{cases}$$

2.Let Y~□x, then Y ~Expo(1)

proof: since

$$P(Y \leq y) = P(X \leq y/\lambda) = 1 - e^{-y}$$

(just plot it into x) and we could check that E[X]=Var[X]=1, so X=Y/λ has E[x]=1/λ, Var[x]=1/λ^2

e.g. Memoryless Property(This property implies that the remaining lifetime distribution does not depend on how much time has already elapsed.)(The exponential distribution is the only continuous distribution that has the memoryless property):

$$P(X \geq s + t | X \geq s) = P(X \geq t)$$

which is actually satisfied by the exponential and we could prove it(though it intuitively makes sence): Here

$$P(X \geq s) = 1 - P(X \leq s) = e^{-\lambda s}$$

$$P(X \geq s+t \mid X \geq s) = \frac{P(X \geq s+t \text{ and } X \geq s)}{P(X \geq s)}$$

Since $X \geq s+t$ implies $X \geq s$, we can simplify the numerator:

$$P(X \geq s+t \mid X \geq s) = \frac{P(X \geq s+t)}{P(X \geq s)}$$

Now, substitute the survival function for the exponential distribution:

$$P(X \geq s+t \mid X \geq s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}}$$

Simplify the expression:

$$P(X \geq s+t \mid X \geq s) = \frac{e^{-\lambda s} \cdot e^{-\lambda t}}{e^{-\lambda s}} = e^{-\lambda t}$$

Notice that $e^{-\lambda t} = P(X \geq t)$:

$$P(X \geq s+t \mid X \geq s) = P(X \geq t)$$

usefulness of it:

$$X \sim Expo(\lambda), E(X|X > a) = a + E(X - a|X > a) = a + 1/\lambda$$

e.g.2The hazard rate (or failure rate) for an exponential distribution is constant over time. For an exponential random variable X with rate parameter λ, the hazard rate is:

$$h(t) = \frac{f_X(t)}{1 - F_X(t)} = \lambda.$$

A constant hazard rate implies that the event is equally likely to occur at any point in time, which is a reasonable assumption for many processes, such as the lifetime of certain electronic components or the occurrence of certain types of random failures.

e.g.3 The exponential distribution is closely related to the Poisson process, which is a process that models the occurrence of events happening independently at a constant average rate. If the times between consecutive events in a Poisson process are independent and identically distributed, then these interarrival times follow an exponential distribution. This relationship makes the exponential distribution a natural choice in contexts where events occur randomly over time, such as phone calls arriving at a switchboard or buses arriving at a bus stop.

Ga If r.v.X~N(0,1), then X^2~Ga(1/2,1/2)

Proof: If $X \sim N(0,1)$, then $X^2 \sim$ Ga$(1/2, 1/2)$.

Let (X) be a random variable such that (X ~N(0,1)).

The probability density function (pdf) of (X) is:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad -\infty < x < \infty.$$

First, we find the pdf of (Y = X^2).

The cumulative distribution function (CDF) of (Y) is given by:

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y).$$

Since (X^2 ≥0), we only consider (y ≥0):

$$F_Y(y) = P(-\sqrt{y} \leq X \leq \sqrt{y}).$$

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Using the CDF of the normal distribution, we have:

$$F_Y(y) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\, dx.$$

The pdf of (Y) is the derivative of the CDF:

$$f_Y(y) = \frac{d}{dy} F_Y(y).$$

$$F_Y(y) = 2\int_{0}^{\sqrt{y}} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\, dx.$$

$$F_Y(y) = 2\int_{0}^{\sqrt{y}} \frac{1}{\sqrt{2\pi}} e^{-u} \frac{du}{\sqrt{2u}} = \frac{2}{\sqrt{2\pi}} \int_{0}^{\sqrt{y}} e^{-u} \frac{du}{\sqrt{2u}}.$$

Thus,

$$f_Y(y) = \frac{d}{dy}\left(\frac{2}{\sqrt{2\pi}} \int_{0}^{\sqrt{y}} e^{-u} \frac{du}{\sqrt{2u}} \right).$$

Differentiating with respect to y gives:

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{y}{2}} \cdot y^{-\frac{1}{2}}.$$

Simplifying, we get:

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} y^{-\frac{1}{2}} e^{-\frac{y}{2}}.$$

$$Y = X^2 \sim \text{Ga}\left(\frac{1}{2}, \frac{1}{2}\right).$$

Chi-square Chi-square is a

⟵ **Proof idea**
Identify the exact statement before proving it.

1.Let (Z_1, Z_2, ..., Z_i) are independent standard normal random variables(i.e. Z~N(0,1)), then the random variable (X) defined by

$$X = Z_1^2 + Z_2^2 + \cdots + Z_i^2$$

(Z_j~iid.N(0,1))

follows a chi-square distribution with (i) degrees of freedom. So chi-square of 1 is the same thing as Gamma of (1/2,1/2) so chi-square of n is Gamma(n/2,1/2)

2.If

$$Z_i = \frac{x_i - \mu}{\sigma}$$

The sum of squared standardized deviations is:

$$\sum_{i=1}^{n} Z_i^2 = \sum_{i=1}^{n} \left(\frac{x_i - \mu}{\sigma}\right)^2 = \frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2$$

Let $x_1, x_2, \cdots, x_n$ be samples of $N\left(\mu, \sigma^2\right)$, μ is a known constant, find the distribution of statistics:

$$T = \sum_{i=1}^{n} \left(x_i - \mu\right)^2$$

sol: $y_i = (x_i - \mu)/\sigma, i = 1, 2, \cdots, n$, 则 $y_1, y_2, \cdots, y_n$ are iid r.v. of $N(0,1)$,so

$$\frac{T}{\sigma^2} = \sum_{i=1}^{n} \left(\frac{x_i - \mu}{\sigma}\right)^2 = \sum_{i=1}^{n} y_i^2 \sim \chi^2(n),$$

(i.e.

$$\sum_{i=1}^{n} Z_i^2 \sigma^2 \sim \chi^2(n)$$

)

Besides, T's PDF is

$$p(t) = \frac{1}{(2\sigma^2)^{n/2}\,\Gamma(n/2)} \mathrm{e}^{-\frac{1}{2\sigma^2}t^{\frac{n}{2}-1}}, \quad t > 0,$$

which is Gamma distribution $Ga\left(\frac{n}{2}, \frac{1}{2\sigma^2}\right)$.

3.chi-square is uesful because of theorems below:let $x_1, x_2, \cdots, x_n$ are samples from $N(\mu, \sigma^2)$, whose sample mean and sample variance are

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \text{ and } s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2,$$

then we can get: (1) $\bar{x}$ and s^2 are independent; (2) $\bar{x} \sim N(\mu, \sigma^2/n)$; (3) $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$.

Proof:

$$p(x_1, x_2, \cdots, x_n) = (2\pi\sigma^2)^{-n/2} \mathrm{e}^{-\sum_{i=1}^{n}\frac{(x_i-\mu)^2}{2\sigma^2}} = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\sum_{i=1}^{n} x_i^2 - 2n\bar{x}\mu + n\mu^2}{2\sigma^2}\right\}$$

denote$X = (x_1, x_2, \cdots, x_n)^{\mathrm{T}}$, then we create an n-dimension orthogonal A and every element in the first row is $1/\sqrt{n}$, such as

$$A = \begin{pmatrix} \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \frac{1}{\sqrt{n}} & \cdots & \frac{1}{\sqrt{n}} \\ \frac{1}{\sqrt{2\cdot 1}} & -\frac{1}{\sqrt{2\cdot 1}} & 0 & \cdots & 0 \\ \frac{1}{\sqrt{3\cdot 2}} & \frac{1}{\sqrt{3\cdot 2}} & -\frac{2}{\sqrt{3\cdot 2}} & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ \frac{1}{\sqrt{n(n-1)}} & \frac{1}{\sqrt{n(n-1)}} & \frac{1}{\sqrt{n(n-1)}} & \cdots & -\frac{n-1}{\sqrt{n(n-1)}} \end{pmatrix},$$

令 $Y = (y_1, y_2, \cdots, y_n)^{\mathrm{T}} = AX$, $|Jacobi| = 1$, and we can find thatt

$$\bar{x} = \frac{1}{\sqrt{n}} y_1,$$

$$\sum_{i=1}^{n} y_i^2 = Y^{\mathrm{T}}Y = X^{\mathrm{T}}A^{\mathrm{T}}AX = \sum_{i=1}^{n} x_i^2,$$

so$y_1, y_2, \cdots, y_n$'s joint density function is

$$p(y_1, y_2, \cdots, y_n) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\sum_{i=1}^{n} y_i^2 - 2\sqrt{n}y_1\mu + n\mu^2}{2\sigma^2}\right\}$$

$$= (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\sum_{i=2}^{n} y_i^2 + (y_1 - \sqrt{n}\mu)^2}{2\sigma^2}\right\}$$

So, $Y = (y_1, y_2, \cdots, y_n)^{\mathrm{T}}$ independently distributed as normal distribution and their variances are all equal toσ^2, but their means are not all the same because $y_2, \cdots, y_n$'s means are $0, y_1$'s is $\sqrt{n}\mu$, which ends our proof of (2).

$$(n-1)s^2 = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - (\sqrt{n}\bar{x})^2 = \sum_{i=1}^{n} y_i^2 - y_1^2 = \sum_{i=2}^{n} y_i^2,$$

This proves conclusion (1). Since $y_2, \cdots, y_n$ are independently and identically distributed as $N(0, \sigma^2)$, we have:

$$\frac{(n-1)s^2}{\sigma^2} = \sum_{i=2}^{n} \left(\frac{y_i}{\sigma}\right)^2 \sim \chi^2(n-1).$$

Theorem is proved. (similar to the proof above this maybe easier to understand: besides $Y_1 = \frac{1}{\sqrt{n}}x_1 + \cdots + \frac{1}{\sqrt{n}}x_n = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i$ and $Y_1 = \sqrt{n} \cdot \frac{1}{n}\sum_{i=1}^{n} X_i = \sqrt{n}\bar{X}$, then $\bar{x} = \frac{1}{\sqrt{n}}y_i$ $iz\,(Y_1, Y_2, \cdots, Y_2)^{\top} = A\,(X_1, \cdots, x_n)^{\top}$ then $\sum_{i=1}^{n} Y_i^2 = (Y_1, \cdots, Y_n)\,(Y_1, \cdots, Y_n)^{\top}$ $= \left[A\,(x_1, \cdots, x_n)^{\top}\right]^{\top}\left[A\,(x_1, \cdots, X_n)^{\top}\right]$ stand: $= (x_1, \cdots, x_n)\,A^{\top}A\,(x_1, \cdots, x_n)^{\top}$ $= (x_1, \cdots, x_n)\,E\,(x_1, \cdots, x_n)^{\top} = \sum_{i=1}^{n} x_i^2$)

$$BS^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right]$$

$$= \frac{1}{n-1}\left[\sum_{i=1}^{n} Y_i^2 - Y_1^2\right] = \frac{1}{n-1}\sum_{i=2}^{n} Y_i^2$$

$$2 \oplus L = (\sqrt{2\pi}\sigma)^{-n} \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n} (x_i - \mu)^2\right].$$

$$= (\sqrt{2\pi}\sigma)^{-n} \exp\left[-\frac{1}{2\sigma^2}\left(\sum_{i=1}^{n} x_i^2 - 2\mu n\bar{x} + \mu^2 n\right]\right.$$

$$= (\sqrt{2\pi}\sigma)^{-n} \exp\left[-\frac{1}{2\sigma^2}\left(\sum_{1=1}^{n} {y_i}^2 - 2\mu n\frac{1}{\sqrt{n}}Y_1 + n\mu^2\right]\right.$$

$$= (\sqrt{2}\pi\sigma)^{-1} exp[-1/2\sigma^2(Y_1 - \sqrt{n}u)^2] \times (\sqrt{2}\pi\sigma)^{-1} exp[-1/2\sigma^2 {Y_2}^2] * ... * (\sqrt{2}\pi\sigma)^{-1} exp[-1/2\sigma^2 {Y_n}^2]$$

So L is $Y_1 ... Y_n$'s joint density function and so they are independent. Besides, we have proved that its mean is $1/\sqrt{n}Y_1$ and S^2=$1/n - 1\Sigma i = 2Y_i^2$, so the normal distribution's mean and variance are independent.

When the random variable $\chi^2 \sim \chi^2(n)$, for a given α (where $0 < \alpha < 1$), the value $\chi^2_{1-\alpha}(n)$ satisfying the probability equation $P\left(\chi^2 \leqslant \chi^2_{1-\alpha}(n)\right) = 1 - \alpha$ is called the $1-\alpha$ quantile of the χ^2 distribution with n degrees of freedom.

Suppose the random variables $X_1 \sim \chi^2(m)$ and $X_2 \sim \chi^2(n)$, and X_1 and X_2 are independent. Then the distribution of $F = \frac{X_1/m}{X_2/n}$ is called the F distribution with m and n degrees of freedom, denoted as $F \sim F(m, n)$. Here, m is called the numerator degrees of freedom and n the denominator degrees of freedom. We derive the density function of the F distribution in two steps. First, we derive the density function of $Z = \frac{X_1}{X_2}$. Let $p_1(x)$ and $p_2(x)$ be the density functions of $\chi^2(m)$ and $\chi^2(n)$ respectively. According to the formula for the distribution of the quotient of independent random variables, the

density function of Z is:

$$p_Z(z) = \int_0^\infty x_2 p_1(zx_2)\, p_2(x_2)\, \mathrm{d}x_2$$
$$= \frac{z^{\frac{m}{2}-1}}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right) 2^{\frac{m+n}{2}}} \int_0^\infty x_2^{\frac{n}{2}-1} e^{-\frac{x_2}{2}(1+z)} \mathrm{d}x_2.$$

Using the transformation $u = \frac{x_2}{2}(1+z)$, we get:

$$p_Z(z) = \frac{z^{\frac{m}{2}-1}(1+z)^{-\frac{m+n}{2}}}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)} \int_0^\infty u^{\frac{n}{2}-1} e^{-u}\, \mathrm{d}u$$

The final integral is the gamma function $\Gamma\left(\frac{n}{2}\right)$, so:

$$p_Z(z) = \frac{\Gamma\left(\frac{m+n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)} z^{\frac{m}{2}-1}(1+z)^{-\frac{m+n}{2}}, \quad z \geq 0.$$

Second, we derive the density function of $F = \frac{n}{m}Z$. Let the value of F be y. For $y \geq 0$, we have:

$$p_F(y) = p_Z\left(\frac{m}{n}y\right) \cdot \frac{m}{n} = \frac{\Gamma\left(\frac{m+n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)} \left(\frac{m}{n}y\right)^{\frac{m}{2}-1} \left(1+\frac{m}{n}y\right)^{-\frac{m+n}{2}} \cdot \frac{m}{n}$$
$$= \frac{\Gamma\left(\frac{m+n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)} \left(\frac{m}{n}\right) \left(\frac{m}{n}y\right)^{\downarrow \frac{2}{2}-1} \left(1+\frac{m}{n}y\right)^{-\frac{m+n}{2}}$$

When the random variable $F \sim F(m,n)$, for a given α (where $0 < \alpha < 1$), the value $F_{1-\alpha}(m,n)$ satisfying the probability equation $P\left(F \leqslant F_{1-\alpha}(m,n)\right) = 1-\alpha$ is called the $1-\alpha$ quantile of the F distribution with m and n degrees of freedom. By the construction of the F distribution, if $F \sim F(m,n)$, then $1/F \sim F(n,m)$. Therefore, for a given α (where $0 < \alpha < 1$),

$$\alpha = P\left(\frac{1}{F} \leqslant F_\alpha(n,m)\right) = P\left(F \geqslant \frac{1}{F_\alpha(n,m)}\right).$$

Thus,

$$P\left(F \leqslant \frac{1}{F_\alpha(n,m)}\right) = 1-\alpha$$

This implies

$$F_\alpha(n,m) = \frac{1}{F_{1-\alpha}(m,n)}.$$

Corollary Suppose $x_1, x_2, \cdots, x_m$ is a sample from $N\left(\mu_1, \sigma_1^2\right)$ and $y_1, y_2, \cdots, y_n$ is a sample from $N\left(\mu_2, \sigma_2^2\right)$, and these two samples are independent. Let:

$$s_x^2 = \frac{1}{m-1}\sum_{i=1}^m \left(x_i - \bar{x}\right)^2, \quad s_y^2 = \frac{1}{n-1}\sum_{i=1}^n \left(y_i - \bar{y}\right)^2,$$

where

$$\bar{x} = \frac{1}{m}\sum_{i=1}^m x_i, \quad \bar{y} = \frac{1}{n}\sum_{i=1}^n y_i$$

then

$$F = \frac{s_x^2/\sigma_1^2}{s_y^2/\sigma_2^2} \sim F(m-1, n-1).$$

In particular, if $\sigma_1^2 = \sigma_2^2$, then $F = \frac{s_x}{s_y^2} \sim F(m-1, n-1)$. Proof: Since the two samples are independent, s_x^2 and s_y^2 are independent. According to a Theorem , we have

$$\frac{(m-1)s_x^2}{\sigma_1^2} \sim \chi^2(m-1), \quad \frac{(n-1)s_y^2}{\sigma_2^2} \sim \chi^2(n-1).$$

By the definition of the F distribution, $F \sim F(m-1, n-1)$. Corollary: Suppose $x_1, x_2, \cdots, x_n$ is a sample from a normal distribution $N(\mu, \sigma^2)$, and let $\bar{x}$ and s^2 denote the sample mean and sample variance of the sample, respectively. Then

$$t = \frac{\sqrt{n}(\bar{x} - \mu)}{s} \sim t(n-1).$$

Proof: From a Theorem we obtain

$$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Then,

$$\frac{\sqrt{n}(\bar{x} - \mu)}{s} = \frac{\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)s^2/\sigma^2}{n-1}}}$$

Since the numerator is a standard normal variable and the denominator's square root contains a χ^2 variable with $n-1$ degrees of freedom divided by its degrees of freedom, and they are independent, by the definition of the t distribution, $t \sim t(n-1)$. The proof is complete.

Corollary: In the notation of Corollary , assume $\sigma_1^2 = \sigma_2^2 = \sigma^2$, and let

$$s_w^2 = \frac{(m-1)s_x^2 + (n-1)s_y^2}{m+n-2} = \frac{\sum_{i=1}^m (x_i - \bar{x})^2 + \sum_{i=1}^n (y_i - \bar{y})^2}{m+n-2}$$

Then

$$\frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{s_w\sqrt{\frac{1}{m} + \frac{1}{n}}} \sim t(m+n-2)$$

Proof: Since $\bar{x} \sim N\left(\mu_1, \frac{\sigma^2}{m}\right), \bar{y} \sim N\left(\mu_2, \frac{\sigma^2}{n}\right)$, and $\bar{x}$ and $\bar{y}$ are independent, we have

$$\bar{x} - \bar{y} \sim N\left(\mu_1 - \mu_2, \left(\frac{1}{m} + \frac{1}{n}\right)\sigma^2\right).$$

Thus,

$$\frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{\frac{1}{m} + \frac{1}{n}}} \sim N(0, 1).$$

By a Theorem , we know that $\frac{(m-1)s_x^2}{\sigma^2} \sim \chi^2(m-1)$ and $\frac{(n-1)s_y^2}{\sigma^2} \sim \chi^2(n-1)$, and they are independent. By additivity, we have

$$\frac{(m+n-2)s_w^2}{\sigma^2} = \frac{(m-1)s_x^2 + (n-1)s_y^2}{\sigma^2} \sim \chi^2(m+n-2).$$

Since $\bar{x} - \bar{y}$ and s_w^2 are independent, by the definition of the t distribution, we get the desired result. □

One interesting example shows the relationship of above distributions used charismatically to solve problems: r.v.: $X_1\ X_2\ X_3\ X_4$ indpendently identically distribute(iid) as $N\left(0.\sigma^2\right)$. $Z = \left(x_1^2+x_2^2\right)/\left(x_1^2+x_2^2+x_3^2+x_4^2\right)$ prove: $Z \sim U(0.1)$.

Solution: $LetY = \dfrac{X_3^2+X_4^2}{X_1^2+X_2^2} = \dfrac{\left[\left(\frac{X_3}{\sigma}\right)^2+\left(\frac{X_4}{\sigma}\right)^2\right]/2}{\left[\left(\frac{X_1}{\sigma}\right)^2+\left(\frac{X_2}{\sigma}\right)^2\right]/2} \sim F(2,2).$

i.e. $f_Y(y) = \dfrac{1}{(1+y)^2}, y>0$

then $P(Z \le z) = P\left(\dfrac{1}{1+Y} \le z\right) = P\left(Y \geqslant \dfrac{1}{z}-1\right)$

$= \displaystyle\int_{\frac{1}{z}-1}^{+\infty} \frac{1}{(1+y)^2} dy = z \quad \text{H } 0<z<1.$

$\therefore Z \sim U(0.1)$

(ps:$

PDF functions's transformation(from books)

Standard normal distribution

Pdf : $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

- The calculation of the Gaussian integral is to transform the integral into the double integral and then use the polar coordinate

similar question the calculation of the n-dimensional ball (in the second picture that I summarized before), which follows the idea of using the double integral to calculate Gaussian integral and the idea of iterative method. Then I feel how charismatic "a multiple integral in a high-dimensional space can be transformed into an iterated integral where each integral is just a definite integral on an interval" that you said is!

The definition of an N-dimensional sphere with the radius 1 is:

$$B^n = \{(x_1, x_2, \cdots, x_n) \in \mathbb{R}^n \mid x_1^2 + \cdots + x_n^2 \le 1\}$$

(Denote $V_n(r)$ as the volume of N-dimensional sphere with radius r)

Derivation of the recursive formula for the volume of an N-dimensional sphere:

$$V_n(1) = \int_{B^n} dx_1 \cdots dx_n$$

By using the recurrence relation:

$$V_n(1) = \int_{-1}^{1} V_{n-1}(\sqrt{1-x_1^2}) dx_1$$

Further elaboration:

$$= \int_{-1}^{1} \int_{-\sqrt{1-x_1^2}}^{\sqrt{1-x_1^2}} V_{n-2}(\sqrt{1-x_1^2-x_2^2}) dx_1 dx_2$$

⟵ My thought
The theorem is easier to remember when its hypothesis feels necessary.

In polar coordinates:

$$= \int_0^{2\pi} \int_0^1 V_{n-2}(\sqrt{1-r^2}) r dr d\theta$$

Let $y = 1 - r^2$, so $dy = -2r\,dr$. Therefore

$$= \int_0^{2\pi} \int_0^1 \frac{1}{2} y^{(n-2)/2} V_{n-2}(1)\,dy\,d\theta.$$

The calculation shows:

$$= 2\pi \left[\frac{1}{2} V_{n-2}(1) \cdot \frac{y^{\frac{n}{2}}}{\frac{n}{2}} \right]_0^1 = \frac{2\pi}{n} V_{n-2}(1)$$

Therefore, the recurrence relation is obtained:

$$V_n(1) = \frac{2\pi}{n} V_{n-2}(1)$$

Consider the even and odd dimensions respectively:

For an even dimension $n = 2k$:

$$V_{2k}(1) = \frac{\pi^k}{k!}$$

$$V_{2k}(R) = \frac{\pi^k R^{2n}}{k!}$$

For odd-numbered dimensions $n = 2k + 1$:

$$V_{2k+1}(1) = \frac{2^{k+1}\pi^k}{(2k+1)(2k-1)\cdots 3 \cdot 1}$$

$$V_{2k+1}(R) = \frac{2^{k+1}\pi^k R^{2n+1}}{(2k+1)(2k-1)\cdots 3 \cdot 1}$$

Generally, the volume of an N-dimensional sphere with radius R is:

$$V_n(R) = \frac{\pi^{\frac{n}{2}} R^n}{\Gamma\left(\frac{n}{2} + 1\right)}$$

- asymptotic behavior when $n \to \infty$

When the dimension $n$approaches infinity, the volume of the n-dimensional unit sphere satisfies:

$$\lim_{n\to\infty} V_n(R) = 0$$

This indicates that in high-dimensional space, the volume of the sphere is concentrated near the spherical surface, and the interior of the sphere becomes “sparse”.–A topology problem

Chi-square distribution

From standard normal distribution $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, we want to get Chi-square distribution $f(x;k) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$

⟵ **Motivation**
From a probabilistic model to a usable numerical quantity: what is observed, what is random?

Assuming that $Z \sim N(0,1)$
$W = Z^2$
So we can deduce the pdf for the variable W

$$\begin{aligned} f_W(w) &= Pr(Z^2 = w) \\ &= f_Z(\sqrt{w}) \left| \frac{d(\sqrt{w})}{dw} \right| + f_Z(-\sqrt{w}) \left| \frac{d(-\sqrt{w})}{dw} \right| \\ &= 2 \cdot \frac{1}{\sqrt{2\pi}} e^{-w/2} \cdot \frac{1}{2\sqrt{w}} = \frac{1}{\sqrt{2\pi w}} e^{-w/2} \end{aligned}$$

$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt$
$\Gamma(1/2)=\sqrt{(\pi)}$,we can know that $W \sim \chi^2(1)$

Now, we can introduce the variable Y.
$Y = \sum_{i=1}^k Z_i^2$ (Z_i is independent of each other)
We want to get the mgf of Y

$$\begin{aligned} M_Y(t) &= E[e^{tY}] \\ &= E[e^{tZ_1^2} e^{tZ_2^2} ... e^{tZ_k^2}] \\ &= E[e^{tZ_1^2}] E[e^{tZ_2^2}] ... E[e^{tZ_k^2}] \\ &= \prod_{i=1}^k M_{Z_i^2}(t) \\ &= (1-2t)^{-r1/2} (1-2t)^{-r2/2} ... (1-2t)^{-rk/2} \\ &= (1-2t)^{-\sum_{i=1}^k ri/2} \end{aligned}$$

Because the mgf of Chi-square function is $(1-2t)^{-r/2}$
r is the degree of freedom of this chi-square function
So $Y \sim \chi^2(r1 + r2 + ... + rk)$
r1,r2...rk represents the degree of freedom of every single sample
Here, df=1 for every sample

Student- t distribution

From standard normal distribution $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, we want to get Student- t distribution $f(t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\,\Gamma(\frac{\nu}{2})} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$, where Γ is the gamma function and ν is the degrees of freedom.

⟵ **Proof idea**
Condition explicitly; it often simplifies the calculation.

Given two independent variables Z and V ($Z \sim N(0,1)$ & $V \sim \chi^2(\nu)$), then we construct a new variable $T = \frac{Z}{\sqrt{(V/\nu)}}$.
The joint pdf is

$$g(z,v) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \frac{1}{2^{\nu/2}\Gamma(\nu/2)} v^{\nu/2-1} e^{-v/2}$$

Cdf of T is given by

$$\begin{aligned} F(t) &= Pr(\frac{Z}{\sqrt{(V/\nu)}} \leq t) \\ &= Pr(Z \leq \sqrt{(V/\nu)}t) \\ &= \int_0^\infty \int_{-\infty}^{\sqrt{(V/\nu)}t} g(z,v)\, dz\, dv \end{aligned}$$

Simplify F(t)

$$F(t) = \frac{1}{\sqrt{\pi}\Gamma(\nu/2)} \int_0^\infty [\int_{-\infty}^{\sqrt{(V/\nu)}t} \frac{e^{-z^2/2}}{2^{\frac{\nu+1}{2}}} dz] v^{\frac{\nu}{2}-1} e^{-\frac{v}{2}} dv$$

To get pdf, we will differentiate F(t)

$$\begin{aligned} f(t) = F'(t) &= \frac{1}{\sqrt{\pi}\Gamma(\nu/2)} \int_0^\infty \frac{e^{-(v/2)(t^2/\nu)}}{2^{\frac{\nu+1}{2}}} \sqrt{\frac{v}{\nu}} v^{\nu/2-1} e^{-\frac{v}{2}} dv \\ &= \frac{1}{\sqrt{\pi\nu}\Gamma(\nu/2)} \int_0^\infty \frac{v^{(\nu+1)/2-1}}{2^{(\nu+1)/2}} e^{-(\nu/2)(1+t^2/\nu)} dv \end{aligned}$$

We need to make the change of variables: $y = (1 + t^2/\nu)v$
And we need to change dv: $\frac{dv}{dy} = \frac{1}{1+t^2/\nu}$

$$f(t) = \frac{\Gamma[(\nu+1)/2]}{\sqrt{\pi\nu}\Gamma(\nu/2)} [\frac{1}{(1+t^2/\nu)^{(\nu+1)/2}}] \int_0^\infty \frac{y^{(\nu+1)/2-1}}{\Gamma[(\nu+1)/2]2^{(\nu+1)/2}} e^{-y/2} dy$$

This part $\int_0^\infty \frac{y^{(\nu+1)/2-1}}{\Gamma[(\nu+1)/2]2^{(\nu+1)/2}} e^{-y/2} dy$ is equal to 1, because this part is the whole area under the chi-square distribution with $\nu + 1$ degrees of freedom. So, the pdf for T can be written as follows

$$f(t) = \frac{\Gamma[(\nu+1)/2]}{\sqrt{\pi\nu}\Gamma(\nu/2)} \frac{1}{(1+t^2/\nu)^{(\nu+1)/2}}$$

F-distribution

We will do some trnsformation on chi-square distribution to get F-distribution

⟵ **My thought** From a probabilistic model to a usable numerical quantity: what is observed, what is random?

Assuming that we have two independent random variables

$$X \sim \chi^2(n_1) \quad and \quad Y \sim \chi^2(n_2)$$

Now,we will define a new variable F

$$F = \frac{(X/n_1)}{(Y/n_2)}$$

This looks a little complex, so let's do some simplification.

$$U = \frac{X}{n_1} \quad \text{and} \quad V = \frac{Y}{n_2}$$

So, $F = \frac{U}{V}$.
Because, X and Y are independent of each other. Obviously, U and V are also independent of each other.

$$f_{U,V}(u,v) = f_U(u) f_V(v)$$

$$f_U(u) = \frac{(n_1 u)^{n_1/2-1} e^{-n_1 u/2}}{2^{n_1/2}\Gamma(n_1/2)}$$

$$f_V(v) = \frac{(n_2 v)^{n_2/2-1} e^{-n_2 v/2}}{2^{n_2/2}\Gamma(n_2/2)}$$

To find the joint density function of F & V, we use Jacobian transformation

$$J = \left|\frac{\partial(U,V)}{\partial(F,V)}\right| = \begin{vmatrix} \frac{\partial U}{\partial F} & \frac{\partial U}{\partial V} \\ \frac{\partial V}{\partial F} & \frac{\partial V}{\partial V} \end{vmatrix} = \begin{vmatrix} V & F \\ 0 & 1 \end{vmatrix} = V$$

$$f_{F,V}(f,v) = f_{U,V}(u,v)\left|\frac{\partial(u,v)}{\partial(f,v)}\right| = f_{U,V}(u,v)v$$

Then, we substitute$f_U(u)$ and $f_V(v)$ into $f_{F,V}(f,v)$

$$f_{F,V}(f,v) = \frac{(n_1 u)^{n_1/2-1} e^{-n_1 u/2}}{2^{n_1/2}\Gamma(n_1/2)} \cdot \frac{(n_2 v)^{n_2/2-1} e^{-n_2 v/2}}{2^{n_2/2}\Gamma(n_2/2)} \cdot v$$

Use the condition u=fv

$$\begin{aligned} f_{F,V}(f,v) &= \frac{(n_1 fv)^{n_1/2-1} e^{-n_1 fv/2}}{2^{n_1/2}\Gamma(n_1/2)} \cdot \frac{(n_2 v)^{n_2/2-1} e^{-n_2 v/2}}{2^{n_2/2}\Gamma(n_2/2)} \cdot v \\ &= \frac{(n_1 fv)^{n_1/2-1} e^{-n_1 fv/2}}{2^{n_1/2}\Gamma(n_1/2)} \cdot \frac{(n_2 v)^{n_2/2-1} e^{-n_2 v/2}}{2^{n_2/2}\Gamma(n_2/2)} \cdot v \\ &= \frac{n_1^{n_1/2-1} f^{n_1/2-1} v^{n_1/2-1} e^{-n_1 fv/2}}{2^{n_1/2}\Gamma(n_1/2)} \cdot \frac{n_2^{n_2/2-1} v^{n_2/2-1} e^{-n_2 v/2}}{2^{n_2/2}\Gamma(n_2/2)} \cdot v \\ &= \frac{n_1^{n_1/2-1} n_2^{n_2/2-1} f^{n_1/2-1} v^{(n_1+n_2)/2-1} e^{-(n_1 f+n_2)v/2}}{2^{(n_1+n_2)/2}\Gamma(n_1/2)\Gamma(n_2/2)} \end{aligned}$$

We will integrate this density function with respect to V

$$\begin{aligned} f_F(f) &= \int_0^\infty f_{F,V}(f,v)dv \\ &= \int_0^\infty \frac{n_1^{n_1/2-1} n_2^{n_2/2-1} f^{n_1/2-1} v^{(n_1+n_2)/2-1} e^{-(n_1 f+n_2)v/2}}{2^{(n_1+n_2)/2}\Gamma(n_1/2)\Gamma(n_2/2)} dv \end{aligned}$$

Let’s do some substitutions to make it look simpler

$$c = \frac{n_1 f + n_2}{2}$$

This is the definition of Gamma function

$$\int_0^\infty v^{(n_1+n_2)/2-1} e^{-cv} dv = \frac{\Gamma((n_1+n_2)/2)}{c^{(n_1+n_2)/2}}$$

Then

$$f_F(f) = \frac{n_1^{n_1/2-1} n_2^{n_2/2-1} f^{n_1/2-1}}{2^{(n_1+n_2)/2}\Gamma(n_1/2)\Gamma(n_2/2)} \cdot \frac{\Gamma((n_1+n_2)/2)}{\left(\frac{n_1 f+n_2}{2}\right)^{(n_1+n_2)/2}}$$

Let’s rewrite it in an approximate F-distribution pdf form

$$f_F(f) = \frac{n_1^{n_1/2-1} n_2^{n_2/2-1} f^{n_1/2-1}\Gamma((n_1+n_2)/2)}{2^{(n_1+n_2)/2}\Gamma(n_1/2)\Gamma(n_2/2)} \cdot \left(\frac{2}{n_1 f + n_2}\right)^{(n_1+n_2)/2}$$

beta and Gamma

⟵ **Motivation** Why does this definition matter later? Try an example and a counterexample.

When considering the product of two Gamma functions

$$(\Gamma(a)\Gamma(b))$$

, we can write it as two independent integrals:

$$\Gamma(a)\Gamma(b) = \left(\int_0^\infty t^{a-1}e^{-t}\,dt\right)\left(\int_0^\infty s^{b-1}e^{-s}\,ds\right)$$

To convert this product into a double integral, we can use a change of variables. Consider using polar coordinates in the two independent integrals, which allows us to use certain integration techniques to simplify them. First, we transform the integration region from Cartesian coordinates to polar coordinates:

$$t = r\cos\theta, \quad s = r\sin\theta$$

The Jacobian determinant is

$$r$$

, so the integral can be rewritten as:

$$\Gamma(a)\Gamma(b) = \int_0^\infty \int_0^\infty t^{a-1}e^{-t}s^{b-1}e^{-s}\,dt\,ds$$

Using the polar coordinate transformation, the integral becomes:

$$= \int_0^{\frac{\pi}{2}} \int_0^\infty (r\cos\theta)^{a-1}e^{-r\cos\theta}(r\sin\theta)^{b-1}e^{-r\sin\theta}r\,dr\,d\theta$$

Separating all

$$r$$

and

$$\theta$$

related terms:

$$= \int_0^{\frac{\pi}{2}} (\cos\theta)^{a-1}(\sin\theta)^{b-1}\,d\theta \int_0^\infty r^{a+b-1}e^{-r(\cos\theta+\sin\theta)}\,dr$$

First, compute the

$$r$$

integral part:

$$\int_0^\infty r^{a+b-1}e^{-r(\cos\theta+\sin\theta)}\,dr = \frac{\Gamma(a+b)}{(\cos\theta+\sin\theta)^{a+b}}$$

Next, compute the

$$\theta$$

integral part:

$$\int_0^{\frac{\pi}{2}} (\cos\theta)^{a-1}(\sin\theta)^{b-1}\,d\theta = B(a,b)$$

where

$$B(a,b)$$

is the Beta function, defined as:

$$B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1}\,dt$$

Therefore, we get:

$$\Gamma(a)\Gamma(b) = \frac{\Gamma(a+b)}{(\cos\theta+\sin\theta)^{a+b}} \cdot B(a,b)$$

Since

$$\cos\theta + \sin\theta = 1$$

(in polar coordinates), we have:

$$\Gamma(a)\Gamma(b) = \Gamma(a+b)B(a,b)$$

Thus, the double integral form of the Gamma function product directly comes from the polar transformation and the application of the Beta function. This is an important mathematical technique used to simplify complex integrals and functional relationships.

Why is the integral result as shown?

⟵ **Proof idea**
Choose an order of integration that respects symmetry.

The given integral,

$$\int_0^\infty r^{a+b-1}e^{-r(\cos\theta+\sin\theta)}\,dr,$$

can be evaluated using the definition of the Gamma function. The Gamma function

$$\Gamma(z)$$

is defined as:

$$\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}\,dt.$$

To see why the integral can be expressed in terms of the Gamma function, let's rewrite the integral in a form that matches the Gamma function's definition. The integral has the form:

$$\int_0^\infty r^{a+b-1}e^{-r(\cos\theta+\sin\theta)}\,dr.$$

Let's set

$$t = r(\cos\theta+\sin\theta)$$

, then

$$r = \frac{t}{\cos\theta+\sin\theta}$$

and

$$dr = \frac{dt}{\cos\theta+\sin\theta}$$

. Substituting these into the integral gives:

$$\int_0^\infty \left(\frac{t}{\cos\theta+\sin\theta}\right)^{a+b-1} e^{-t}\cdot\frac{dt}{\cos\theta+\sin\theta}.$$

Simplifying inside the integral:

$$\int_0^\infty \frac{t^{a+b-1}}{(\cos\theta+\sin\theta)^{a+b}}e^{-t}\,dt.$$

Since

$$(\cos\theta+\sin\theta)^{a+b}$$

is a constant with respect to

$$t$$

, it can be factored out of the integral:

$$\frac{1}{(\cos\theta+\sin\theta)^{a+b}}\int_0^\infty t^{a+b-1}e^{-t}\,dt.$$

The integral

$$\int_0^\infty t^{a+b-1}e^{-t}\,dt$$

is recognized as the Gamma function

$$\Gamma(a+b)$$

. Thus, we have:

$$\frac{1}{(\cos\theta+\sin\theta)^{a+b}}\Gamma(a+b).$$

Therefore,

$$\int_0^\infty r^{a+b-1}e^{-r(\cos\theta+\sin\theta)}\,dr=\frac{\Gamma(a+b)}{(\cos\theta+\sin\theta)^{a+b}}.$$

This demonstrates why the integral is evaluated as shown:

$$\int_0^\infty r^{a+b-1}e^{-r(\cos\theta+\sin\theta)}\,dr=\frac{\Gamma(a+b)}{(\cos\theta+\sin\theta)^{a+b}}.$$

(method 2 for integral calculation: x=uv,y=u(1-v) J=-u)

Three Main Sampling distribution

Chi-square distribution

Suppose $X_1,\cdots,X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$.the distribution of the statistic

$$X_1^2+\cdots+X_n^2$$

is called a chi-square distribution with n degrees of freedom, denoted by $\chi^2(n)$.

Besides, random variable $X_i^2 \sim \text{Gamma}\left(\frac{1}{2},\frac{1}{2}\right)$ corresponds to the chi-squared distribution with 1 degree of freedom, denoted as χ_1^2.

⟵ **My thought**
From a probabilistic model to a usable numerical quantity: what is observed, what is random?

This is derived by the MGF:

Since

$$M_{X_1^2+\cdots+X_n^2}(t) = M_{X_1^2}(t) \times \cdots \times M_{X_n^2}(t) = \begin{cases} \infty & t \geq \frac{1}{2} \\ (1-2t)^{-\frac{n}{2}} & t < \frac{1}{2} \end{cases}$$

This is the MGF of the Gamma $\left(\frac{n}{2}, \frac{1}{2}\right)$ distribution, so $X_1^2 + \cdots + X_n^2 \sim$ Gamma $\left(\frac{n}{2}, \frac{1}{2}\right)$. This is called the chi-squared distribution with **n** degree of freedom, denoted χ_n^2.

Properties

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

- If $W_1, \ldots, W_n$ are independent χ^2 random variables with, respectively, $v_1, \cdots, v_n$ degrees of freedom, then the random variable $W_1 + \cdots + W_n$ follows a χ^2-distribution with $v_1 + \cdots + v_n$ degree of freedom.
- The random variable $\frac{(\bar{X}-\mu)^2}{\sigma^2/n}$ follows a χ^2-distribution with 1 degree of freedom when X follows a normal distribution with mean μ and variance σ^2.

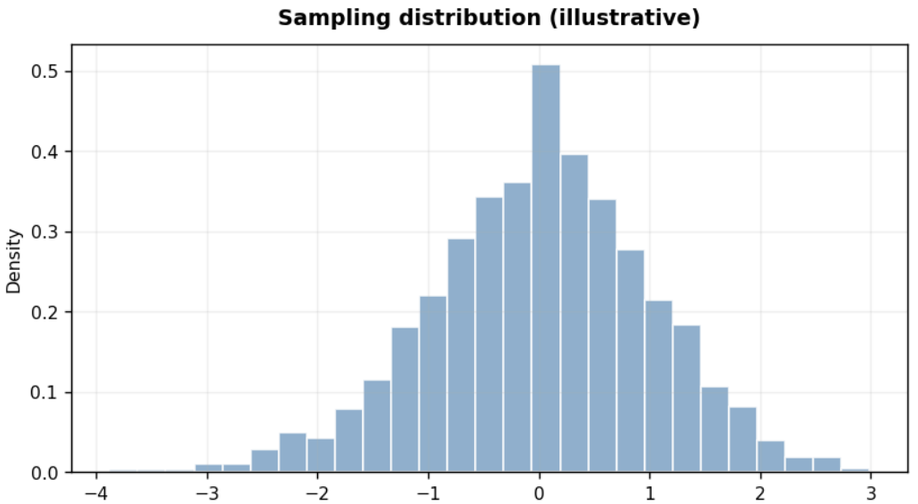


Figure 5: Output from code example 5. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: ggplot, geom_line.

Code to plot examples of Chi-square distribution

Application Chi-square distribution is primarily used in testing:

- Goodness-of-fit
- Independence in contingency tables

Student's t-distribution

⟵ **Proof idea**
Condition explicitly; it often simplifies the calculation.

Construction The statistic $T = \frac{\bar{X}-\mu}{S/\sqrt{n}}$ follows a t-distribution with $v = n-1$ degrees of freedom when $X_1, \cdots, X_n$ are i.i.d. normal RVs.

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \quad \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1) \quad \frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$$

If we know the population variance σ^2, we can easily do inference using the statistic $\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$. However, σ^2 is usually unknown in practice.

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \quad \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1) \quad \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

We can construct the t-statistic using the sample variance S^2 :

$$T = \frac{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)s^2}{\sigma^2}/(n-1)}} = \frac{\bar{X}-\mu}{S/\sqrt{n}}$$

Notice the sample mean $\bar{X}$ and the sample variance S^2 are independent (the proof is beyond the scope of this course). So the T is now a ratio of a standard normal variable and the square root of a χ^2RV divided by its degrees of freedom. This is the definition of a t-distribution with $n-1$ degrees of freedom.

Properties The t-distribution is primarily used in contexts where the underlying population is assumed to be normally distributed, especially when the sample size is small. Used extensively in problems that deal with inference about population mean μ when population variance σ^2 is unknown; problems where one is trying to determine if means from two samples are significantly different when population variances σ_1^2 and σ_2^2 are unknown.

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

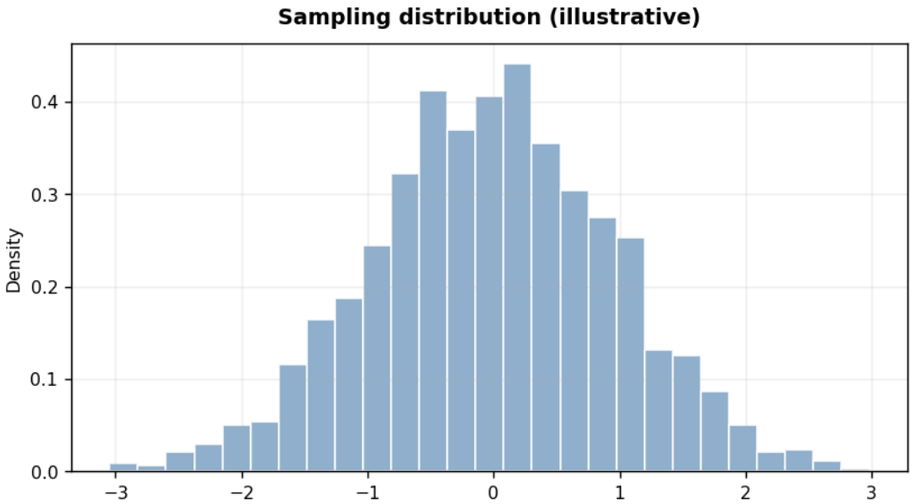


Figure 6: Output from code example 6. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: ggplot, geom_line, dnorm.

Code to plot examples of t-distribution

F-distribution

Let U and V be two independent RVs following χ^2 distributions with ν_1 and ν_2 degrees of freedom, respectively. Then the distribution of the random variable $F = \frac{U/\nu_1}{V/\nu_2}$ is known as F-distribution.

Example If S_1^2 and S_2^2 are the variances of independent RVs of size n_1 and n_2 taken from normal populations with variances σ_1^2 and σ_2^2 respectively, then

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2}$$

follows an F-distribution with $\nu_1 = n_1 - 1$ and $\nu_2 = n_2 - 1$ degrees of freedom.

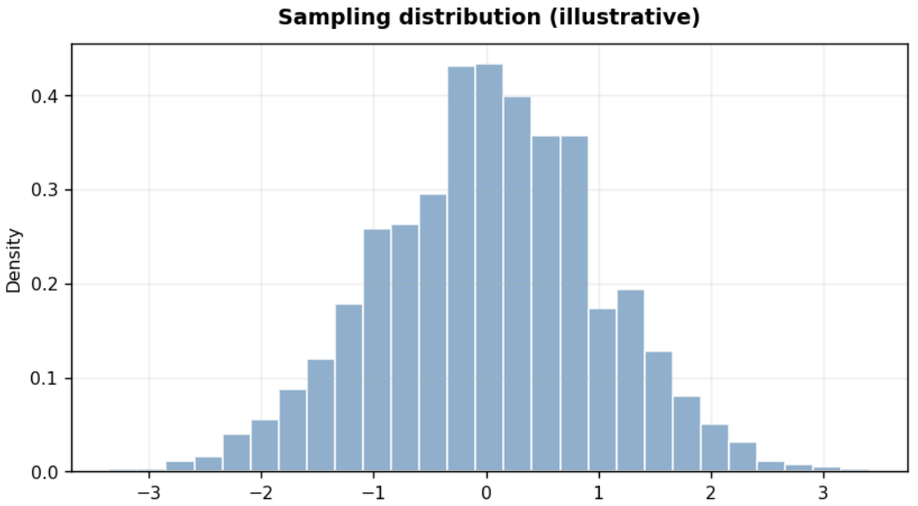


Figure 7: Output from code example 7. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: ggplot, geom_line.

Code to plot examples of F-distribution

Application Analysis of variance (ANOVA)

Estimation of Population Characteristic

Point Estimation

A point estimate of a population characteristic is a single number that is based on sample data and represents a plausible value of the characteristic.

Interval Estimation—Confidence interval(CI)

⟵ **Proof idea**
Often the log-likelihood is simpler than the likelihood.

An interval estimate of a parameter θ is an interval of the form $\hat{\theta}_L < \theta < \hat{\theta}_U$, where $\hat{\theta}_L$ and $\hat{\theta}_U$ depend on the value of $\hat{\theta}$ for a particular sample and also on the sampling distribution of $\hat{\Theta}$.

Introduction

- If we were to construct a 95% confidence interval for some population characteristics (population proportion p or population mean μ), we would be using a method that is successful 95% of the time.
- This is also about the question relevant to “How to choose a sample size”

Definition

A $100(1-\alpha)\%$ confidence interval is an interval of the form $\hat{\theta}_L < \theta < \hat{\theta}_U$, where $\hat{\theta}_L$ and $\hat{\theta}_U$ are respectively values of $\widehat{\Theta}_L$ and $\widehat{\Theta}_U$ obtained for a particular sample, based on

$$P\left(\widehat{\Theta}_L < \theta < \widehat{\Theta}_U\right) = 1-\alpha \quad ; \quad 0 < \alpha < 1$$

in the estimation of population parameter θ.

Interpretation

For confidence level of 95% CI for any normal distribution: About 95% of the values are within 1.96 standard deviations of the mean. (Recall the concept of Z-scores)

That is, if Estimate $\pm$ $(Z \times \sigma)$ was used to generate an interval estimate over and over again with different samples, in the long run 95% of the resulting intervals would include the actual value of the characteristic being estimated.

- The confidence level 95% refers to the method used to construct the interval rather than to any particular interval, such as the one we obtained.

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

eg

Proportion

- check to make sure that the three necessary conditions are met:

$n\hat{p} \geq 10, n(1-\hat{p}) \geq 10$

$\frac{\hat{p}-p}{\sigma/\sqrt{n}} = 1.96)$

- sample size question

Using sample proportion with 95% confidence interval, and we want ME no more than 5%, we may be interested in solving n from the following equation:

0.05=1.96 $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

CI on Mean Here, $\bar{x}$ is the sample mean from a simple random sample.

μ is the population mean which we are interested in estimating.

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

CI on μ with σ known n ≥ 30 or the population is normal — Use z-statistics CI: $\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$, for example, 95% CI is $\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$

One-side Confidence Bound on μ with σ known n ≥ 30 or the population is normal — Use z-statistics Upper one-side bound: $\mu < \bar{x} + z_{\alpha}\frac{\sigma}{\sqrt{n}}$

Lower one-side bound: $\mu > \bar{x} - z_{\alpha}\frac{\sigma}{\sqrt{n}}$

For example, 95% Confidence bound on μ is $\bar{x} \pm 1.645\frac{\sigma}{\sqrt{n}}$

CI on μ with σ unknown and the population is normal — Use t-statistics (use s as the estimate for σ (t-statistics with df = n-1)) CI: $\bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}$, for example, 95% CI is $\bar{x} \pm t_{0.025}\frac{s}{\sqrt{n}}$ and df = n-1

Remark: The distribution of t is more spread out than the standard normal distribution but when n ≥ 30, t and z are very close to each other.

CI for $\mu_1 - \mu_2$, both σ_1^2 and σ_2^2 are known CI of $\mu_1 - \mu_2$: $(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

CI for $\mu_1 - \mu_2$, both σ_1^2 and σ_2^2 are unknown but assumed equal CI of $\mu_1 - \mu_2$: $(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2}s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ with df = $n_1 + n_2 - 2$ where $s_p = \sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}}$

CI for paired observations Previous, we have two independent samples, now we have two dependent samples. We can use the difference between the two samples to construct a confidence interval.

CI of μ_d: $\bar{d} \pm t_{\alpha/2}\frac{s_d}{\sqrt{n}}$ with df = n-1 where s_d is the sample standard deviation of the differences $d_i = x_{1i} - x_{2i}$ and $\bar{d}$ is the sample mean of the differences.

Estimating σ a $100(1-\alpha)\%$ CI for σ^2 is $\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right)$

where S^2 is the sample variance and $\chi^2_{\alpha/2,n-1}$ and $\chi^2_{1-\alpha/2,n-1}$ are the critical values of the chi-square distribution with $n-1$ degrees of freedom.

Estimating σ_1^2/σ_2^2 A $100(1-\alpha)\%$ CI for $\frac{\sigma_1^2}{\sigma_2^2}$ using F-statistics with $f_{1-\alpha/2}(n_1-1, n_2-1) = 1/f_{\alpha/2}(n_1-1, n_2-1)$ is

$$\frac{s_1^2}{s_2^2}\frac{1}{f_{\alpha/2}(n_1-1, n_2-1)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} f_{\alpha/2}(n_2-1, n_1-1)$$

where $f_{\alpha/2}(v_1, v_2)$ is an F-value with v_1 and v_2 degrees of freedom, leaving an area of $\alpha/2$ to the right, and $f_{\alpha/2}(v_2, v_1)$ is a similar F-value with v_2 and v_1 degrees of freedom.

Hypothesis Testing

A hypothesis test compares an observed statistic with its null reference distribution and a chosen rejection region.

⟵ **Proof idea** Identify the exact statement before proving it.

- z-test

Suppose $X_1, \cdots, X_n \xrightarrow[\sim]{\text{iid}} \mathcal{N}(\mu, \sigma^2)$, where μ is unknown and where $\sigma^2 = \sigma_0^2$ is known.

Suppose we wish to test $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$. Then we can use the test statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma_0/\sqrt{n}}$$

If H_0 is true then $Z \sim \mathcal{N}(0,1)$. Let

$$z_{\text{obs}} = \frac{\bar{x} - \mu_0}{\sigma_0/\sqrt{n}}$$

A large value of z_{obs} casts doubt on the validity of H_0 and indicates a departure from H_0 in the direction of H_1. So the p-value for testing H_0 against H_1 is

$$\begin{aligned} p &= P(Z \geq Z_{\text{obs}} \mid H_0) \\ &= P(\mathcal{N}(0,1) \geq Z_{\text{obs}}) \\ &= 1 - \Phi(Z_{\text{obs}}) \end{aligned}$$

The z-test of $H_0 : \mu = \mu_0$ against the alternative $H_1 : \mu < \mu_0$ is similar but this time a small, i.e. very negative, value of Z_{obs} casts doubt on H_0. So the p-value is

$$\begin{aligned} p &= P(Z \leq Z_{\text{obs}} \mid H_0) \\ &= P(\mathcal{N}(0,1) \leq Z_{\text{obs}}) \\ &= \Phi(Z_{\text{obs}}) \end{aligned}$$

Finally, consider testing $H_0 : \mu = \mu_0$ against the alternative $H_1 : \mu \neq \mu_0$. Let $z_0 = |z_{\text{obs}}|$. A large value of z_0 indicates a departure from H_0, so the p-value is

$$\begin{aligned} p &= P(|Z| \geq z_0 \mid H_0) \\ &= P(\mathcal{N}(0,1) \geq z_0) + P(\mathcal{N}(0,1) \leq -z_0) \\ &= 2(1 - \phi(z_0)) \end{aligned}$$

- t-test

We can use the test statistic

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

If H_0 is true then $T \sim t_{n-1}$. Let $t_{\text{obs}} = t(x) = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$ and $t_0 = |t_{\text{obs}}|$. Similarly, we have □ the p-value is $P\left(t_{n-1} \geq t_{\text{obs}}\right)$ - the p-value is $P\left(t_{n-1} \leq t_{\text{obs}}\right)$ the p-value is $2P\left(t_{n-1} \geq t_0\right)$

Example of one-sample t-test

A marine biologist is studying a species of fish known to have an average length of 20 cm in ocean populations. A new population in a freshwater lake is being analyzed to determine if the environmental differences have altered the fish's average length. The biologist measures the lengths of 10 randomly selected fish, yielding the following data:

22, 23, 21, 24, 22, 20, 25, 19, 23, 22

Assuming the data satisfy the assumption of normality, please address the following using a significance level of 0.1:

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

a

- null hypothesis: The mean length of fish is 20 cm ($H_0 : \mu = 20$).
- alternative hypothesis: The mean length of fish is not 20 cm ($H_1 : \mu \neq 20$).

b

Since the data is supposed to be normally distributed, the sampling distribution of the sample mean follows t-distribution. The t-test statistic is calculated as follows:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

The t-test statistic is calculated as follows:

The t-test statistic is 3.705882...

c

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Using pt() function, the p-value is calculated as follows:

The p-value is approximately 0.1. Since the p-value is less than 0.1, we reject the null hypothesis.

Therefore, there is sufficient evidence to conclude that the population mean is not equal to 20 which means the environmental differences have altered the fish's average length.

d

Using qt() function to find the critical value for a two-tailed test with 90% confidence level:

The critical value for a two-tailed test with 90% confidence level is about 1.833113.

Since the t-test statistic 3.705882 is greater than the critical value 1.833113, which is in the critical region. Therefore, we reject the null hypothesis.

Also, we could use confidence interval to verify the result. The 90% confidence interval for the population mean is calculated as follows:

So the 90% confidence interval for the population mean is about (21.1, 23.2).

The confidence interval does not contain the hypothesized value 20. Therefore, we reject the null hypothesis.

Therefore, there is sufficient evidence to conclude that the population mean is not equal to 20 which means the environmental differences have altered the fish's average length.

Example of unpaired t-test

Suppose σ_1^2 and σ_2^2 are unknown but assumed equal. We want to test the null hypothesis $H_0 : \mu_1 = \mu_2$ against the alternative hypothesis $H_1 : \mu_1 \neq \mu_2$. The test statistic is given by

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where s_p is the pooled sample standard deviation, given by

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

The unpaired two-sample t procedure combines within-group variation through the pooled standard deviation.

⟵ **Proof idea**
Identify the exact statement before proving it.

Example of one-sample Variance Test

χ^2-test for variance. Suppose $X_1, \cdots, X_n$ are i.i.d. normal random variables with mean μ and variance σ^2. We want to test the null hypothesis $H_0 : \sigma^2 = \sigma_0^2$ against the alternative hypothesis $H_1 : \sigma^2 \neq \sigma_0^2$. The test statistic is given by

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

Example of two-sample Variance Test

H_0	Test Statistic	H_1	Rejection Region
$\sigma_1^2 = \sigma_2^2$	$f = \frac{s_1^2}{s_2^2}$	$\sigma_1^2 < \sigma_2^2$	$f < f_\alpha(\nu_1, \nu_2)$
		$\sigma_1^2 > \sigma_2^2$	$f > f_{1-\alpha}(\nu_1, \nu_2)$
	$\sigma_1^2 \neq \sigma_2^2$	$f < f_{\alpha/2}(\nu_1, \nu_2)$ or $f > f_{1-\alpha/2}(\nu_1, \nu_2)$	
$\nu_1 = n_1 - 1$ and $\nu_2 = n_2 - 1$ are two degree of freedom.			

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

$$F = \frac{\frac{(n_1-1)S_1^2}{\sigma_1^2} / (n_1 - 1)}{\frac{(n_2-1)S_2^2}{\sigma_2^2} / (n_2 - 1)} = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2}$$

If $\sigma_1^2 = \sigma_2^2$, we have

$$F = \frac{S_1^2}{S_2^2} \sim F_{n_1-1, n_2-1}$$

ANOVA– Analysis of Variance

- one-way ANOVA

we need to test the null hypothesis that the group population means are all the same against the alternative that at least one group population mean differs from the others. That is,

$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$ against $H_1 :$ at least one μ_i differs from the others.

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

ANOVA Table

Source	DF	Sum Sq	Mean Sq	F value	p value
Factor	m-1	11.84 (SS between)	2.9587 (MSB)	8.074 (MSB/MSW)	$5.38\text{e}-05$ (p-value)
Error	n-m	16.49 (SS Within)	0.3664 (MSW)		
Total	n-1	28.33 (SS Total)			

Source means "the source of the variation in the data." the possible sources for a one-factor study are Factor, Residuals, and Total.

Factor means "the variability due to the factor of interest." In the drug example, the factor was the different drug. In the learning example on the previous page, the factor was the method of learning. Sometimes the row heading is labeled as Between.

Error (or Residuals) means "the variability within the groups" or "unexplained random error." Sometimes the row heading is labeled as Within.

Total means "the total variation in the data from the grand mean".

DF means "the degrees of freedom in the source."

Sum Sq means "the sum of squares due to the source."

Mean Sq means "the mean sum of squares due to the source."

F value means "the F-statistic."

P value means "the P-value."

SS(Total)=SS(Between)+SS(Within), where

SS(Between) is the sum of squares between the group means and the grand mean. As the name suggests, it quantifies the variability between the groups of interest.

SS(Within) is the sum of squares between the data and the group means. It quantifies the variability within the groups of interest.

SS(Total) is the sum of squares between the n data points and the grand mean. As the name suggests, it quantifies the total variability in the observed data.

- two-way ANOVA

We can extend the idea of a one-way ANOVA, which tests the effects of one factor on a response variable, to a two-way ANOVA which tests the effects of two factors and their interaction on a response variable.

Source	DF	Sum Sq	MSW	F
Cells	$ab-1$	$\sum_{i=1}^{a}\sum_{j=1}^{b} n\left(\bar{X}_{ij}-\bar{X}_{\cdot\cdot\cdot\cdot\cdot}\right)^2$		
A	a-1	bn $\sum_{i=1}^{a}\left(\bar{X}_{i\cdot\cdot}-\bar{X}_{\cdot\cdot\cdot\cdot}\right)^2$	SS(A)	MS(Error
B	b-1	an $\sum_{j=1}^{b}\left(\bar{X}_{\cdot j\cdot}-\bar{X}_{\cdot\cdot\cdot\cdot}\right)^2$	SS(B)	MS(B)
A × B	$(a-1)(b-1)$	SS(Cells)-SS(A)-SS(B)	SS(AB) DF(A × B)	MS(Error MS(Error
Error	$ab(n-1)$	$\sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{l=1}^{n}\left(X_{ijl}-\bar{X}_{ij\cdot}\right)^2$	SS(Error)	
Total	$abn-1$	$\sum_{i=1}^{a}\sum_{j=1}^{b} n\left(X_{ijl}-\bar{X}_{\cdot\cdot\cdot\cdot}\right)^2$		

- $F=\frac{\text{MS(A)}}{\text{MS(Error)}}$, for H_0 : no effect of factor A on response variable,
- $F=\frac{\text{MS(B)}}{\text{MS(Error)}}$, for H_0 : no effect of factor B on response variable,
- $F=\frac{\text{MS(A}\times\text{B)}}{\text{MS(Error)}}$, for H_0 : no effect of interaction on response variable.

We reject any H_0 if $F \geq F_{\text{critical}}$; otherwise, we do not reject H_0.

Example of two-way ANOVA

Two-way ANOVA. In this question, we will use the built-in R data set ToothGrowth to perform two-way ANOVA test. ToothGrowth includes information from a study on the effects of vitamin C on tooth growth in Guinea pigs. The response is the length of odontoblasts (cells responsible for tooth growth) in 60 guinea pigs. Each animal received one of three dose levels of vitamin C (0.5, 1, and 2 mg/day) by one of two delivery methods, orange juice or ascorbic acid (a form of vitamin C and coded as VC). Assuming the data satisfy the assumptions of normality and equal variance, please address the following using a significance level of 0.05

⟵ **Proof idea**
Identify the exact statement before proving it.

a

- The effects of vitamin C on tooth growth in guinea pigs:
- null hypothesis: H_0: mean tooth growth for all doses of vitamin C are equal
- alternative hypothesis: H_1: at least one of the means of all doses of vitamin C is different from the others
- The effects of delivery method on tooth growth in guinea pigs:
- null hypothesis: H_0: mean tooth growth for the delivery method of orange juice and ascorbic acid are equal.
- alternative hypothesis H_1: mean tooth growth for the delivery method of orange juice and ascorbic acid are different.
- The interaction effects of the dose of vitamin C and delivery method on tooth growth in guinea pigs:
- null hypothesis: H_0 : there is no interaction between the dose of vitamin C and delivery method on tooth growth in guinea pigs, meaning that the relationship between vitamin C and tooth growth is the same for both delivery methods (similarly, the relationship between delivery method and tooth growth is the same for all doses of vitamin C).
- alternative hypothesis: H_1: there is an interaction between the dose vitamin C and delivery method on tooth growth in guinea pigs, meaning that the relationship between vitamin C and tooth growth is different for both delivery methods (similarly, the relationship between delivery method and tooth growth depends on the dose of vitamin C).

b We can plot the relationship one by one using two plots

or just one:

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

c Since all p-values are less than 0.05, we reject all null hypotheses. Therefore, there is sufficient evidence to conclude that the dose of vitamin C, delivery method, and their interaction have significant effects on tooth growth in guinea pigs.

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

d

Non-parametric tests

Application of testing the goodness of fit

Testing whether there is a “good fit” between the observed data and the assumed probability model amounts to testing:

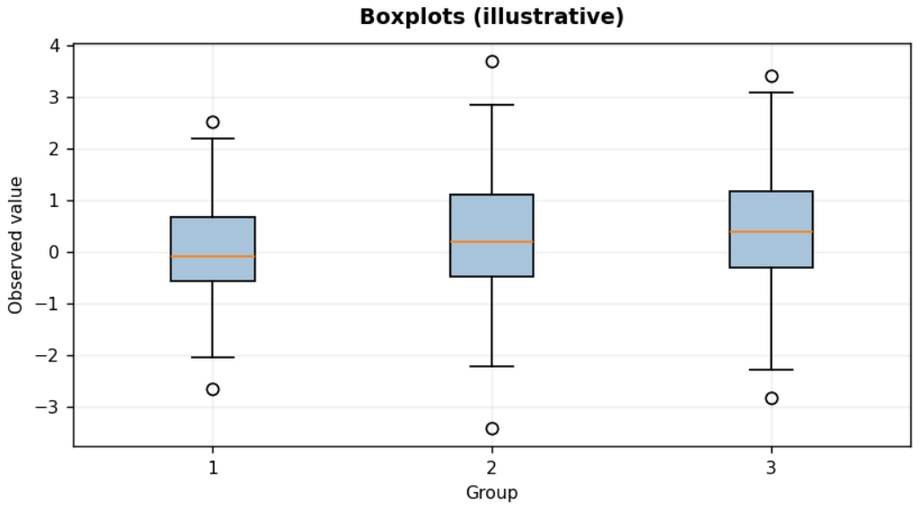


Figure 8: Output from code example 12. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: ggplot, geom_boxplot.

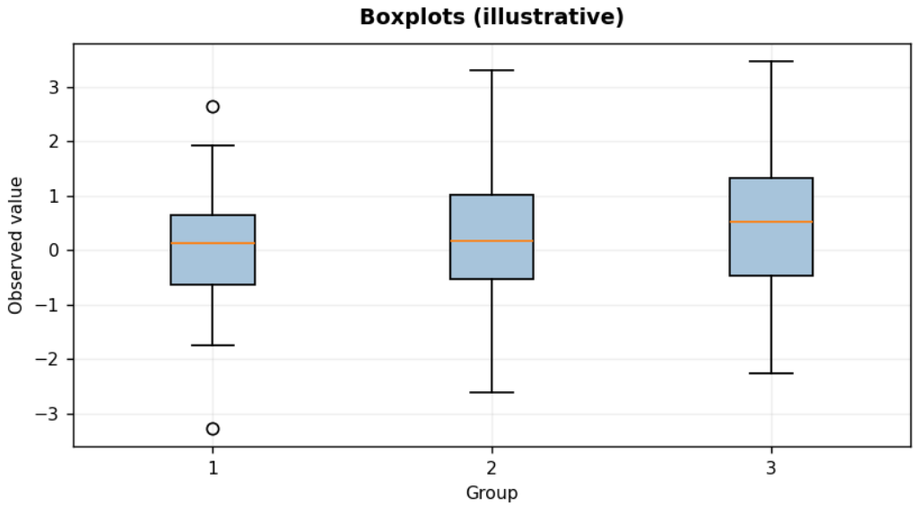


Figure 9: Output from code example 12. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: ggplot, geom_boxplot.

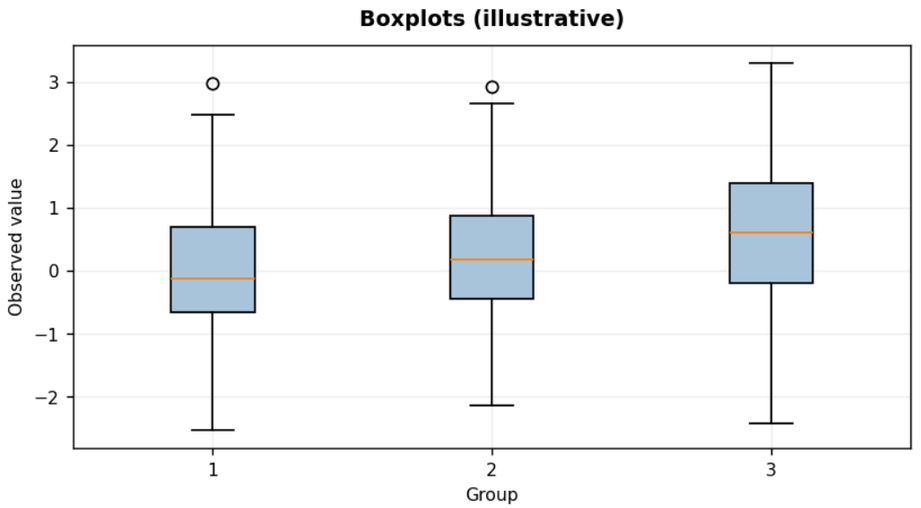


Figure 10: Output from code example 13. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: ggplot, geom_boxplot.

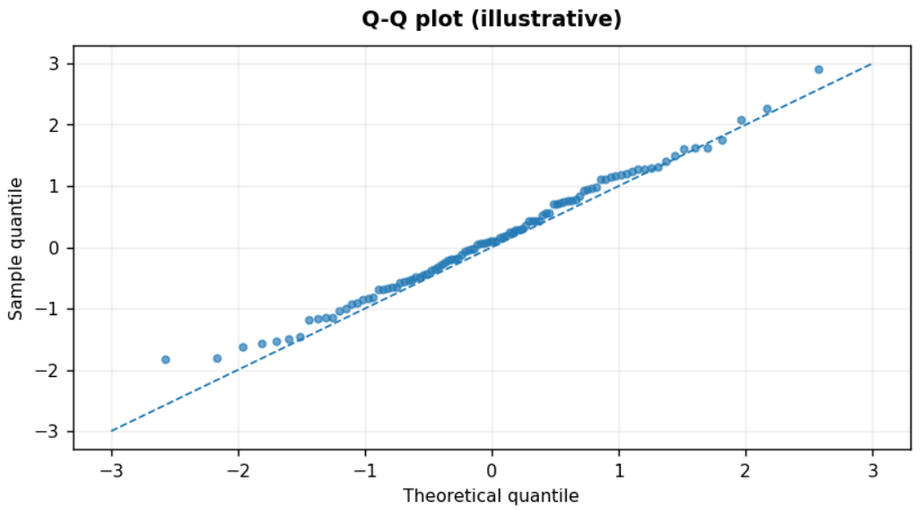


Figure 11: Output from code example 15. Illustrative reconstruction based on nearby code; not the exact original output. R packages: car; functions: plotting / simulation functions.

Construction of test statistics with an example of 2 categories Population is 60% female and 40% male. Then, if a sample of 100 students yields 53 females and 47 males, can we conclude that the sample is (random and) representative of the population? That is, how "good" do the data "fit" the assumed probability model of 60% female and 40% male?

Here, let Y_1 denote the number of females selected, $Y_1 \sim B(n, p_1)$ and let Y_2 denote males selected, $Y_2 = (n - Y_1) \sim B(n, p_2) = B(n, 1 - p_1)$.

for samples satisfying the general rule of thumb (the expected number of successes must be at least 5 and the expected number of failures must be at least 5), we can use the normal approximation to the binomial distribution. The test statistic is given by

$$Z = \frac{Y_1 - np_1}{\sqrt{np_1(1-p_1)}} \sim N(0,1)$$

which is at least approximately normally distributed.

and

$$Z^2 = Q_1 = \frac{(Y_1 - np_1)^2}{np_1(1-p_1)} \sim \chi^2(1)$$

which is an approximate chi-square distribution with one degree of freedom.

Now we can multiply Q_1 by 1 = $(1 - p_1) + p_1$ to get

$$Q_1 = \frac{(Y_1 - np_1)^2(1-p_1)}{np_1(1-p_1)} + \frac{(Y_1 - np_1)^2 p_1}{np_1(1-p_1)} \sim \chi^2(1)$$

Since $Y_1 = n - Y_2$ and $p_1 = 1 - p_2$, after simplifying, we have

$$Q_1 = \frac{(Y_1 - np_1)^2}{np_1} + \frac{(-(Y_2 - np_2))^2}{np_2} \sim \chi^2(1)$$

which is $Q_1 = \sum_{i=1}^{2} \frac{(Y_i - np_i)^2}{np_i} = \sum_{i=1}^{2} \frac{(\text{ Observed } - \text{ Expected })^2}{\text{Expected}} \sim \chi^2(1)$

Hence, it is observed that if the observed counts are very different from the expected counts, then the test statistic will be large. So we reject the null hypothesis if Q_1 is large and how large is large is determined by the critical value of the chi-square distribution with one degree of freedom.

The statistics Q_1 is called the chi-square goodness of fit statistic.

Going back to the example,

- H_0:$p_F = 0.6$
- H_1:$p_F \neq 0.6$

we can calculate the test statistic using a significant level of $\alpha = 0.05$ ($\chi^2_{0.05,1} = 3.84$)as follows:

$$Q_1 = \frac{(53-60)^2}{60} + \frac{(47-40)^2}{40} = 2.04$$

Since $Q_1 = 2.04 < 3.84$, we do not reject the null hypothesis. Therefore, we conclude that the sample is (random and) representative of the population.

This can be extended to k categories

Categories	1	2	⋯	$k-1$	k

Construction of test statistics with an example of k categories For categories more than 2, i.e.

⟵ **Proof idea** Identify the exact statement before proving it.

Categories	1	2	⋯	$k-1$	k
Observed	Y_1	Y_2	⋯	Y_{k-1}	$n-Y_1-Y_2-\cdots-Y_{k-1}$
Expected	np_1	np_2	⋯	np_{k-1}	np_k

Karl Pearson showed that the chi-square statistic Q_{k-1} defined as:

$$Q_{k-1} = \sum_{i=1}^{k} \frac{(Y_i - np_i)^2}{np_i}$$

follows approximately a chi-square random variable with $k-1$ degrees of freedom. Let's try it out on an example.

- Example:

Categories	Brown	Yellow	Orange	Green	Coffee	Total
Observed y_i	224	119	130	48	59	580
Assumed $H_0\,(p_i)$	0.4	0.2	0.2	0.1	0.1	1.0
Expected np_i	232	116	116	58	58	580

$Q_4 = \frac{(224-232)^2}{232} + \frac{(119-116)^2}{116} + \frac{(130-116)^2}{116} + \frac{(48-58)^2}{58} + \frac{(59-58)^2}{58} = 3.784$

Because there are $k = 5$ categories, we have to compare our chisquare statistic Q_4 to a chi-square distribution with $k-1 = 5-1 = 4$ degrees of freedom:

$$Q_4 = 3.784 < \chi^2_{4,0.05} = 9.488$$

we fail to reject the null hypothesis.

Application of testing for homogeneity

⟵ **My thought** The theorem is easier to remember when its hypothesis feels necessary.

This is to look at a method for testing whether two or more multinomial distributions are equal.

- Example:

Test the hypothesis that the acceptances of males and females are ditributed equally among the four schools,

(Acceptances)	Bus	Eng	L Arts	Sci	(FIXED) Total
Male	240 (20%)	480 (40%)	120 (10%)	360 (30%)	1200
Female	240 (30%)	80 (10%)	320 (40%)	160 (20%)	800
Total	480 (24%)	560 (28%)	440 (22%)	520 (26%)	2000

Here,

$H_0 : p_{MB} = p_{FB}, p_{ME} = p_{FE}, p_{ML} = p_{FL}$, and $p_{MS} = p_{FS}$

$H_1 : p_{MB} \neq p_{FB}$ or $p_{ME} \neq p_{FE}$ or $p_{ML} \neq p_{FL}$, or $p_{MS} \neq p_{FS}$

where:

- p_{Mj} is the proportion of males accepted into school $j = B, E, L, S$.
- p_{Fj} is the proportion of females accepted into school $j = B, E, L, S$.

In conducting such a hypothesis test, we're comparing the proportions of two multinomial distributions.

#(Acc)	Bus ($j=1$)	Eng ($j=2$)	L Arts ($j=3$)	Sci $(j=4)$	(FIXED) Total
M$(i=1)$	$y_{11}\,(\hat{p}_{11})$	$y_{12}\,(\hat{p}_{12})$	$y_{13}\,(\hat{p}_{13})$	$y_{14}\,(\hat{p}_{14})$	$n_1=\sum_{j=1}^{k} y_{1j}$
F $(i=2)$	$y_{21}\,(\hat{p}_{21})$	$y_{22}\,(\hat{p}_{22})$	$y_{23}\,(\hat{p}_{23})$	$y_{24}\,(\hat{p}_{24})$	$n_2=\sum_{j=1}^{k} y_{2j}$
Total	$y_{11}+y_{21}\,(\hat{p}_1)$	$y_{12}+y_{22}\,(\hat{p}_2)$	$y_{13}+y_{23}\,(\hat{p}_3)$	$y_{14}+y_{24}\,(\hat{p}_4)$	n_1+n_2

The chi-square test statistic for testing the equality of two multinomial distributions:

$$Q=\sum_{i=1}^{2}\sum_{j=1}^{k}\frac{\left(y_{ij}-n_i\hat{p}_j\right)^2}{n_i\hat{p}_j}$$

follows an approximate chi-square distribution with $k-1$ degrees of freedom. Reject the null hypothesis of equal proportions if Q is large (since if male and female distributed nearly equally, the expected number of each should be $n_i\hat{p}_j$):

$$Q\geq\chi^2_{\alpha,k-1}$$

(omit the derive of the above Q)

Generally,

$$Q=\sum_{i=1}^{h}\sum_{j=1}^{k}\frac{\left(y_{ij}-n_i\hat{p}_j\right)^2}{n_i\hat{p}_j}\sim\chi^2_{(h-1)(k-1)}$$

Further example The head of a surgery department at a university medical center was concerned that surgical residents in training applied unnecessary blood transfusions at a different rate than the more experienced attending physicians. Therefore, he ordered a study of the 49 Attending Physicians and 71 Residents in Training with privileges at the hospital. For each of the 120 surgeons, the number of blood transfusions prescribed unnecessarily in a one-year period was recorded. Based on the number recorded, a surgeon was identified as either prescribing unnecessary blood transfusions Frequently, Occasionally, Rarely, or Never. Here's a summary table (or "contingency table") of the resulting data:

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Physician	Frequent	Occasionally	Rarely	Never	Total
Attending	6.942	12.658	22.05	7.35	49
Resident	10.058	18.342	31.95	10.65	71
Total	17	31	54	18	120

Here,

$H_0: p_{RF}=p_{AF}, p_{RO}=p_{AO}, p_{RR}=p_{AR}$, and $p_{RN}=p_{AN}$

$H_1: p_{RF}\neq p_{AF}$ or $p_{RO}\neq p_{AO}$ or $p_{RR}\neq p_{AR}$, or $p_{RN}\neq p_{AN}$

We should also calculate the expected counts under the null hypothesis. The expected counts are calculated as follows:

Physician	Frequent	Occasionally	Rarely	Never	Total
Attending	6.942	12.658	22.05	7.35	49
Resident	10.058	18.342	31.95	10.65	71
Total	17	31	54	18	120

where, for example, $6.942=\frac{17}{120}\times 49$ and $10.058=\frac{17}{120}\times 71$. Now that we have the observed and expected counts, calculating the chisquare statistic is a straightforward exercise:

$$Q=\frac{(2-6.942)^2}{6.942}+\cdots+\frac{(5-10.65)^2}{10.65}=31.88$$

The chi-square test tells us to reject the null hypothesis, at the 0.05 level, if Q is greater than a chi-square random variable with 3 degrees of freedom, that is, if $Q = 31.88 > 7.815$, we reject the null hypothesis.

Application of testing for independence

⟵ **Proof idea** Identify the exact statement before proving it.

This is to look at whether two or more categorical variables are independent.

(previously,the the sampling scheme involves: Taking two random (and therefore independent) samples with n1 and n2 fixed in advance and observing into which of the k categories the first random samples fall, and observing into which of the k categories the second random samples fall.)

lets consider a different example to illustrate an alternative sampling scheme. Suppose 395 people are randomly selected, and are "cross-classified" into one of eight cells, depending into which age category they fall and whether or not they support legalizing marijuana:

(the sampling scheme involves: Taking one random sample of size n, with n fixed in advance, and then "cross-classifying" each subject into one and only one of the mutually exclusive and exhaustive $A_i \cap B_j$ cells.)

Marijuana Support		Variable B (Age)				
Variable A	OBSERVED	$(18-24)B_1$	(25-34) B_12	(35-49) B_3	$(50-64)B_4$	Total
	(YES) A_1	60	54	46	41	201
	(NO) A_2	40	44	53	57	194
	Total	100	98	99	98	$n = 395$

Here,

H0 : Variable A is independent of variable B, that is $P(A_i \cap B_j) = PA_i \times B_j$ for all i and j

H1: Variable A is not independent of variable B.

Generally,

Suppose we have k (column) levels of Variable B indexed by the letter j, and h (row) levels of Variable A indexed by the letter i. Then, we can summarize the data and probability model in tabular format, as follows:

Variable B					
Variable A	$B_1(j=1)$	$B_2(j=2)$	$B_3(j=3)$	$B_4(j=4)$	Total
$A_1(i=1)$	$Y_{11}\,(p_{11})$	$Y_{12}\,(p_{12})$	$Y_{13}\,(p_{13})$	$Y_{14}\,(p_{14})$	$(p_{1.})$
$A_2(i=2)$	$Y_{21}\,(p_{21})$	$Y_{22}\,(p_{22})$	$Y_{23}\,(p_{23})$	$Y_{24}\,(p_{24})$	$(p_{2.})$
Total	$(p_{.1})$	$(p_{.2})$	$(p_{.3})$	$(p_{.4})$	n

where $p_{ij} = Y_{ij}/n, p_{i}. = \sum_{j=1}^{k} p_{ij}$, and $p_{.j} = \sum_{i=1}^{h} p_{ij}$

$Q = \sum_{j=1}^{k} \sum_{i=1}^{h} \frac{\left(y_{ij} - \frac{y_{i\cdot} y_{\cdot j}}{n}\right)^2}{\frac{y_{i\cdot} y_{\cdot j}}{n}} \sim \chi^2_{(h-1)(k-1)}$

⟵ **My thought** The theorem is easier to remember when its hypothesis feels necessary.

Are chi-square statistic for homogeneity and the chi-square statistic for independence equivalent? Although their chi-square statistics are equivalent, the two tests are not equivalent since their sampling experiment designs are different.

Here's the table of expected counts:

Bicycle Riding Interest		Variable B (Age)				
Variable A	EXPECTED	18-24	25-34	35-49	50-64	Total
	YES	50.886	49.868	50.377	49.868	201

Bicycle Riding Interest		Variable B (Age)				
	NO	49.114	48.132	48.623	48.132	194
	Total	100	98	99	98	395

$$Q = \frac{(60 - 50.886)^2}{50.886} + \cdots + \frac{(57 - 48.132)^2}{48.132} = 8.006$$

The chi-square test tells us to reject the null hypothesis, at the 0.05 level, since Q is greater than a chi-square random variable with 3 degrees of freedom, that is, $Q = 8.006 > 7.815$.

⟵ **Motivation** Why does this definition matter later? Try an example and a counterexample.

Summary Parametric tests make assumptions that aspects of the data follow some sort of theoretical probability distribution. Non-parametric tests or distribution free methods do not, and are used when the distributional assumptions for a parametric test are not met. While this is an advantage, it often comes at a cost of power (in the sense they are less likely to be able to detect a difference when a true difference exists).

Most non-parametric tests are just hypothesis tests; there is no estimation of a confidence interval.

Most non-parametric methods are based on ranking the values of a variable in ascending order and then calculating a test statistic based on the sums of these ranks.

Non-parametric tests include:

- Two-sample independent t-test - Wilcoxon rank-sum test or Mann-Whitney U test
- Paired t-test - Wilcoxon signed-rank test
- One-way ANOVA - Kruskal-Wallace Test
- Normality tests - Shapiro-Wilk test and Kolmogorov-Smirnov test

Linear Regression

Simple linear regression

⟵ **Proof idea** Distinguish a model for the conditional mean from the noise model.

Brief introduction

Some people think simple methods is bad and like complicated methods , but actually simple is very good–SLR works very well in lots of situations.

SLR is used to answer:

is there a relationship between..

How strong the relationship is

which variable contribute to this relationship

How accurate could we predict the response variable

Is the relationship linear?

Is there a synergy among independent variables?

This is a model with two random variables, X and Y, where we are trying to predict Y from X. Here are the model's assumptions:

- The distribution of X is arbitrary, possibly is even non-random;
- If X=x, then $Y = \beta_0 + \beta_1 x + \epsilon$ for some constants β_0, β_1 and some random noise variable ϵ
- ϵ has mean 0, a constant variance σ^2, and is uncorrelated with X and uncorrelated across observations $\text{Cov}(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$

Using Least Squares, we can estimate $\hat{\beta}_0$ and $\hat{\beta}_1$, which are unbiased estimates of β_0 and β_1.

- Gaussian-Noise Simple Linear Regression Model

Now we further assume that the distribution of ϵ is normal, i.e. $\epsilon \sim N(0, \sigma^2)$, independent of X.

They tell us, exactly, the probability distribution for Y given X, and so will let us get exact distributions for predictions and for other inferential statistics.

Maximum Likelihood Estimation (MLE)

Introduction to MLE Likelihood is a fundamental concept in statistics that measures how well a particular set of parameters (e.g., the mean of a distribution) explains observed data. Think of it as a "score" that tells you which parameter values make your data most plausible.

⟵ **My thought** Inference uses data to distinguish plausible parameters; assumptions still matter.

Compared to probability, which answers: "What's the chance of seeing this data if we assume specific parameters?" , likelihood answers: "Given this data, how plausible are these parameters?"

If the parameters are b_0, b_1, s^2 (reserving the Greek letters for their true values), then $Y \mid X = x \sim N\left(b_0 + b_1 x, s^2\right)$, and Y_i and Y_j are independent given X_i and X_j, so the overall likelihood is

$$\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi s^2}} e^{-\frac{(y_i - (b_0 + b_i x_i))^2}{2s^2}}$$

As usual, we work with the log-likelihood, which gives us the same information but replaces products with sums:

$$L\left(b_0, b_1, s^2\right) = -\frac{n}{2} \ln\left(2\pi s^2\right) - \frac{1}{2s^2} \sum_{i=1}^{n} \left(y_i - (b_0 + b_1 x_i)\right)^2$$

maximize it:

$$\frac{\partial L}{\partial b_0} = -\frac{1}{2s^2} \sum_{i=1}^{n} 2\left(y_i - (b_0 + b_1 x_i)\right)(-1)$$

$$\frac{\partial L}{\partial b_1} = -\frac{1}{2s^2} \sum_{i=1}^{n} 2\left(y_i - (b_0 + b_1 x_i)\right)(-x_i)$$

Same result of MLE as least squares in linear regression Notice that when we set these derivatives to zero, all the multiplicative constants - in particular, the prefactor of $1/2s^2$ - go away. We are left with

⟵ **Motivation** Why minimize distance? Orthogonality turns a nonlinear-looking optimization into normal equations.

$$\sum_{i=1}^{n} \left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right)\right) = 0$$

$$\sum_{i=1}^{n} \left(y_i - \left(\hat{\beta}_0 + \hat{\beta}_0 x_i\right)\right) x_i = 0$$

These are, up to a factor of $1/n$, exactly the equations we got from the method of least squares. That means that the least squares solution is the maximum likelihood estimate under the Gaussian noise model.

Maximum likelihood estimates of the regression curve coincide with least-squares estimates when the noise around the curve is additive, Gaussian, of constant variance, and both independent of X and of other noise terms. If any of those assumptions fail, maximum likelihood and least squares estimates can diverge.

Hypothesis testing for estimates with unknown σ^2

Residual sum of squares (RSS) and t-statistics construction in linear regression It can be shown that (the proof is beyond the scope of this course)

⟵ **Proof idea** Often the log-likelihood is simpler than the likelihood.

$$\frac{RSS}{\sigma^2} = \frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-2}$$

This allows us to construct a t-value

$$t = \frac{\hat{\beta} - \beta}{s_{\hat{\beta}}} \sim t_{n-2}$$

Under the normality assumption of the error terms, the estimator of the slope coefficient will itself be normally distributed with mean β_i and variance Var $[\beta_i]$ For $\hat{\beta}_1$, its mean is β_1 and its variance is $\sigma^2 / \sum (x_i - \bar{x})^2$. When σ^2 is known, we know

$$\frac{\hat{\beta}_1 - \beta_1}{\frac{\sigma}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2}}}$$

follows standard normal distribution. However, in practice, σ^2 is often unknown. We then divide this standard normal distributed term by

$$\sqrt{\frac{(n-2)\hat{\sigma}^2}{(n-2)\sigma^2}} = \frac{\hat{\sigma}}{\sigma}$$

Therefore, when we write

$$s_{\hat{\beta}_1} = \frac{\hat{\sigma}}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2}}$$

we construct a t-statistic for $\hat{\beta}_1$ with degrees of freedom $n - 2$. This then allows us to construct a $100(1-\alpha)\%$ confidence interval for β_1 :

$$\hat{\beta}_1 \pm t_{n-2,\alpha/2} \times s_{\hat{\beta}_1}$$

We can also do similar calculation to get the t-statistic and confidence interval for β_0.

Hyphothesis Testing

$$t = \hat{\beta} - \beta / s_{\hat{\beta}_1} \sim t_{n-2}$$

$H_0 : \beta_i = 0$
$H_1 : \beta_i \neq 0$

Codes of linear regression with CI

Linear Regression and ANOVA

R^2-the fraction of variability explained by the regression

$$R^2 = 1 - \frac{SSR}{SSTO}$$

Multiple linear regression (MLR)

$y = X\beta + \epsilon$, where y is a n × 1 row vector, X is a n × (k + 1) matrix, and β is a (k + 1) × 1 column vector for all n observations.

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

⟵ **Motivation**
A linear predictor connects geometry to prediction, but the link function changes the response scale.

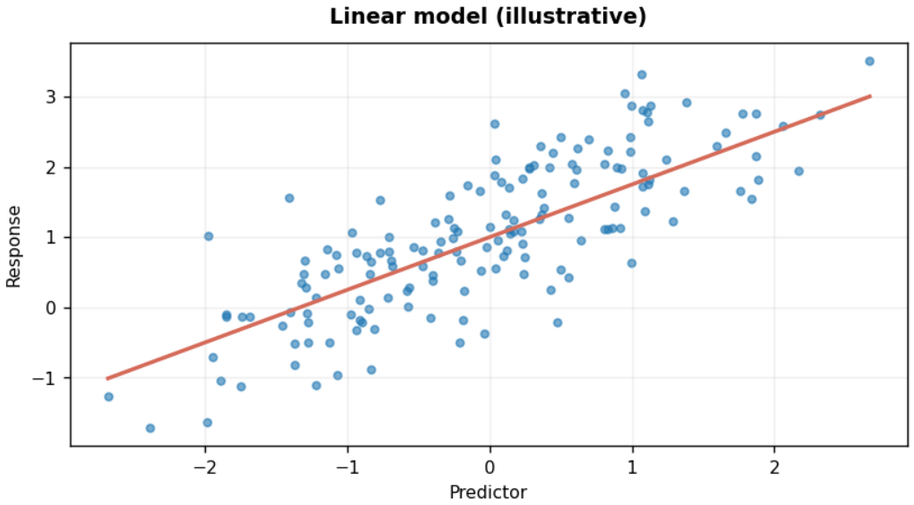


Figure 12: Output from code example 16. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: lm, predict, plot, abline, ggplot, geom_point, geom_smooth.

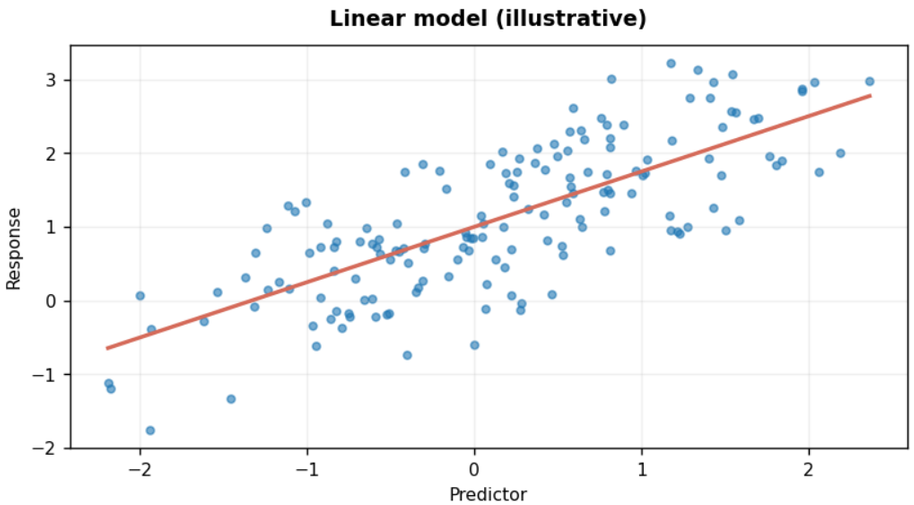


Figure 13: Output from code example 16. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: lm, predict, plot, abline, ggplot, geom_point, geom_smooth.

A potential problem in practice –multicollinearity

When multicollinearity exists, any of the following pitfalls can be exacerbated:

- The estimated regression coefficient of any one variable depends on which other predictors are included in the model
- The precision of the estimated regression coefficients decreases as more predictors are added to the model
- The marginal contribution of any one predictor variable in reducing the error sum of squares depends on which other predictors are already in the model
- Typothesis tests for βj = 0 may yield different conclusions depending on which predictors are in the model

Perfect multicollinearity Perfect multicollinearity refers to a situation where the predictive variables have an exact linear relationship. When there is perfect collinearity, the design matrix X has less than full rank, and therefore the moment matrix X′X cannot be inverted. In this situation, the parameter estimates of the regression are not well-defined, as the system of equations has infinitely many solutions.

⟵ **Proof idea** Identify the exact statement before proving it.

Imperfect multicollinearity Imperfect multicollinearity refers to a situation where the predictive variables have a nearly exact linear relationship.

$R^2 = r^2$ where r is the Pearson correlation coefficient.

Adjusted R-squared

Adjusted R-squared is a modified version of R-squared that adjusts for the number of predictors in a model. It provides a more accurate measure of the model's explanatory power, penalizing for the addition of irrelevant predictors. This helps in comparing models with different numbers of predictors.

Logistic Regression

This is a regression of categorical outcome.

In regression analysis with a categorical outcome, such as predicting a binary variable (yes or no), simple linear regression is not ideal. This is because:

- The predicted values may fall outside the range of 0 to 1, which is not meaningful for probabilities.
- Small changes in the predictors can lead to relatively small fluctuations in the predicted probabilities near the 0.5 mark (natural threshold), which is actually where decision-making is most critical.

⟵ **My thought** A linear predictor connects geometry to prediction, but the link function changes the response scale.

So we need S-curve to satisfy above things.

An example of a not well-predicted logistic model (since stock is not easy to predict)

Odds

Odds are another way of quantifying the probability of an event, commonly used in gambling (and logistic regression).

⟵ **Motivation** Why does this definition matter later? Try an example and a counterexample.

For some event E,

$$\text{odds}(E) = \frac{P(E)}{P\left(E^c\right)} = \frac{P(E)}{1 - P(E)}$$

The odds ratio (OR) is the ratio of the odds of an event occurring in one group to the odds of it occuring in another group. If the event in each of the groups are p_1 (first group) and p_2 (second group), then the odds ratio is:

$$\text{OR} = \frac{p_1 / (1 - p_1)}{p_2 / (1 - p_2)}$$

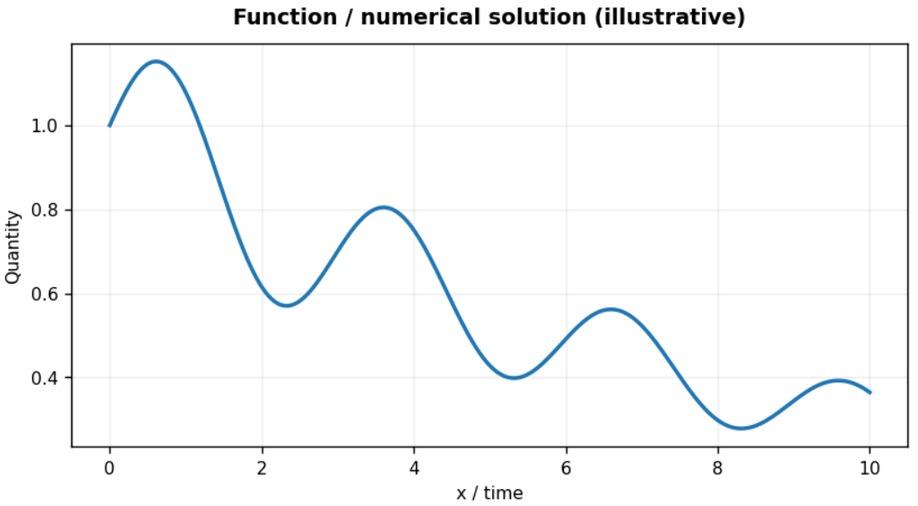


Figure 14: Output from code example 17. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: curve.

Generalized Linear Models

Under the hood, we're still using a linear model $(\beta_0 + \beta_1 x)$, but now it's embedded in a function that ensures valid probabilities. This is the essence of logistic regression - a generalized linear model (GLM) designed for binary outcomes.

Given predictor X and d an outcom Y, a GLM is defined by three components: - A random component, that specifies a distribution for $Y \mid X$ - A systematic component, that relates a parameter η to the predictor X

$$\eta = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

- A link function, that connects the random and systematic component

Random Component

⟵ **Proof idea** Condition explicitly; it often simplifies the calculation.

The random component specifies a distribution for the outcome variable (conditional on X). In the case of linear regression, we assume that $Y \mid X \sim \mathcal{N}(\mu, \sigma^2)$, for some mean μ and variance σ^2. In the case of logistic regression, we assume that $Y \mid X \sim \text{Bern}(p)$ for some probability p.

In a generalized model, we are allowed to assume that $Y \mid X$ has a probability density function or probability mass function of the form

$$f(y; \theta, \phi) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right)$$

Here θ, ϕ are parameters, and a, b, c are functions. Any density of the above form is called an exponential family density. The parameter θ is called the natural parameter, and the parameter ϕ the dispersion parameter.

Exponential Family

⟵ **My thought** The theorem is easier to remember when its hypothesis feels necessary.

Exponential families include many of the most common distributions. For example: - Exponential

$$f(y; \lambda) = \lambda e^{-\lambda y} = \exp(-y\lambda + \ln \lambda)$$

where $\theta = -\lambda, \phi = 1, b(\theta) = \ln \lambda, a(\phi) = 1$, and $c(y, \phi) = 0$ - Poisson

$$f(y; \lambda) = \frac{e^{-\lambda}\lambda^y}{y!} = \exp(y \ln \lambda - \lambda - \ln(y!))$$

where $\theta = \ln \lambda, \phi = 1, b(\theta) = e^\theta = \lambda, a(\phi) = 1$, and $c(y, \phi) = -\lambda - \ln(y!)$

Systematic Component and Link Component

The systematic component relates a parameter η to the predictors X. In a GLM, this is always done via

$$\eta = X\beta = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

We will denote the expectation of the distribution in random component as μ, i.e., $\mathbb{E}[Y \mid X] = \mu$. It will be our goal to estimate μ. Finally, the link component connects the random and systematic components, via a link function g. In particular, this link function provides a connection between μ and η, as in

$$g(\mu) = \eta \quad \text{or} \quad \mu = g^{-1}(\eta)$$

Example

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Gaussian-noise Linear Regression

- Random Component: $Y \mid X \sim \mathcal{N}(\mu, \sigma^2)$ and $\mathbb{E}[Y \mid X] = \mu$
- Systematic Component: $\eta = X\beta$
- Link Component: $g(\mu) = \mu$, so that $\mu = \eta = X\beta$

Bernoulli

Suppose that $Y \in \{0, 1\}$, and we model the distribution of $Y \mid X$ as Bernoulli with success probability p. Then the probability mass function (not a density, since Y is discrete) is

$$f(y) = p^y (1-p)^{1-y}$$

We can rewrite to fit the exponential family form as

$$\begin{aligned} f(y) &= \exp(y \log p + (1-y)\log(1-p)) \\ &= \exp(y \log(p/(1-p)) + \log(1-p)) \end{aligned}$$

$$f(y; \theta, \phi) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right)$$

Here we would identify $\theta = \log(p/(1-p))$ as the natural parameter. Note that the mean here is $\mu = p$, and using the inverse of the above relationship, we can directly write the mean p as a function of θ, as in $p = e^\theta / (1 + e^\theta)$. Hence $b(\theta) = \log(1-p) = -\log(1 + e^\theta)$. There is no dispersion parameter, so we can set $a(\phi) = 1$. Also, $c(y, \phi) = 0$.

Link Function

⟵ **Proof idea**
Identify the exact statement before proving it.

$$g(\mu) = \Phi^{-1}(\mu)$$

where Φ is the standard normal CDF.

Logistic

The three GLM criteria give us:

- $y_i \sim \text{Bern}(p_i)$
- $\eta = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$
- $\text{logit}(p) = \log \frac{p}{1-p} = \eta$

From which we know,

$$p_i = \frac{\exp(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_k X_{ik})}{1 + \exp(\beta_0 + \beta_1 x_{i1} + \cdots + \beta_k X_{ik})}$$

Survival Analysis

Survival analysis is used to analyze data in which the time until the event is of interest. The response is often referred to as a failure time, survival time, or event time.

Some defination

Hazard ratios; ratios of hazard functions between different groups (e.g., exposed vs. unexposed) while adjusting for confounders.

censoring - which occurs when the survival time is only partially known

- Fixed type I censoring occurs when a study is designed to end after C years of follow-up. In this case, everyone who does not have an event observed during the course of the study is censored at C years.
- In random type I censoring, the study is designed to end after C years, but censored subjects do not all have the same censoring time. This is the main type of right-censoring we will be concerned with.
- In type II censoring, a study ends when there is a pre-specified number of events.

Kaplan-Meier estimate

⟵ **My thought**
Inference uses data to distinguish plausible parameters; assumptions still matter.

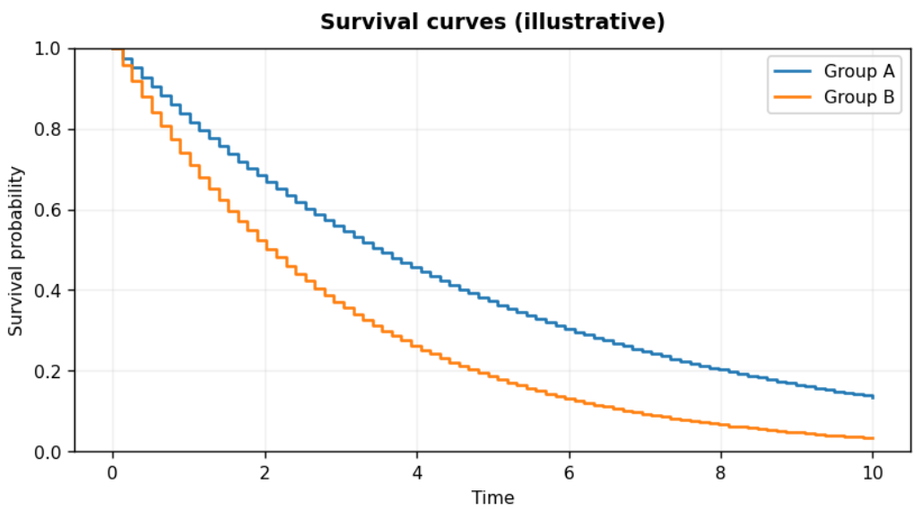


Figure 15: Output from code example 19. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ISLR2, survival; functions: survfit, plot.

Cox-proportional harzards model

S(t) = P(T>t)=1-F(t)

$h(t) = \lim_{\Delta t \to 0} \frac{P(t<T \leq t+\Delta t | T>t)}{\Delta t}$

Final review of R codes

Calculation

Vectors

Rmd knowledge

{r, echo = FALSE} –Hidden Code

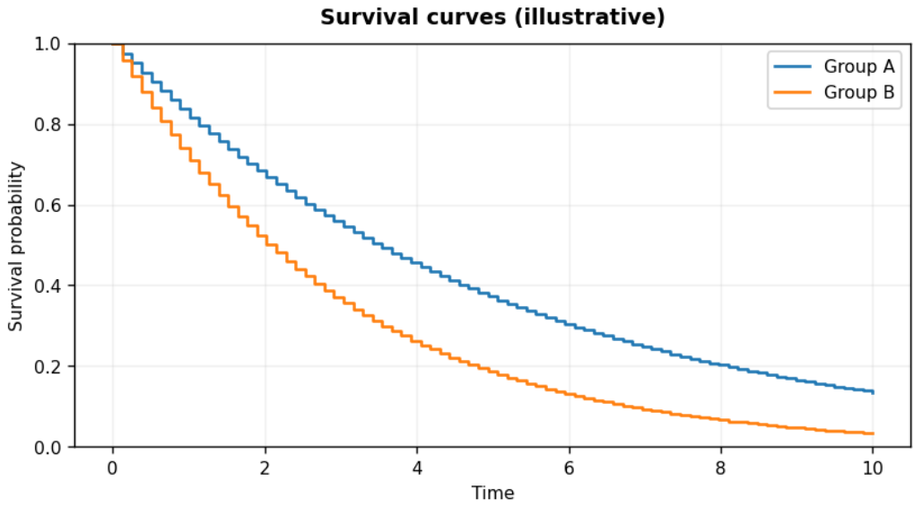


Figure 16: Output from code example 20. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: coxph, survfit, plot.

{r, eval = FALSE} –Do not run this code

{r, message = FALSE} –Do not show the message

{r, warning = FALSE} –Do not show the warning

{r,results='hide'} –Do not show the results

Probability in R

⟵ **Motivation**
From a probabilistic model to a usable numerical quantity: what is observed, what is random?

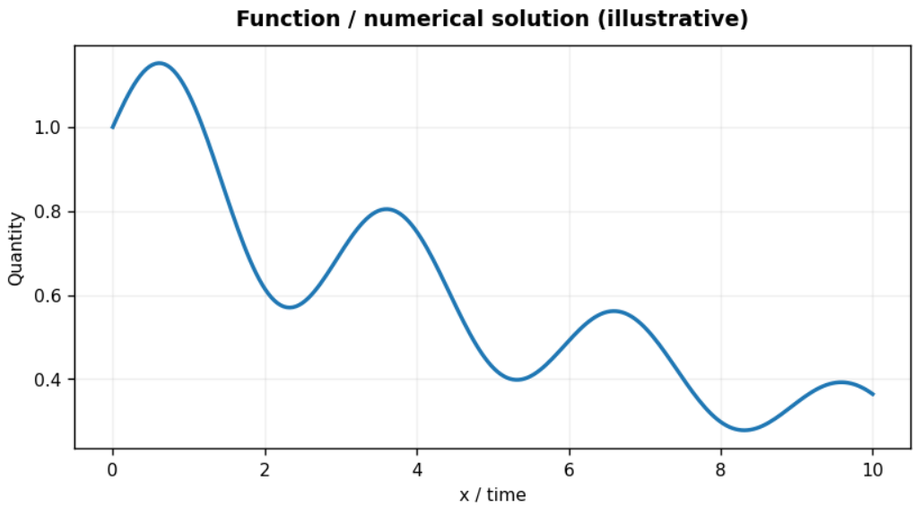


Figure 17: Output from code example 24. Illustrative reconstruction based on nearby code; not the exact original output. R packages: mosaic; functions: plotting / simulation functions.

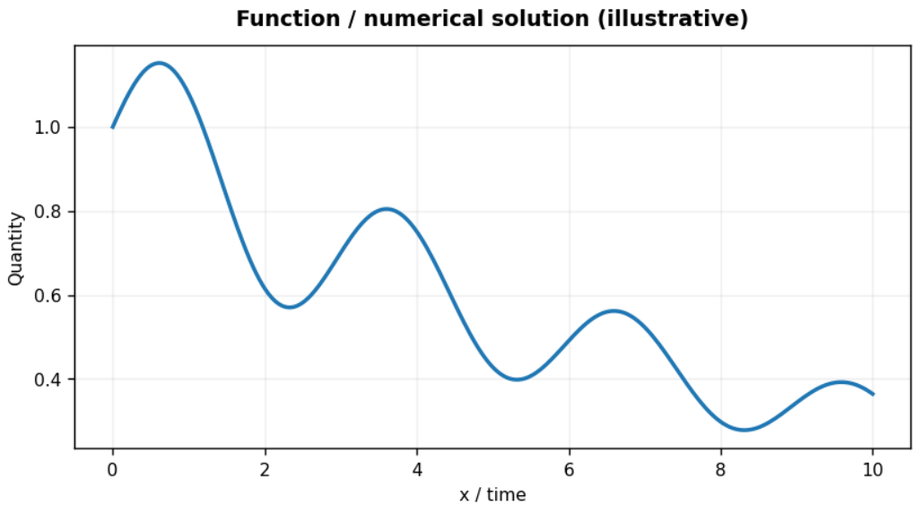


Figure 18: Output from code example 24. Illustrative reconstruction based on nearby code; not the exact original output. R packages: mosaic; functions: plotting / simulation functions.

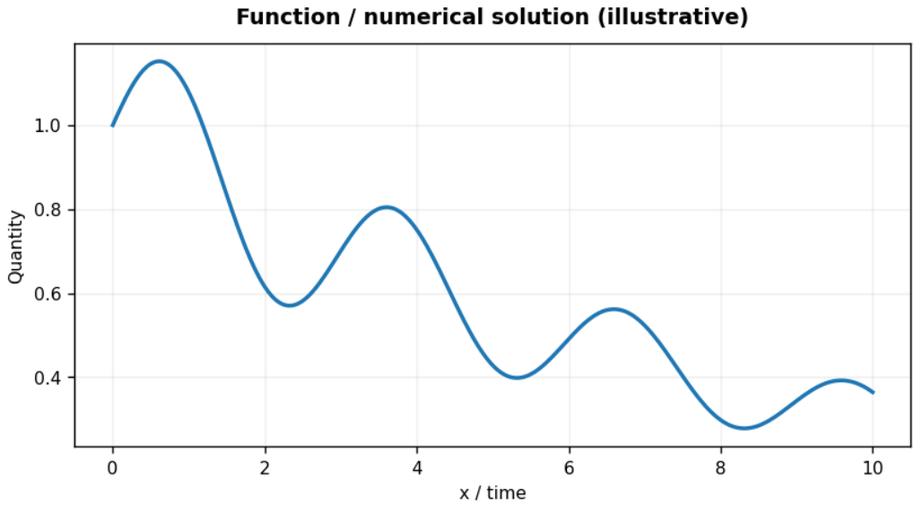


Figure 19: Output from code example 24. Illustrative reconstruction based on nearby code; not the exact original output. R packages: mosaic; functions: plotting / simulation functions.

R basic

Data class

⟵ **Proof idea**
Identify the exact statement before proving it.

Numeric

Integer

Character

Logical

Factor

Date and time

The following symbols can be used with the `format()` function to print dates.

- %d day as a number (0-31) 01-31
- %a abbreviated weekday Mon
- %A unabbreviated weekday Monday
- %m month (00-12) 00-12
- %b abbreviated month Jan
- %B unabbreviated month January
- %y 2-digit year 07
- %Y 4-digit year 2007

Data frame

List

Matrix

Example 1

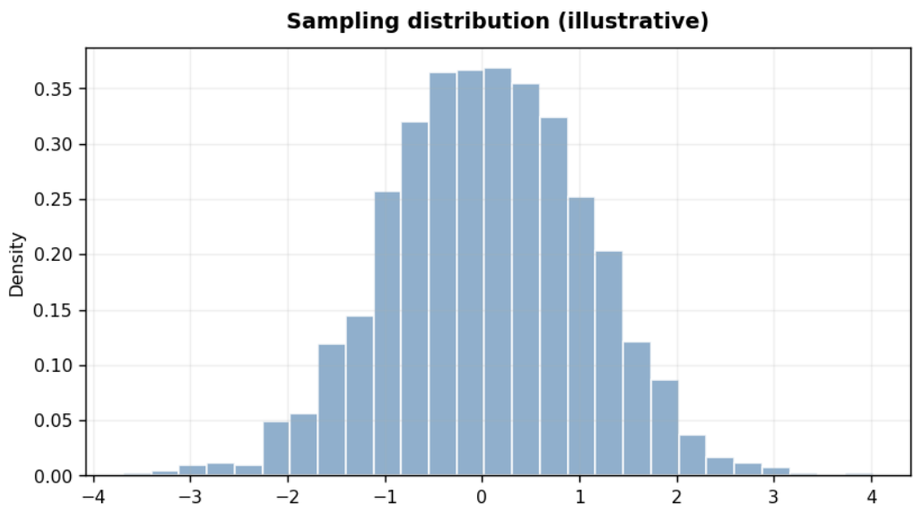


Figure 20: Output from code example 34. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: hist.

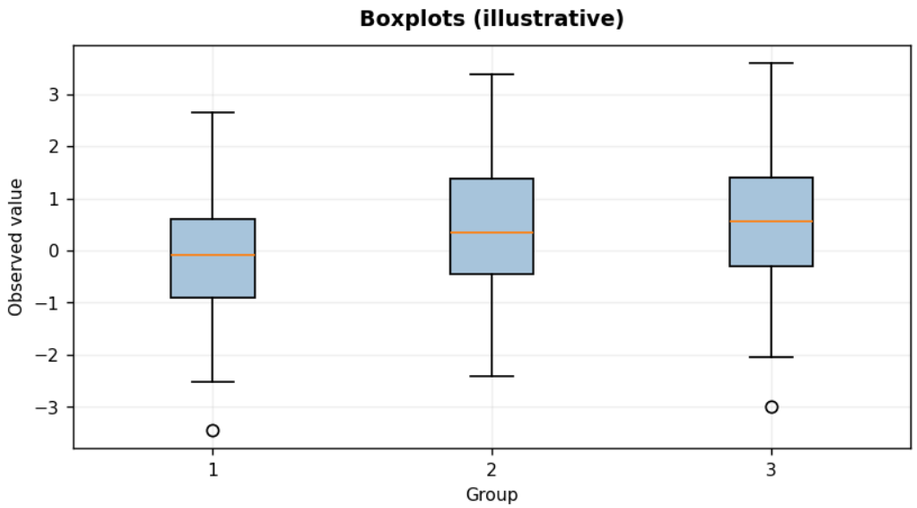


Figure 21: Output from code example 35. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: boxplot, ggplot, geom_boxplot.

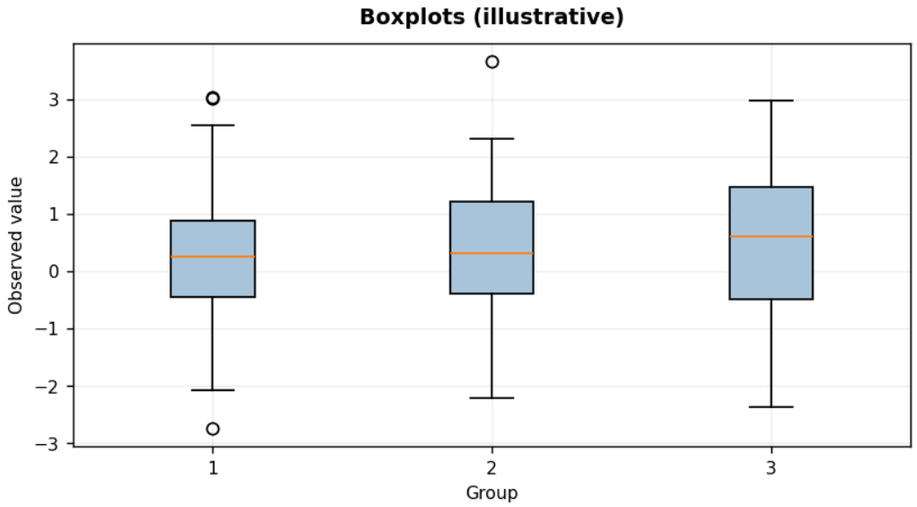


Figure 22: Output from code example 35. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: boxplot, ggplot, geom_boxplot.

Data visualization (mainly: ggplot2)

Box plots

Histogram plots

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

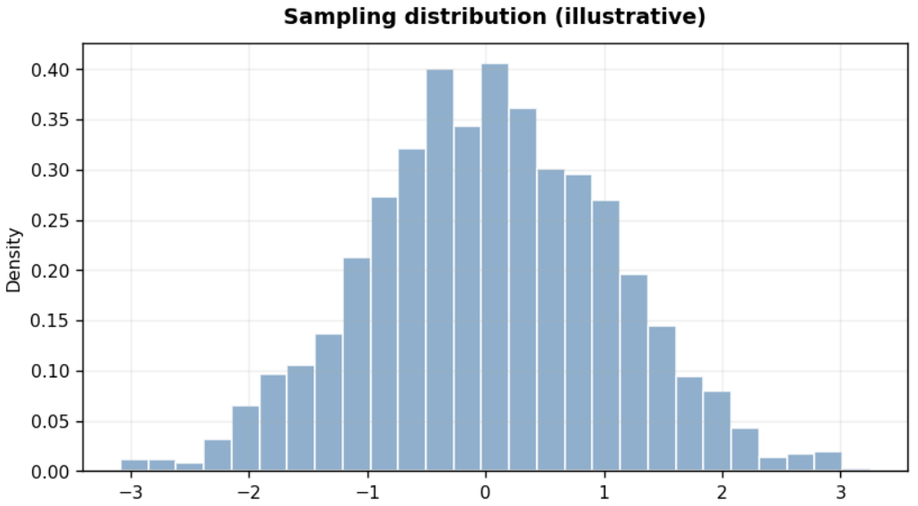


Figure 23: Output from code example 36. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram.

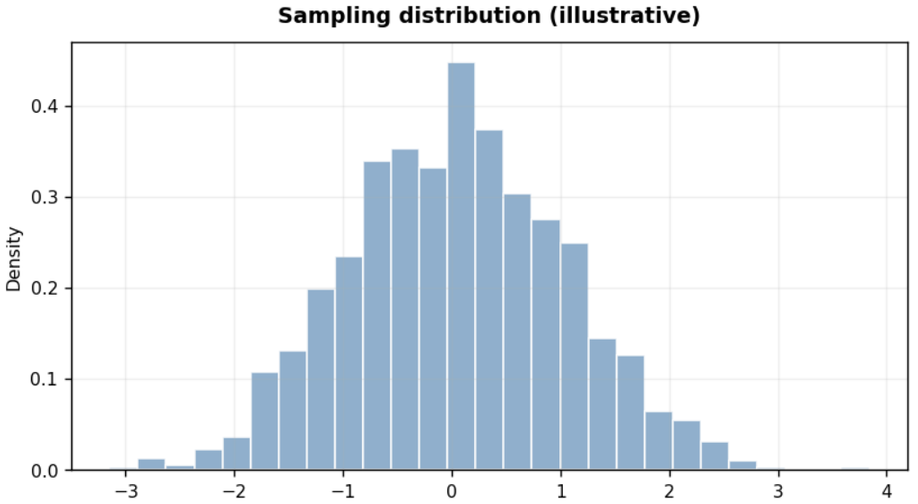


Figure 24: Output from code example 36. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram.

Density plots

- mutiple plots in one figure

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

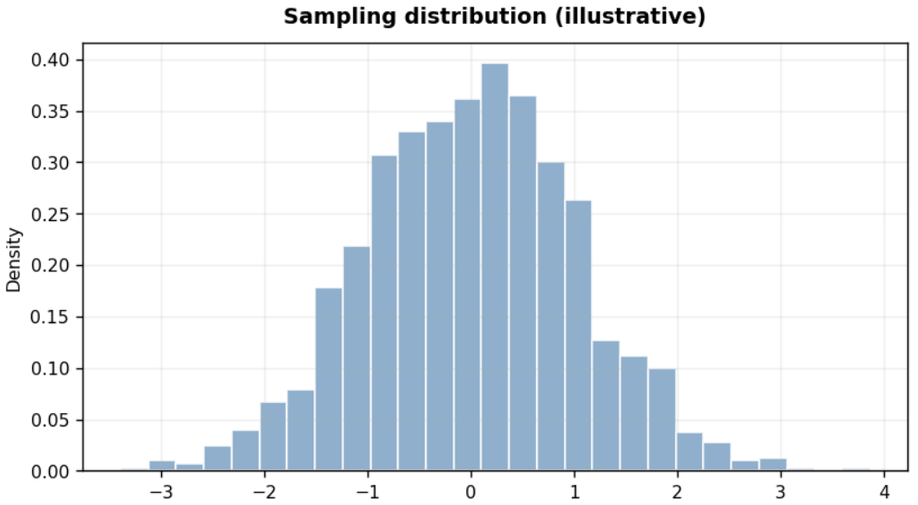


Figure 25: Output from code example 36. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram.

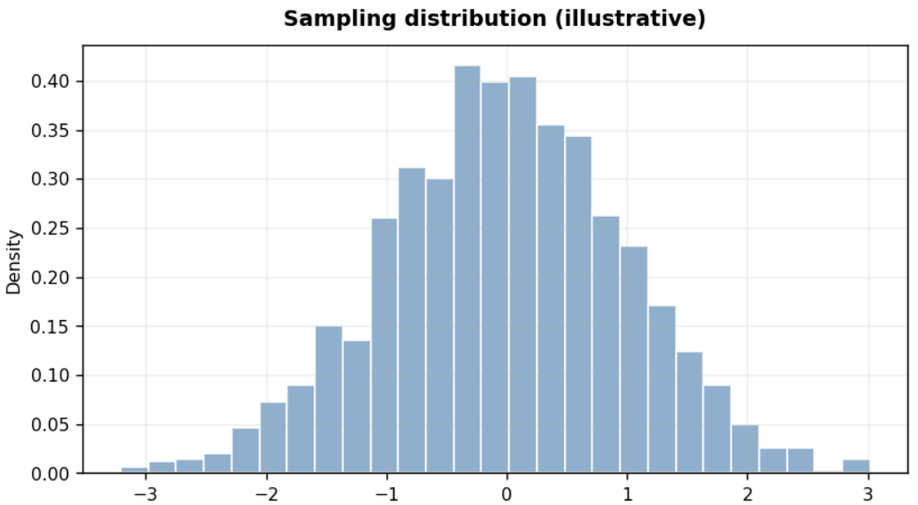


Figure 26: Output from code example 36. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram.

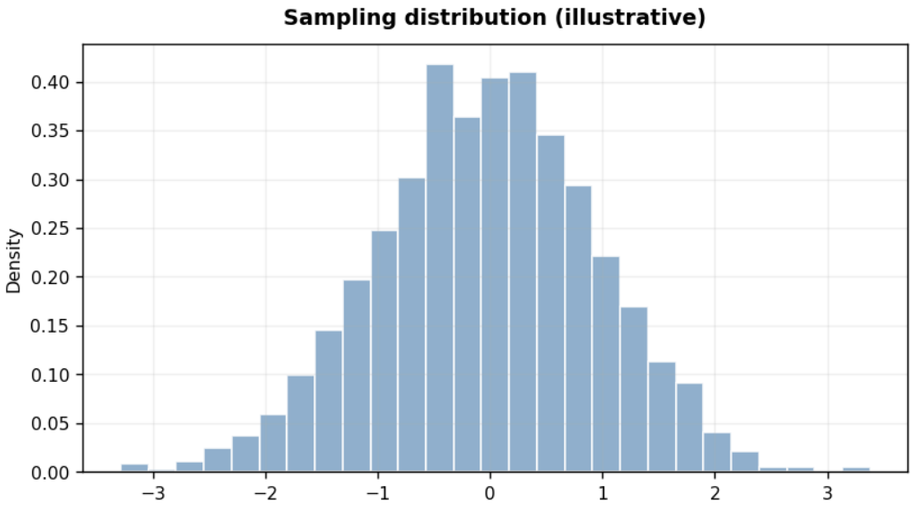


Figure 27: Output from code example 36. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram.

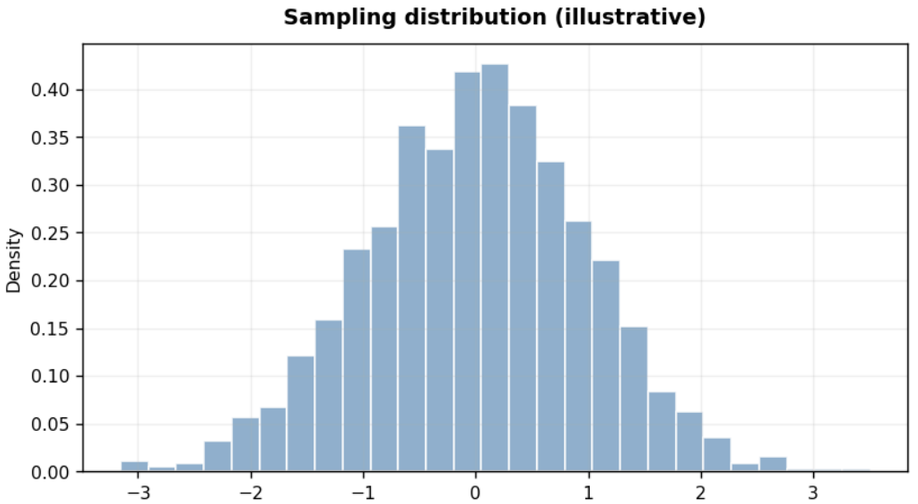


Figure 28: Output from code example 37. Illustrative reconstruction based on nearby code; not the exact original output. R packages: gridExtra; functions: ggplot, geom_histogram, geom_density.

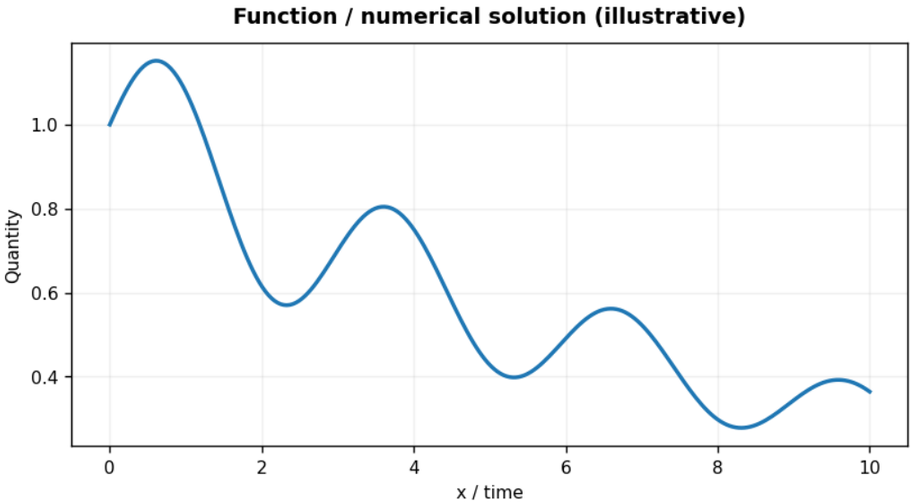


Figure 29: Output from code example 38. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_density.

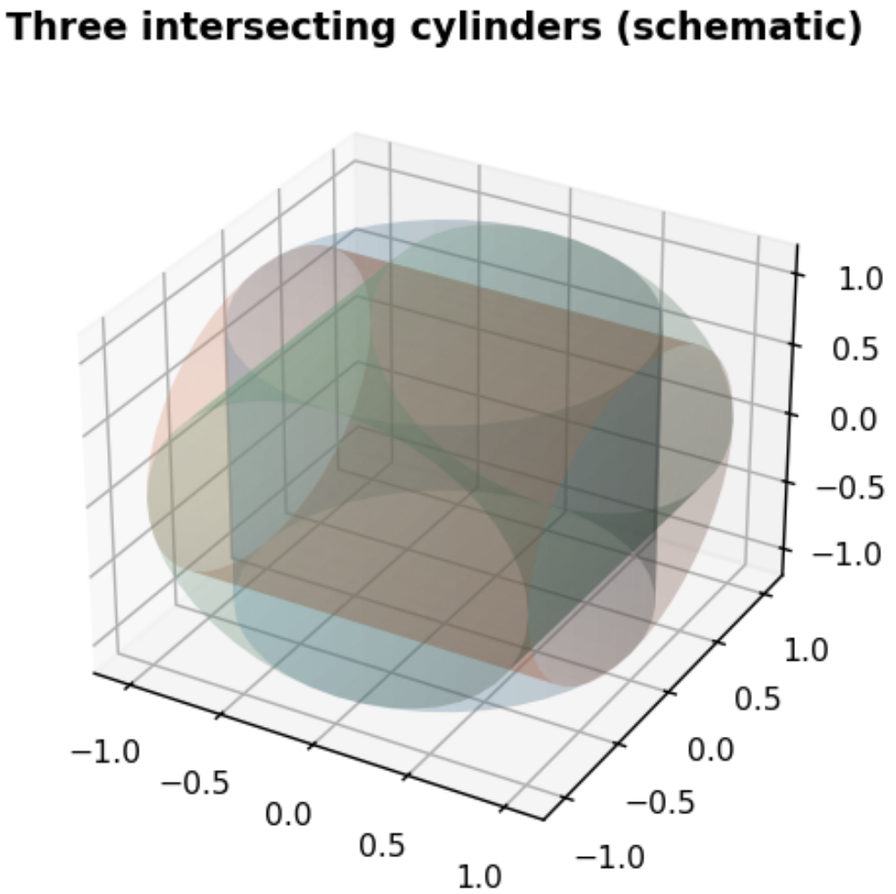


Figure 30: Output from code example 39. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_point.

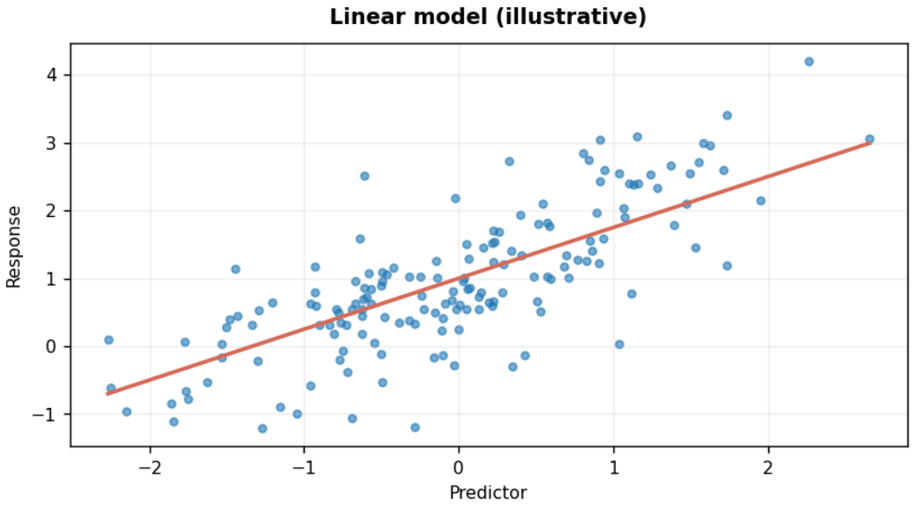


Figure 31: Output from code example 40. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_point, geom_smooth.

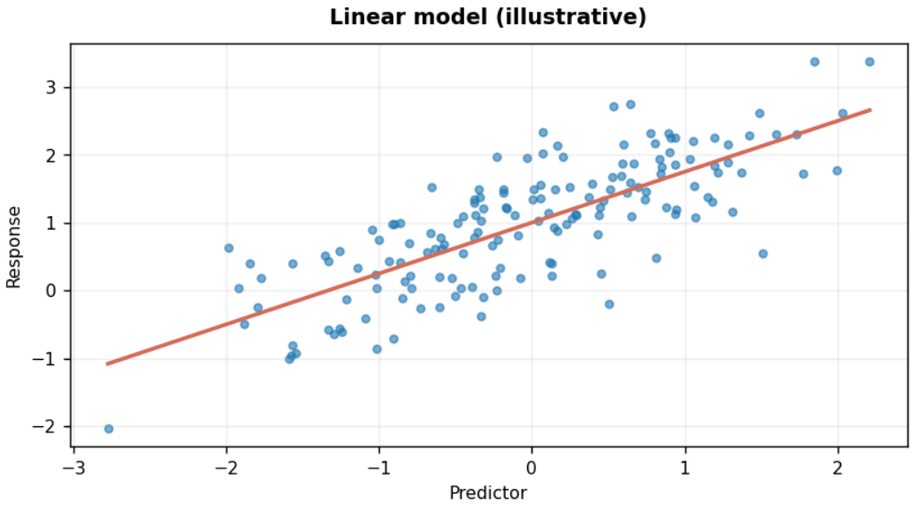


Figure 32: Output from code example 40. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_point, geom_smooth.

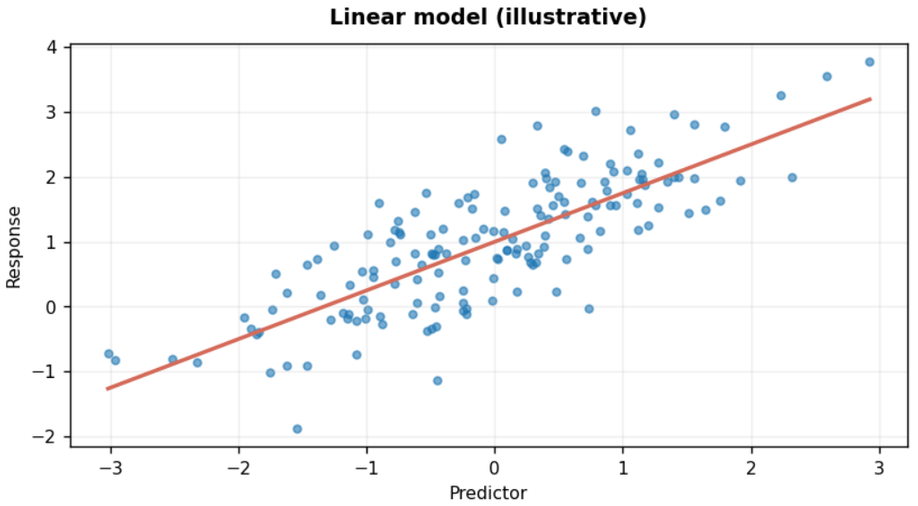


Figure 33: Output from code example 41. Illustrative reconstruction based on nearby code; not the exact original output. R packages: palmerpenguins; functions: ggplot, geom_point, geom_smooth.

Scatter plots

Scatter plots with regression line

Heat map

⟵ **Proof idea**
Distinguish a model for the conditional mean from the noise model.

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

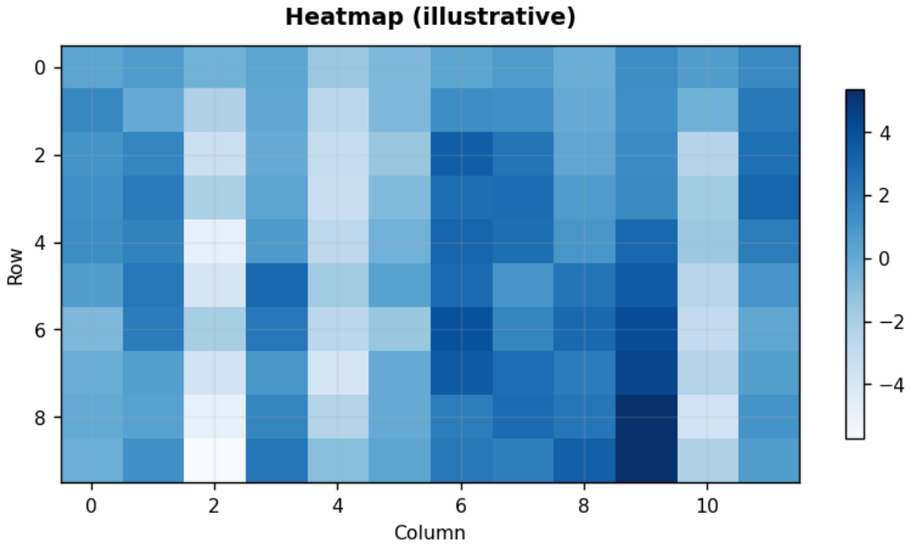


Figure 34: Output from code example 42. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_tile, ggtitle, geom_text.

Create heat map using facet_grid to show the data in different panels by depth

Faceting (make many panels of graphics where each panel represents the same relationship between variables, but something changes between each pane)

- Another example

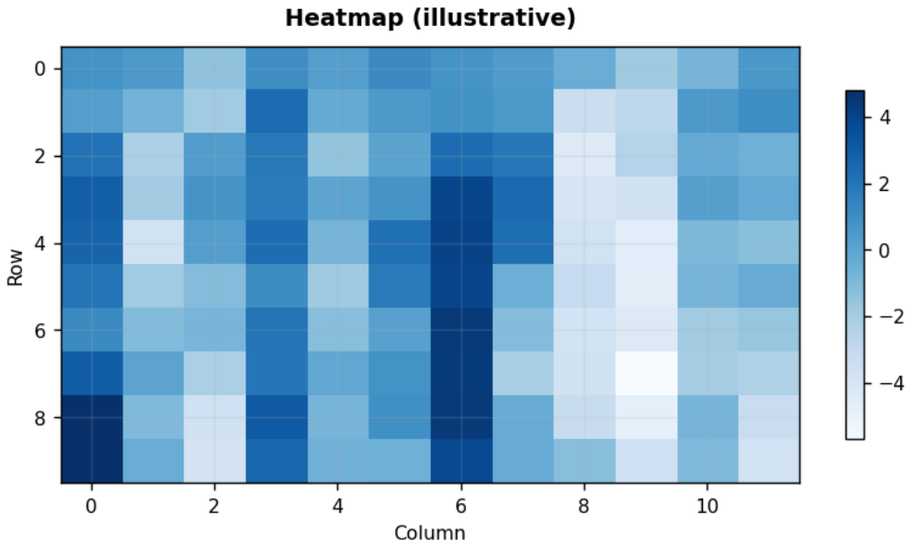


Figure 35: Output from code example 43. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_tile, ggtitle, geom_text.

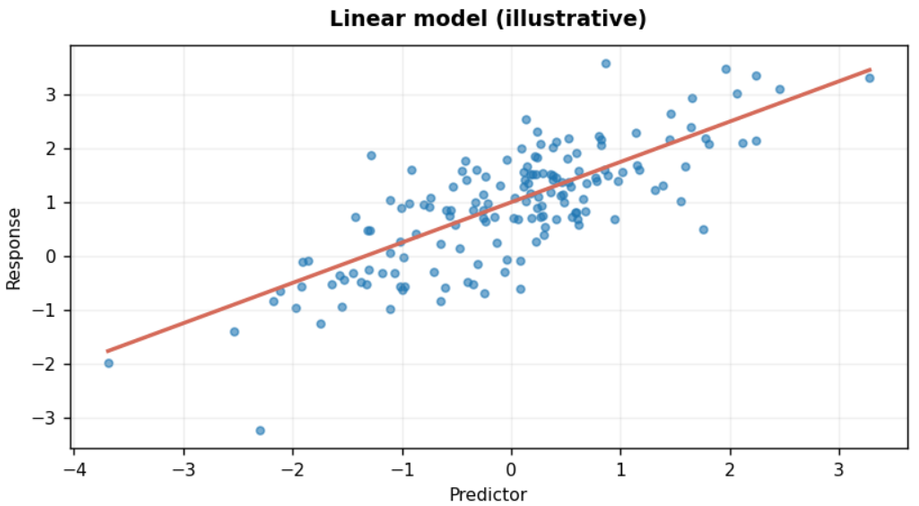


Figure 36: Output from code example 44. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_point.

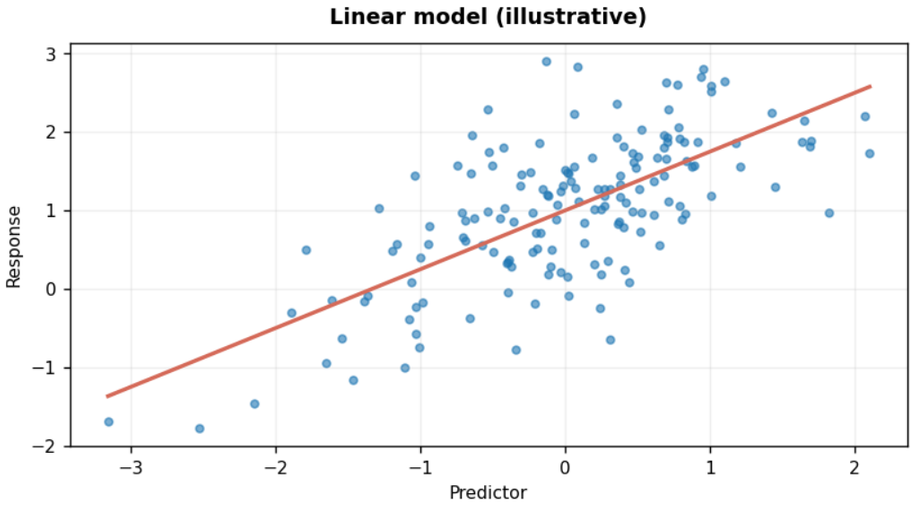


Figure 37: Output from code example 45. Illustrative reconstruction based on nearby code; not the exact original output. R packages: reshape; functions: ggplot, geom_point.

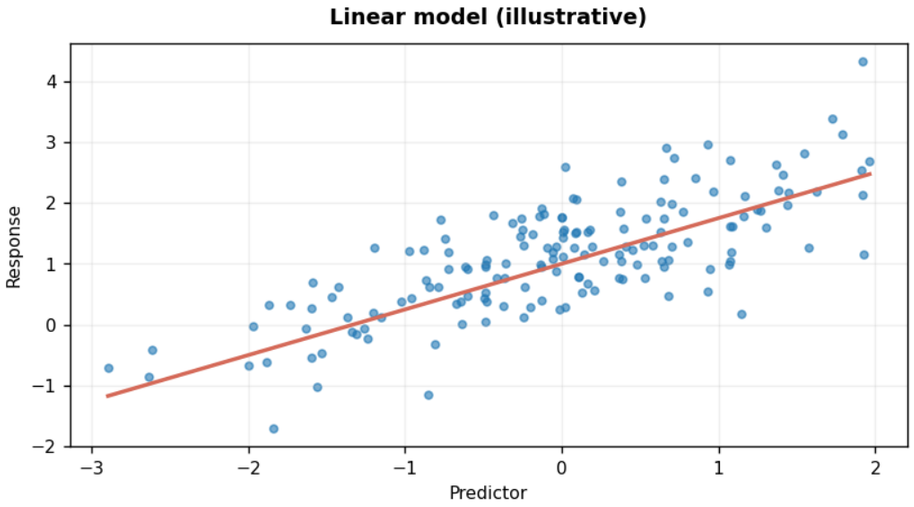


Figure 38: Output from code example 45. Illustrative reconstruction based on nearby code; not the exact original output. R packages: reshape; functions: ggplot, geom_point.

- Multi-plot

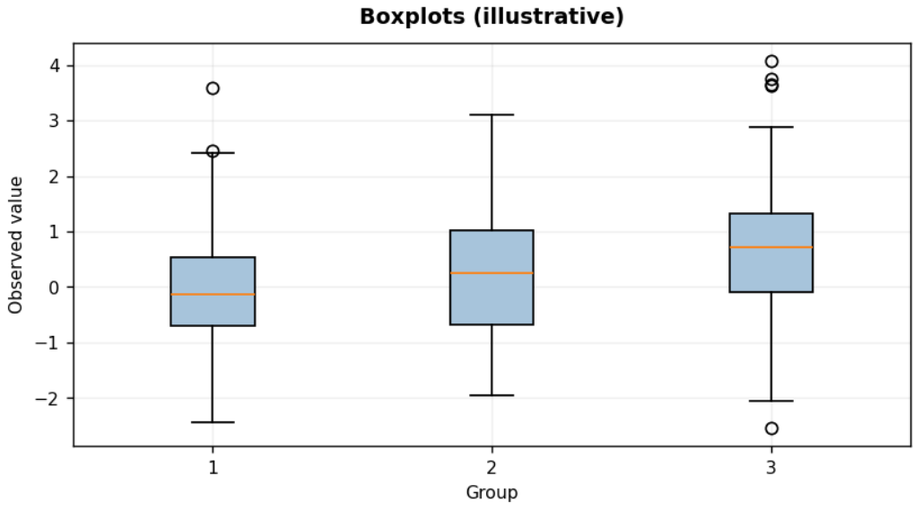


Figure 39: Output from code example 46. Illustrative reconstruction based on nearby code; not the exact original output. R packages: Rmisc, ggpubr; functions: ggplot, geom_line, ggtitle, geom_point, geom_smooth, geom_density, plot.

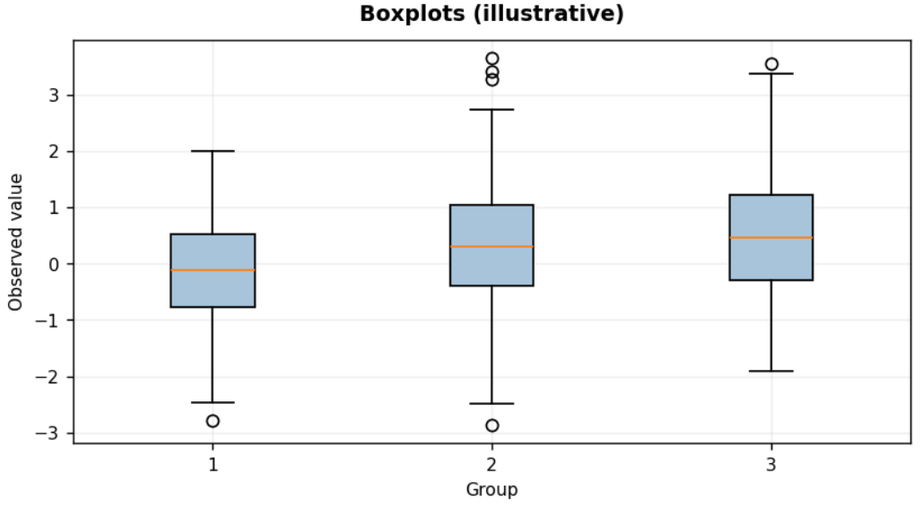


Figure 40: Output from code example 46. Illustrative reconstruction based on nearby code; not the exact original output. R packages: Rmisc, ggpubr; functions: ggplot, geom_line, ggtitle, geom_point, geom_smooth, geom_density, plot.

Data manipulation

apply

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

There are several variants of the apply() function, and the most frequently used ones are lapply() and sapply(). These two functions apply a given function to each element of a list or vector and returns a corresponding list or vector of results.

Tibbles

A tibble, or tbl_df, is a modern reimagining of the data.frame, keeping what time has proven to be effective, and throwing out what is not. Tibbles are data.frames that are lazy and surly: they do less (i.e. they don t change variable names or types, and don t do partial matching) and complain more (e.g. when a variable does not exist). This forces you to confront problems earlier, typically leading to cleaner, more expressive code. Tibbles also have an enhanced print() method which makes them easier to use with large datasets containing complex objects.

%>%

⟵ **Proof idea**
Identify the exact statement before proving it.

The pipe operator %>% is used to pass the result of one function to the next function in a chain, making the code more readable and concise. For example, if we wanted to start with x, and first apply function f(), then g(), and then h(), the usual R command would be h(g(f(x))) which is hard to read because you have to start reading at the innermost set of parentheses. Using the pipe command %>%, this sequence of operations becomes x %>% f() %>% g() %>% h().

select

filter

slice

This function is used to select rows by their position in the data frame. It can be used to select specific rows or a range of rows.

filter rows based on row number:

arrange

This function is used to sort the rows of a data frame by one or more columns. The default sorting of the number in the dataset is in ascending order, but you can use the `desc()` function to sort in descending order.

mutate

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

This function is used to create new columns or modify existing columns in a data frame. It allows you to perform calculations and transformations on the data.

summarise

This function is used to create a summary table. It reduces the data frame to a single row containing summary statistics.

group_by

This function is used to group the data frame by one or more columns. It is often used in combination with `summarise()` to calculate summary statistics for each group.

examples

Find the flight with the longest departure delay among flights from the same origin and destination (use `filter()`). Relocate the origin, destination, and departure delay to the first three columns and sort by origin and dest.

Find the flight with the longest departure delay among flights from the same origin and destination (use `top_n()` or `slice_max()`). Relocate the origin, destination, and departure delay to the first three columns and sort by origin and dest.

How do departure delays vary at different times of the day? Summarize the averaged departure delays by hours and create an new column named as `dep_delay_level` which `cut()` the averaged departure delays into three levels (low, median, and high).

How do departure delays vary at different times of the day? Illustrate your answer with a `geom_smooth()` plot.

how many data points are in the data set

ps:the comparison between whether to use %>% or not

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Control flow

while loop

for loop

Fibonacci sequence

bootstrap estimate of a sampling distribution

- bootstrap estimate of a sampling distribution

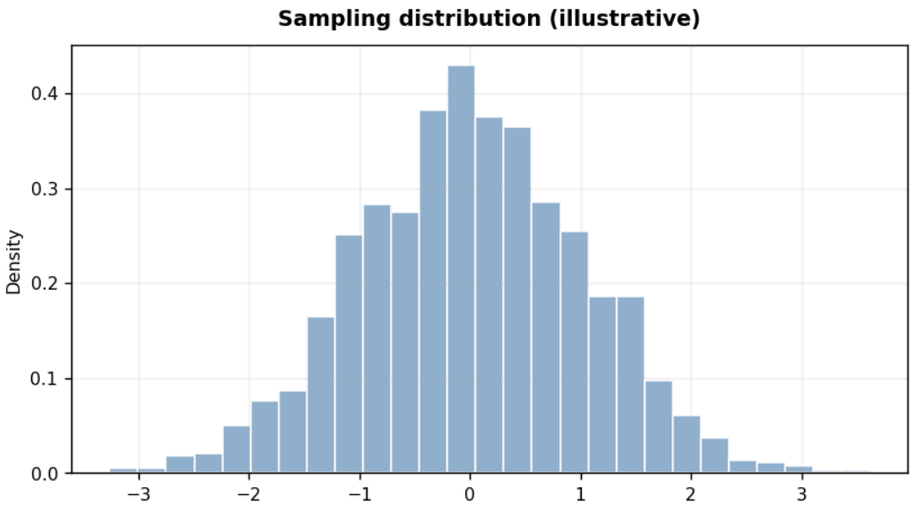


Figure 41: Output from code example 56. Illustrative reconstruction based on nearby code; not the exact original output. R packages: dplyr, ggplot2; functions: ggplot, geom_histogram.

Functions

Functions condtruction

copy of x

Ellipses

Data class

The following symbols can be used with the `format()` function to print dates.

- %d day as a number (0-31) 01-31
- %a abbreviated weekday Mon
- %A unabbreviated weekday Monday
- %m month (00-12) 00-12
- %b abbreviated month Jan
- %B unabbreviated month January
- %y 2-digit year 07

- %Y 4-digit year 2007

Practice of Lab 5

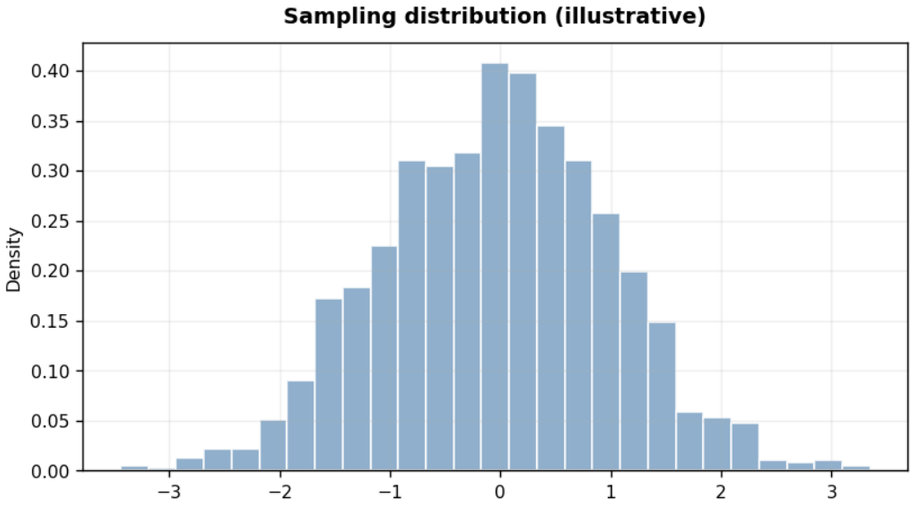


Figure 42: Output from code example 61. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: hist.

LAB 5 –data visualization

⟵ **Proof idea**
Identify the exact statement before proving it.

- Example (https://github.com/jcoliver/learn-r/blob/gh-pages/data/mine-microbe-class-data.csv)

3. Check the structure of the data frame.

The data frame has 8 rows (“obs.”) and 13 columns (“variables”). The first three columns have information about the observation (Site, Depth, Sample.name), and the remaining columns have the abundance for each of 10 classes of bacteria.

We ultimately want a heatmap where the different sites are shown along the x-axis, the classes of bacteria are shown along the y-axis, and the shading of the cell reflects the abundance. This latter value, the abundance of a bacteria at specific depths at different site, is really just a third dimension. However, instead of creating a 3-dimensional plot that can be difficult to visualize, we instead use shading for our “z-axis”. To this end, we need our data formatted so we have a column corresponding to each of these three dimensions:

- X: Sample identity
- Y: Bacterial class
- Z: Abundance

The challenge is that our data are not formatted like this. While the Sample.name column corresponds to what we would like for our x-axis, we do not have columns that correspond to what is needed for the y- and z-axes. All the data are in our data frame, but we need to take a table that looks like this:

Sample.name Actinobacteria Cytophagia 1-S 373871 8052 2-S 332856 28561 … … …

And transform it to one with a column for bacterial class and a column for abundance, like this:

Sample.name Class Abundance 1-S Actinobacteria 373871 1-S Cytophagia 8052 2-S Actinobacteria 332856 … … …

In the later labs, you will learn convenient package and function to achieve this. But for now, we just use some basic function to do so.

4. Remove the first two columns.
5. Use “colnames()” to get a vector of class names and assign it to the object named by “class.name”

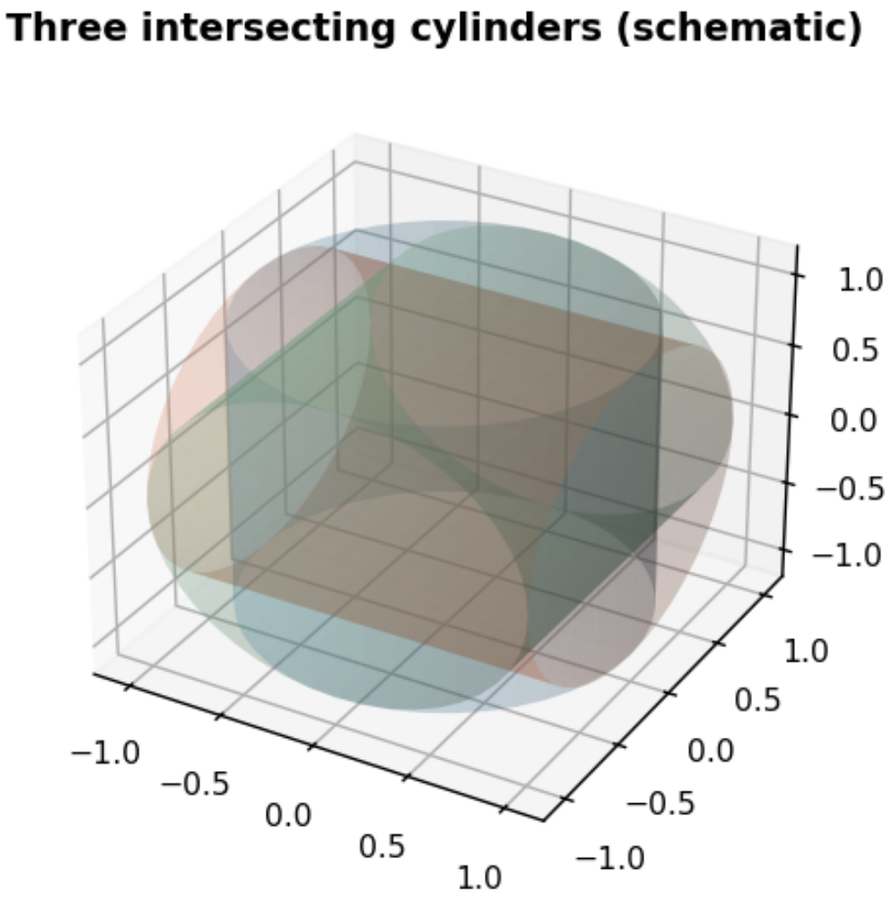


Figure 43: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

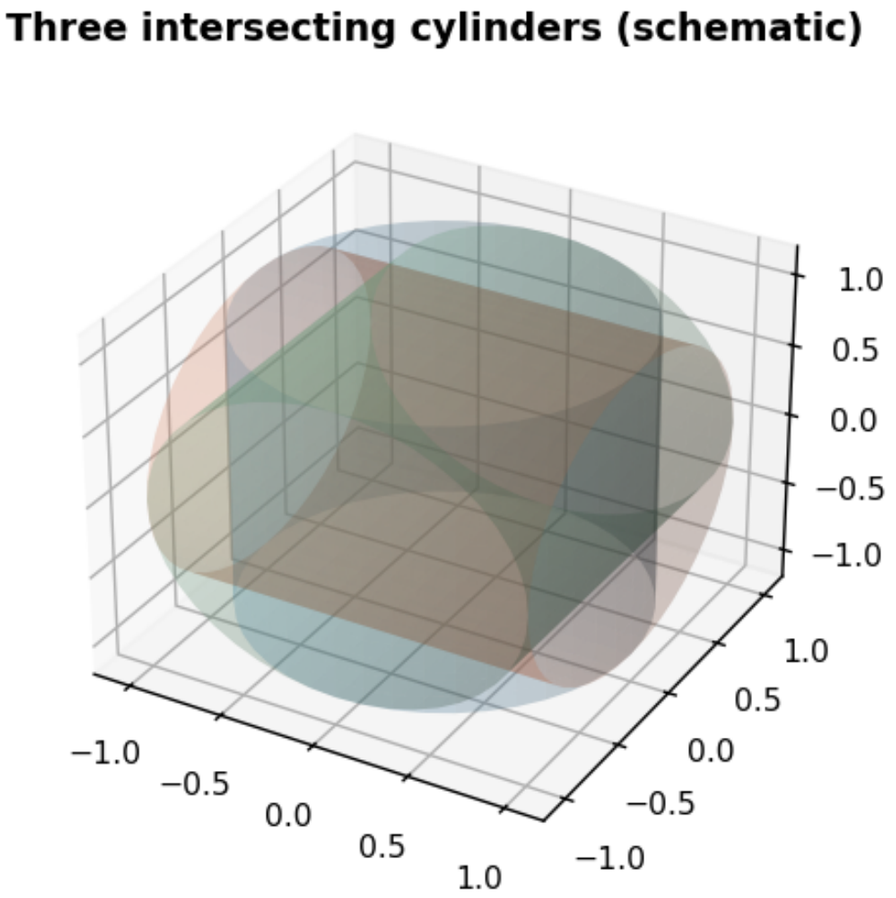


Figure 44: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

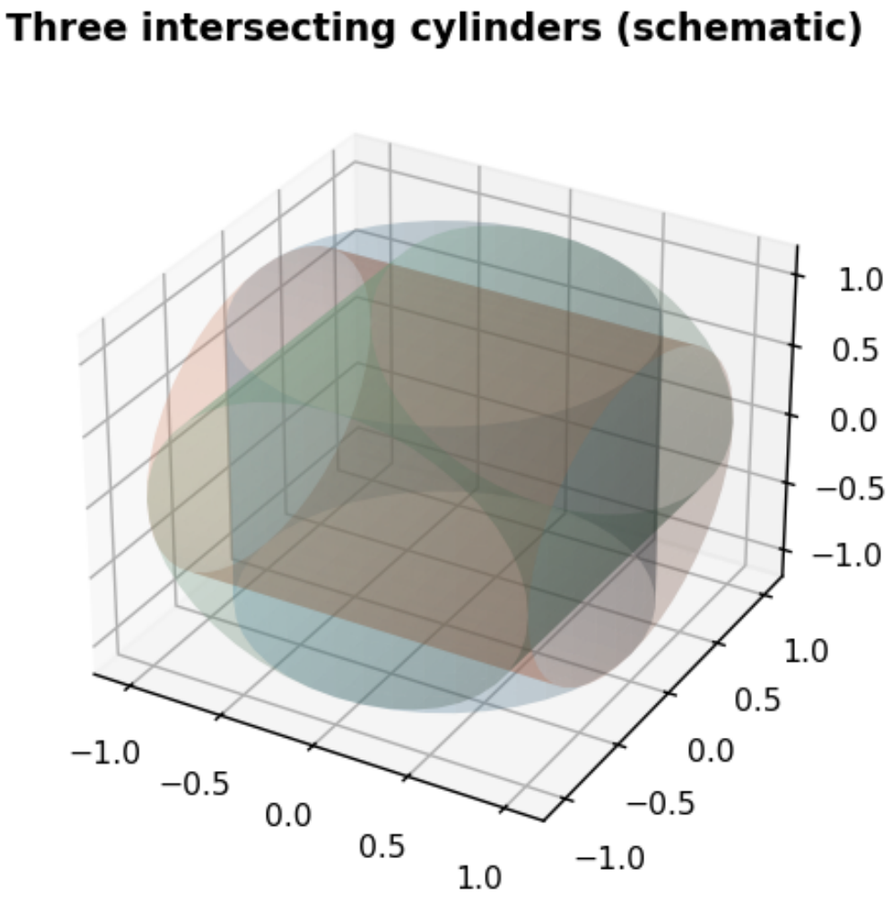


Figure 45: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

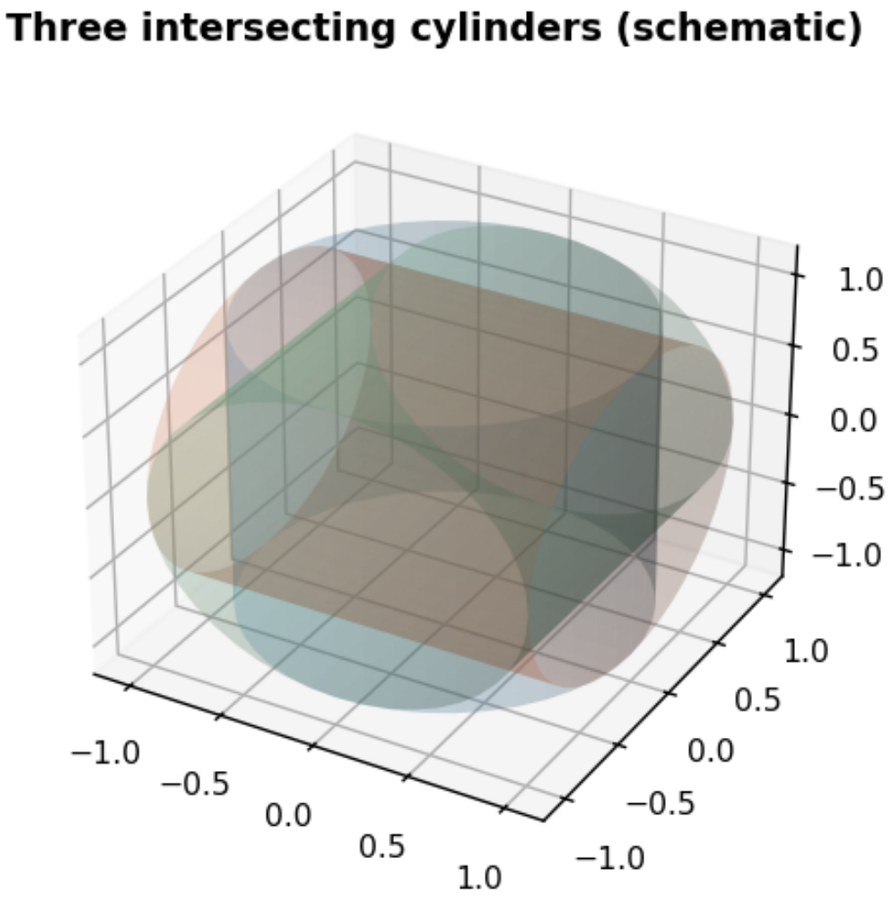


Figure 46: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

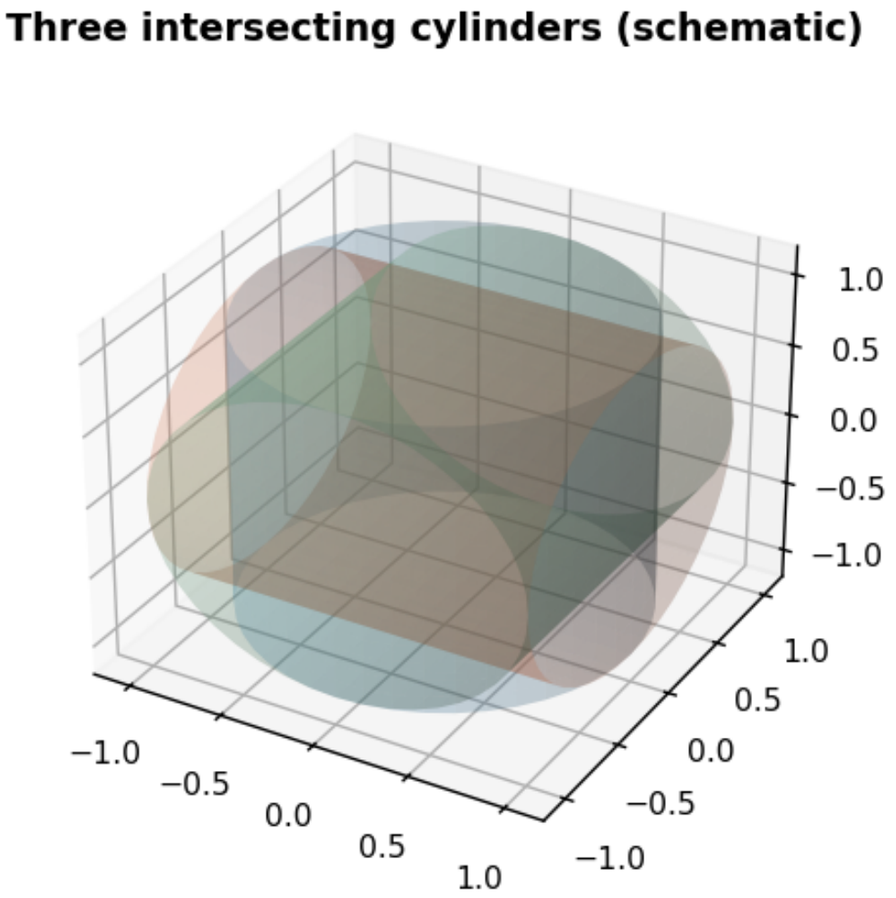


Figure 47: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

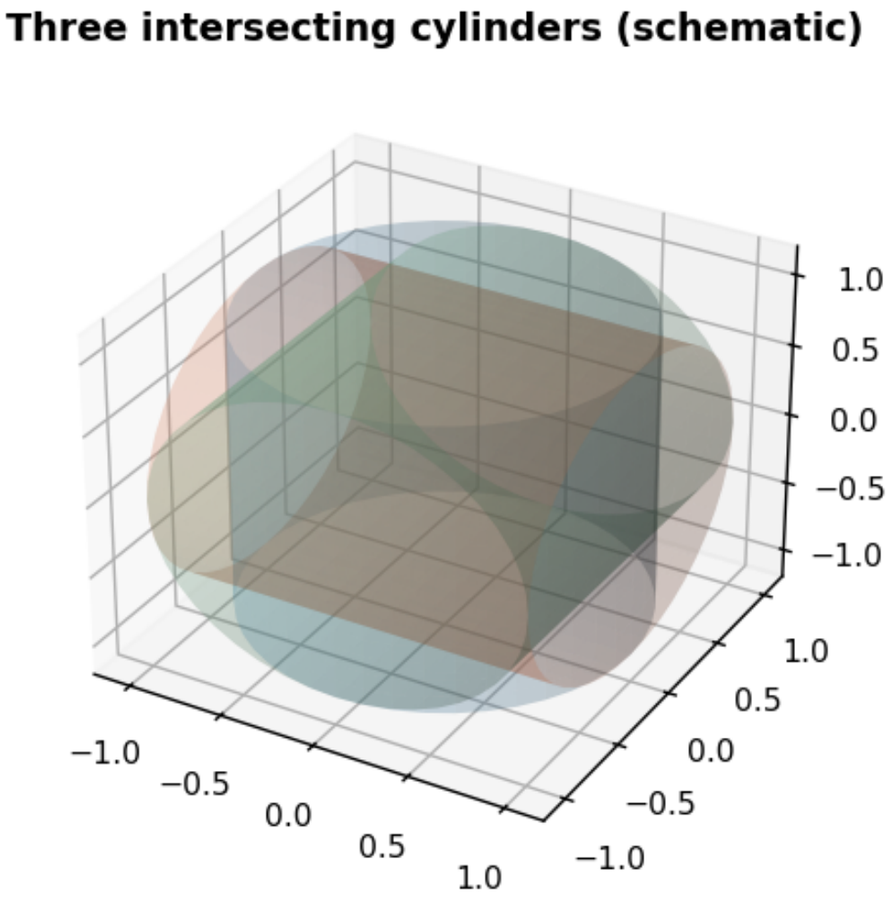


Figure 48: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

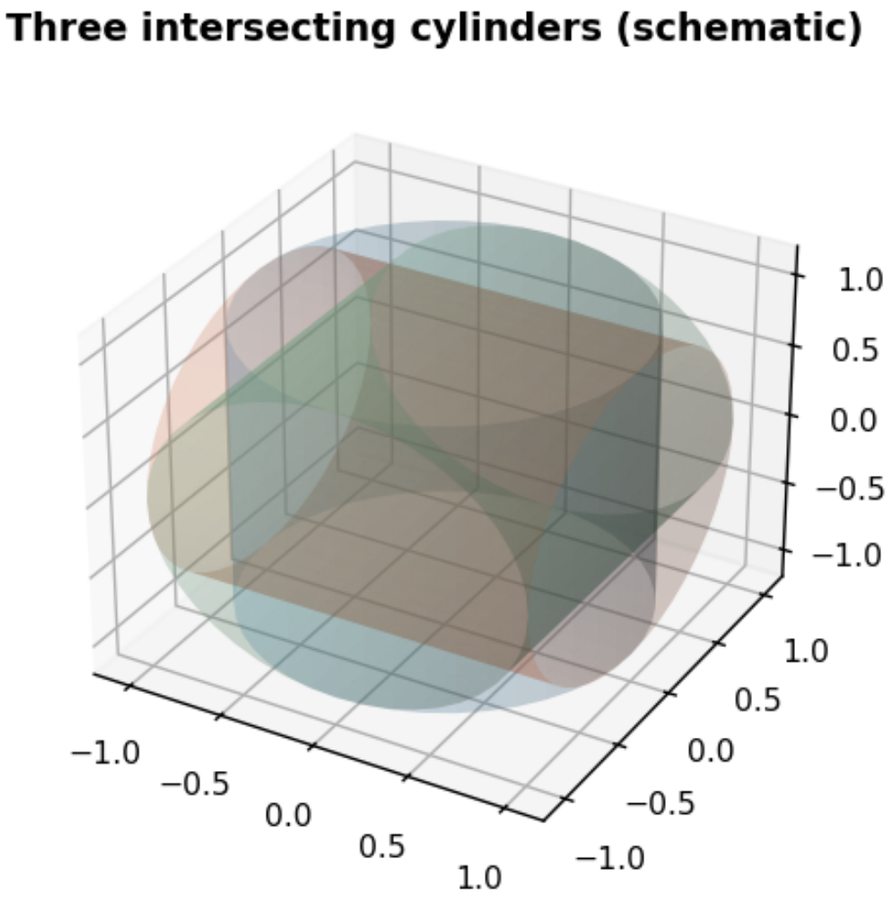


Figure 49: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

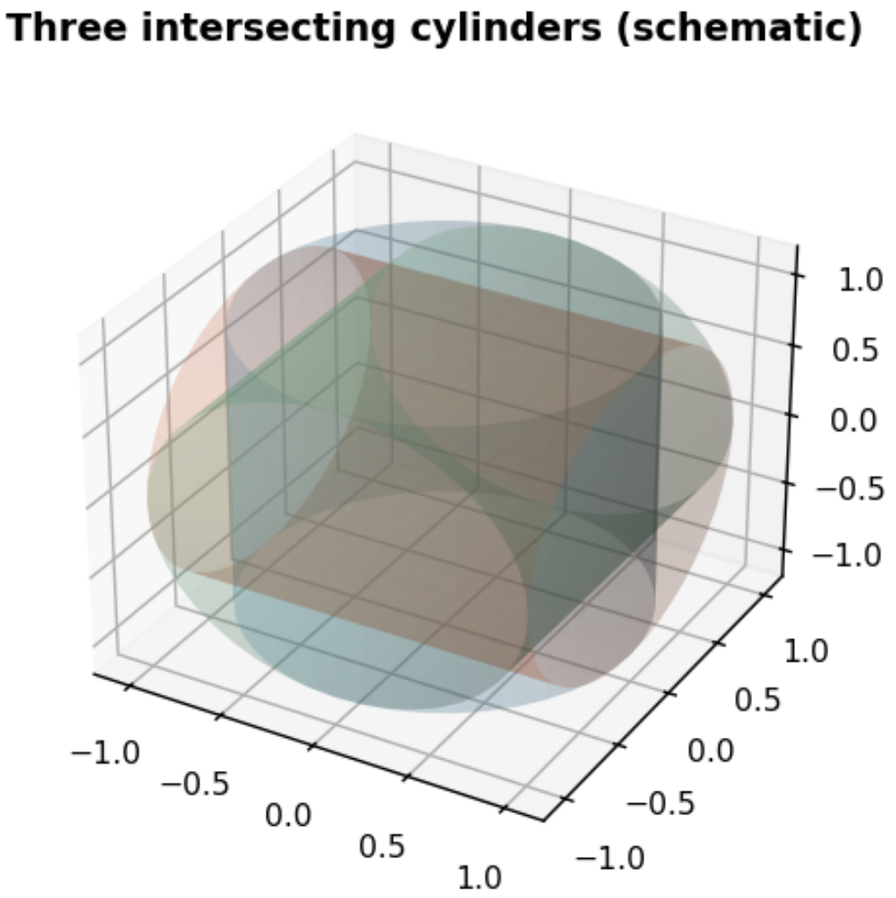


Figure 50: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

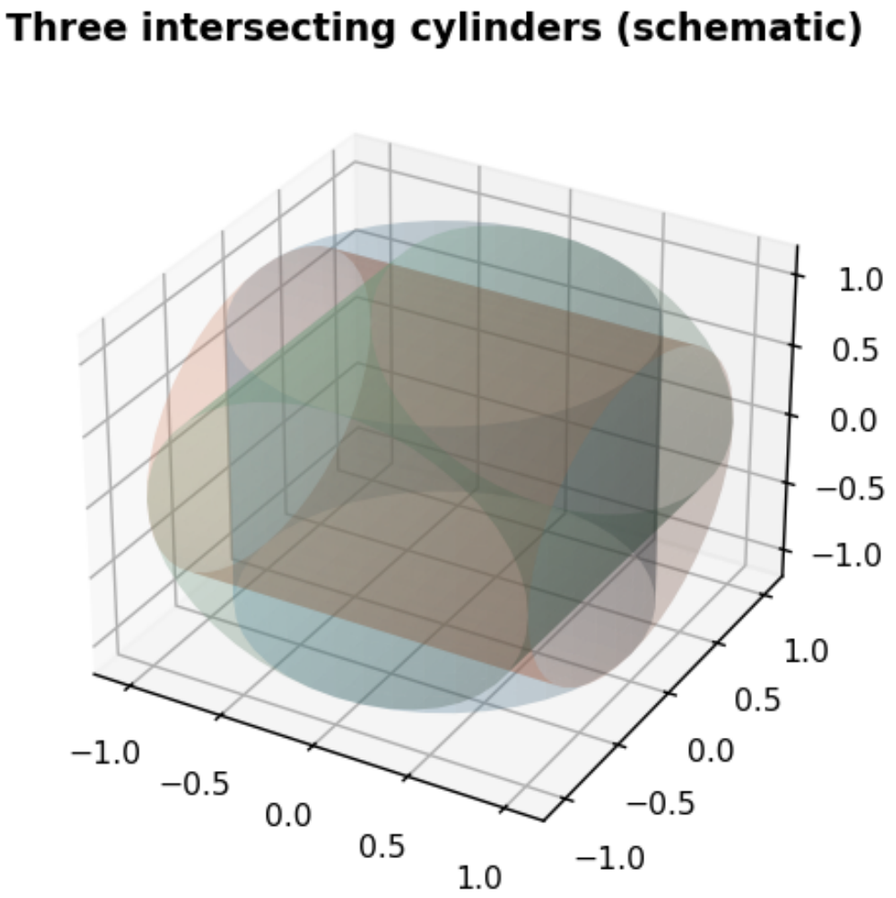


Figure 51: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

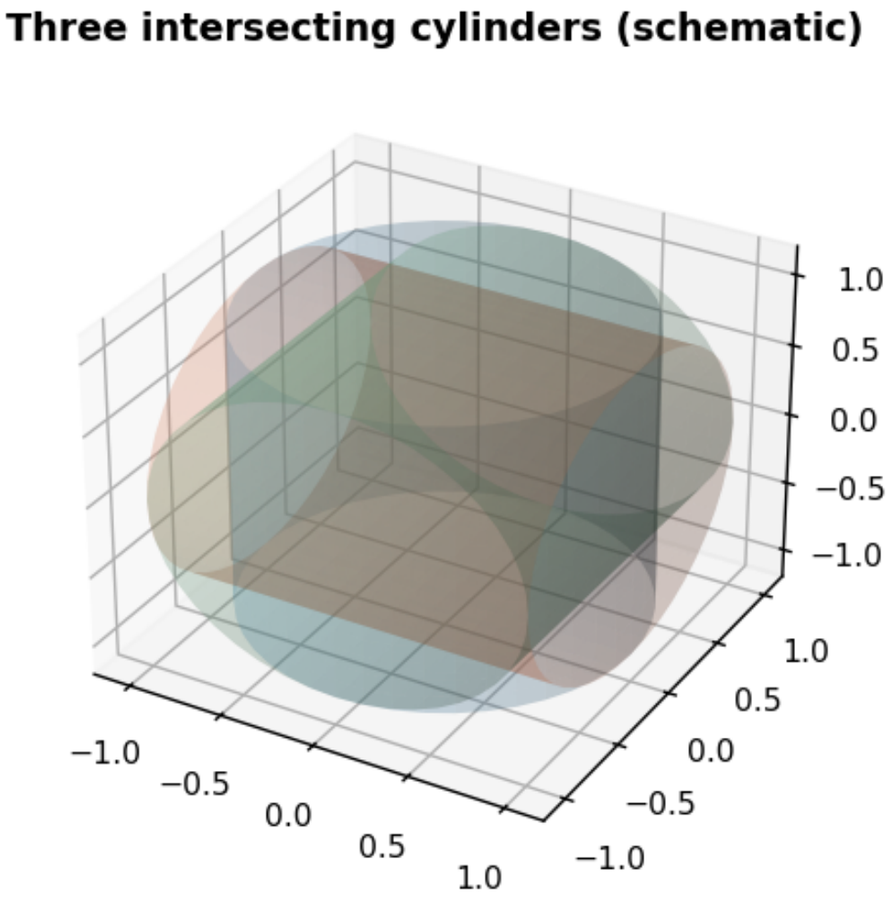


Figure 52: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

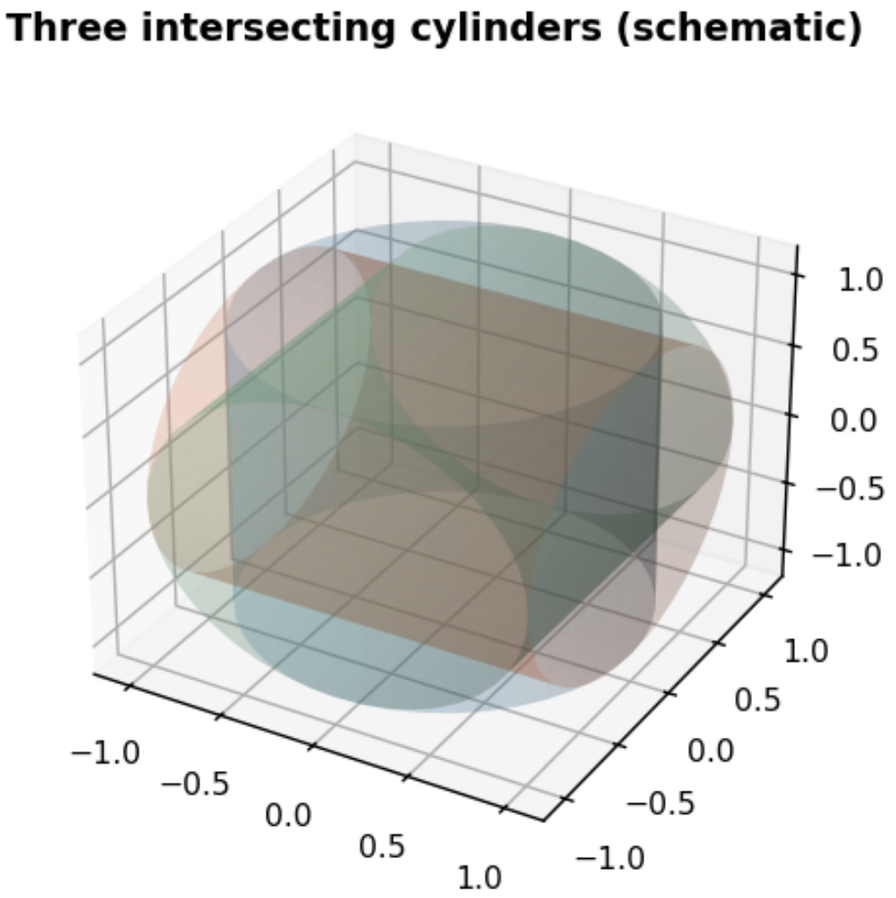


Figure 53: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

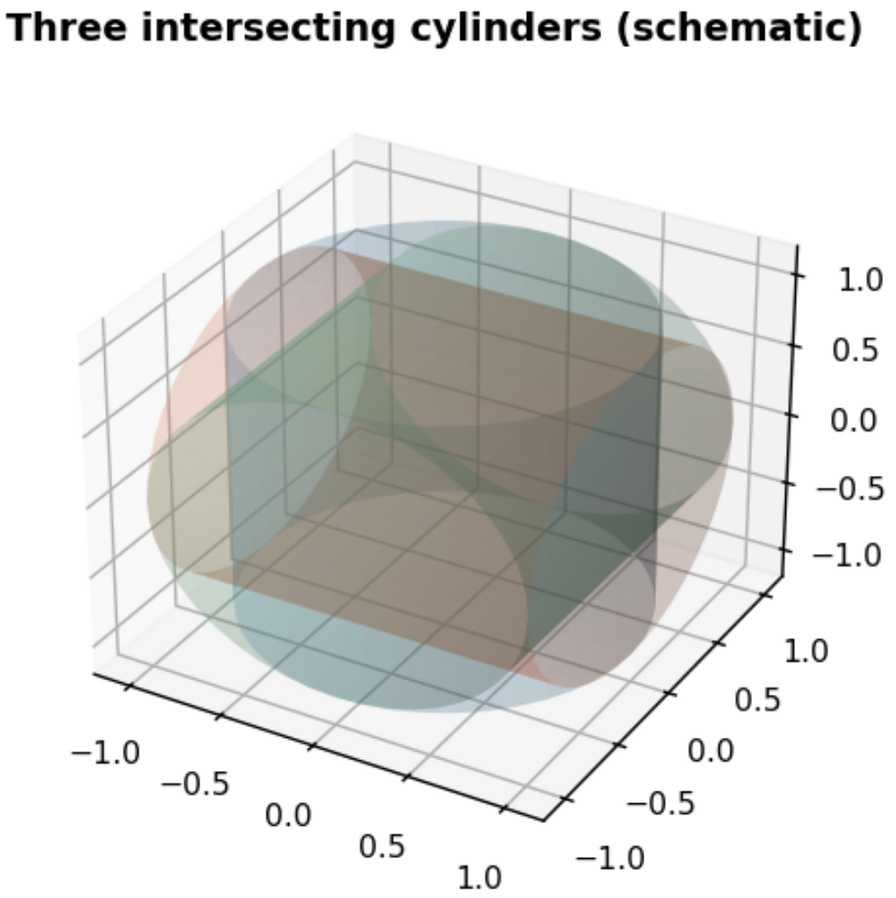


Figure 54: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

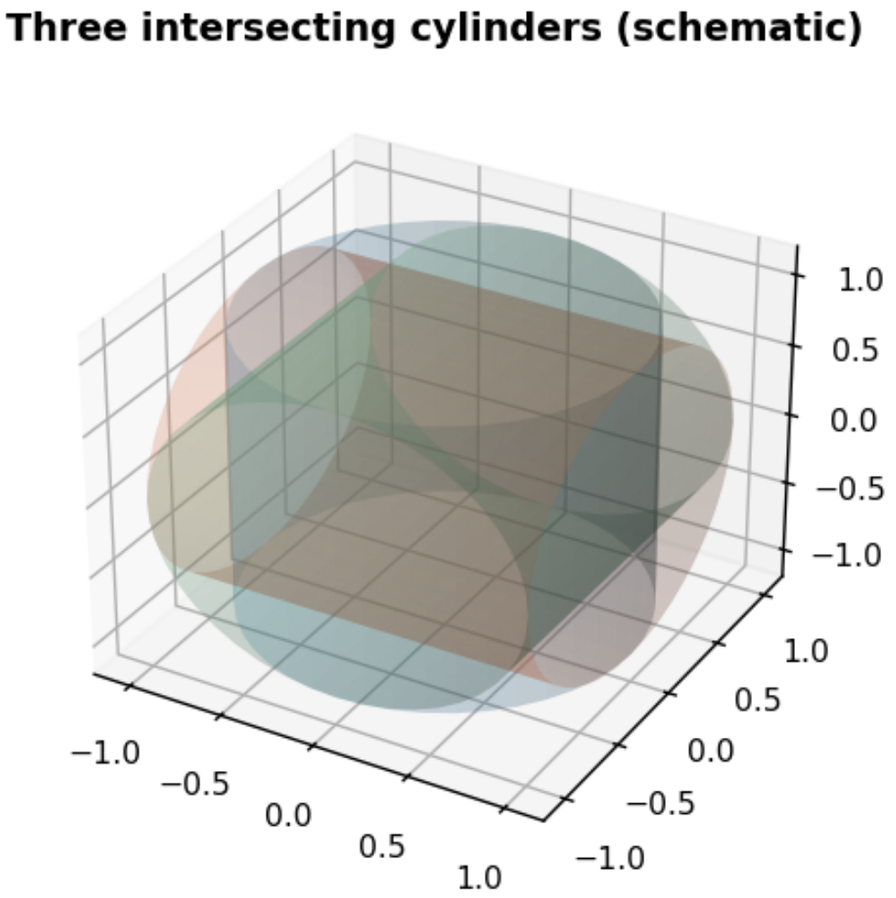


Figure 55: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

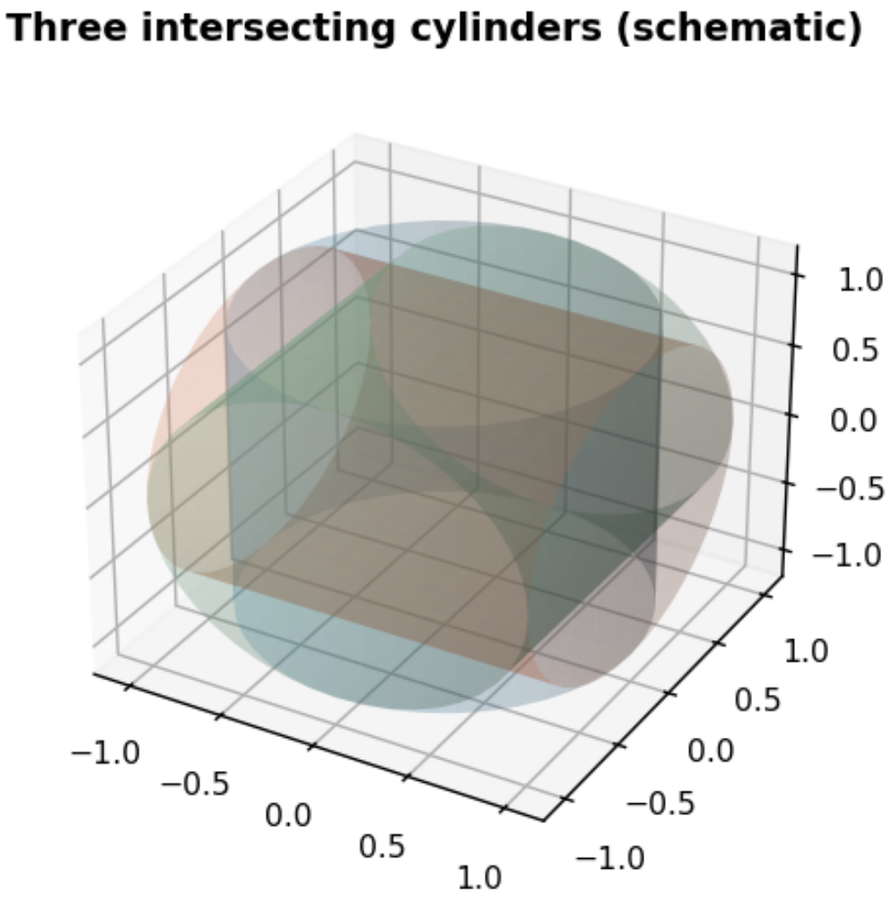


Figure 56: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

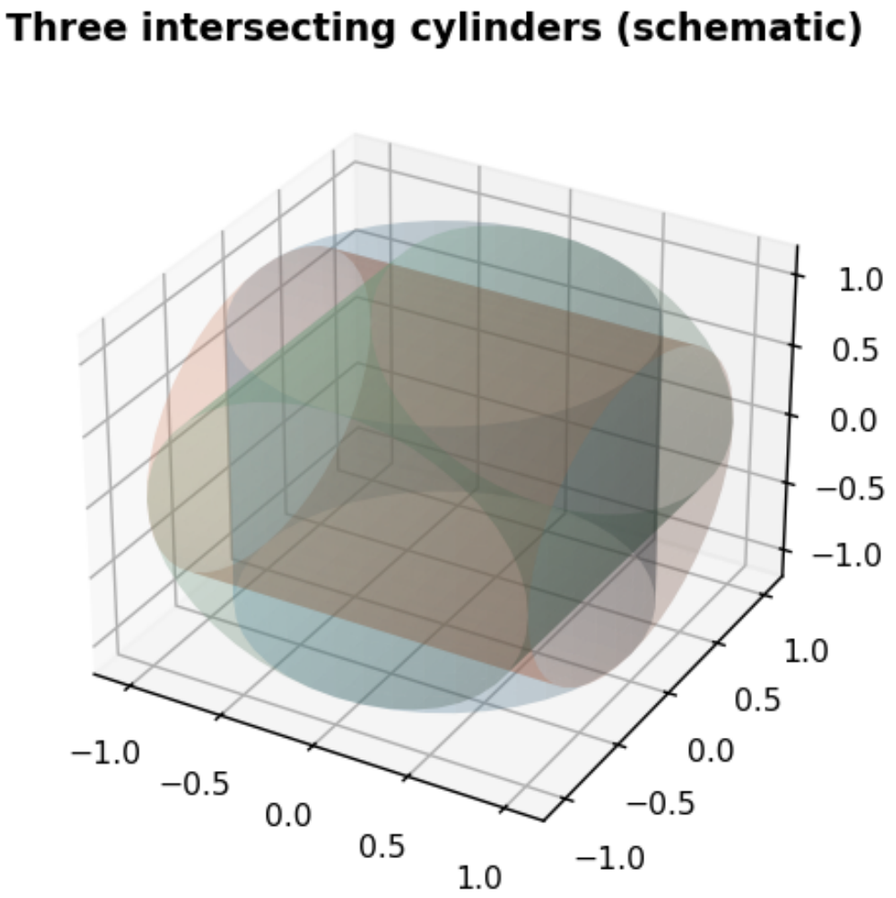


Figure 57: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

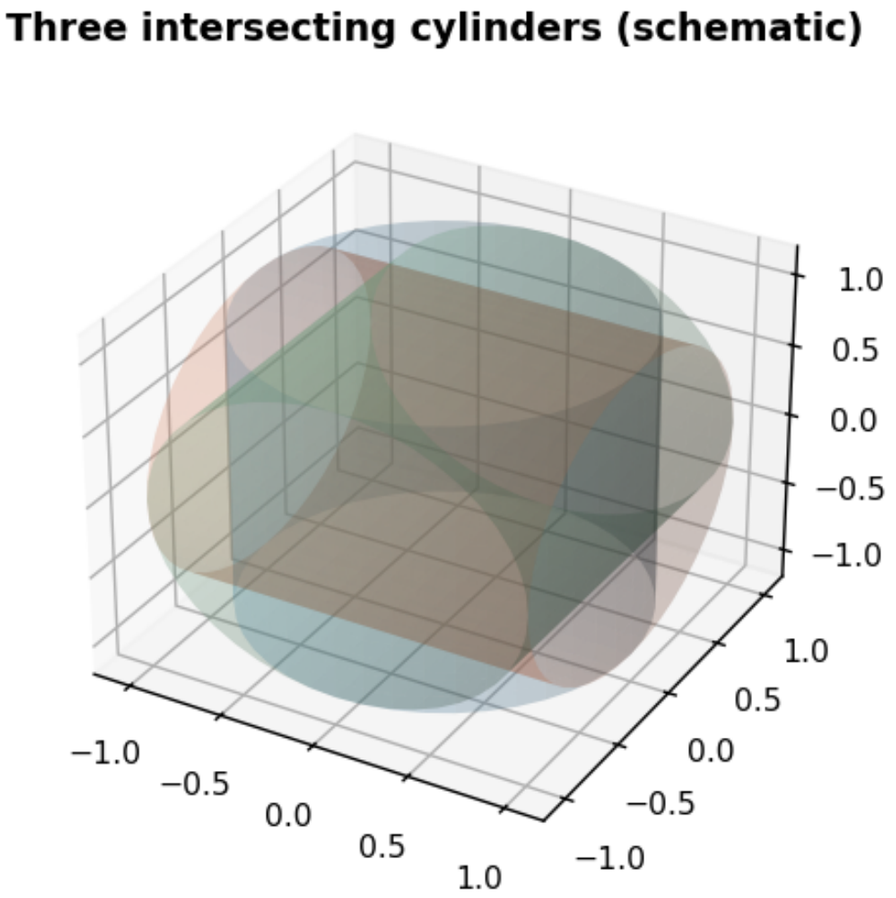


Figure 58: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

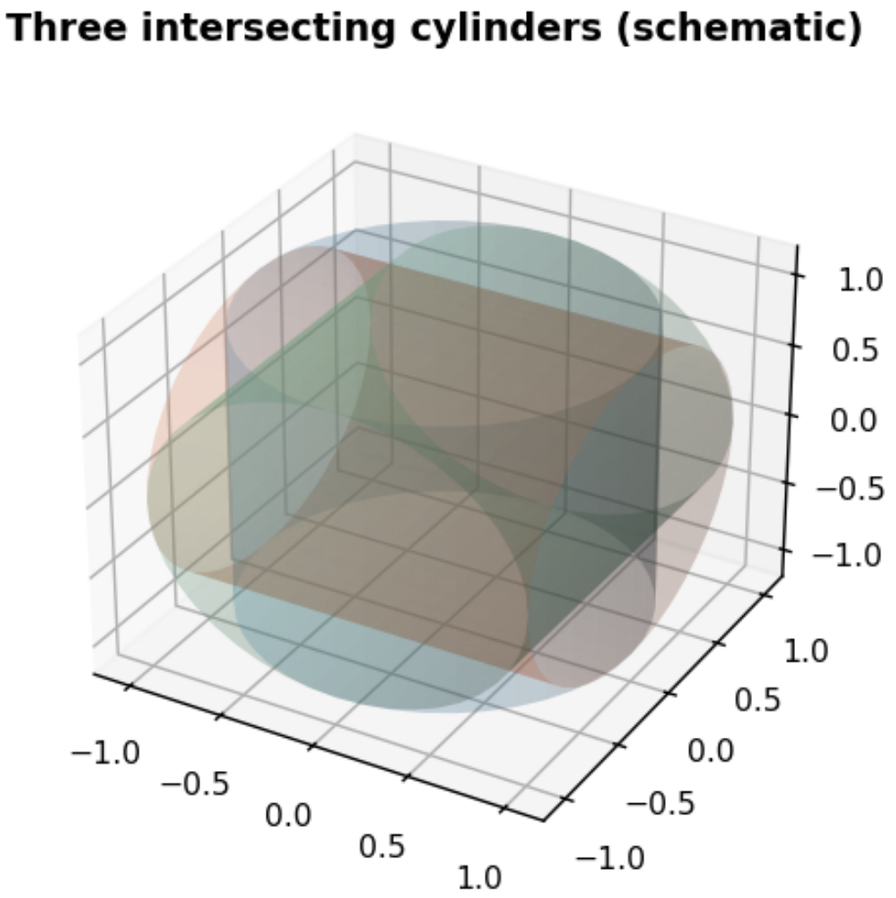


Figure 59: Output from code example 62. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2, gridExtra, colorspace, grDevices, MDplot; functions: boxplot, hist, plot, abline, lm, ggplot, geom_bar, geom_col, geom_histogram, geom_density, geom_point, geom_sm.

6. For Sample.name 1-S, use the as.numeric() function to convert its corresponding row to a vector. You should first exclude the 1-S in that row. Store this vector using an object named “abundance”.
7. Combine the `Sample.name` 1-S, `class.name` vector, and `abundance` vector into a 10-row, 3-column (Sample.name , Class, Abundance) data.frame as shown in the table above.
8. Repeat question 6 and 7 for all Sample.name, and `rbind()` the results into a data frame named “mine.table”.

This can also be realized by

better look of lables

Now the data are ready - on to the plot!

9. To plot a heatmap, we are going to use the `ggplot2` package. Install and library `ggplot2`.

For this plot, we are going to first create the heatmap object with the `ggplot` function, then print the plot. We create the object by assigning the output of the `ggplot` call to the variable `mine.heatmap`, then entering the name of this object to print it to the screen.

- Plot heatmap with x=Sample.name and y=Class. Fill the heatmap using Abundance value.
- Choose right `geom_` function to perform heatmap

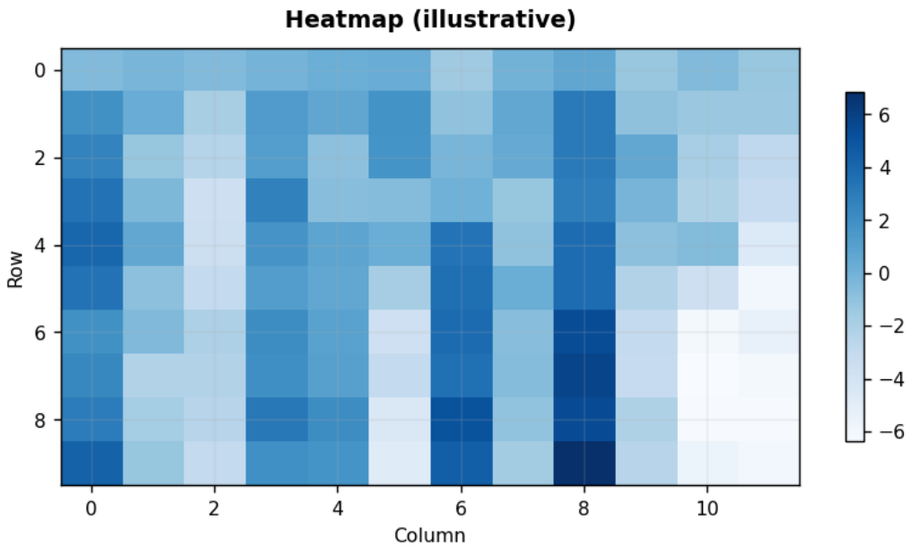


Figure 60: Output from code example 70. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_tile.

11. First let’s deal with a scaling issue. Most values are dark blue, but this is partly caused by the distribution of values - there are a few really high values in the Abundance column. We can transform the data for display by taking the square root of abundance and plotting that. To do this, we need to add another column to our data frame and update our call to ggplot to reference this new column, `Sqrt.abundance`.
12. Change the shading so low values are light and high values are dark. Use white (#FFFFFF) for the low values and a dark blue (#012345) for high values. You can also pick whatever color you like.
13. Add a title “Microbe Class Abundance” to this plot.
14. Reverse the order of the y-axis, so ‘Actinobacteria’ is at the top. Use `rev()` function to get the reverse of the `levels` of the Class. Use the obtained character vector as the `limits` of `scale_y_discrete()` function.
15. Use “geom_text()” to add abundance values on top of the heatmap. Maybe the text is too big to display on top of the heatmap? what can we do? Consider removing decimals and resizing the text.

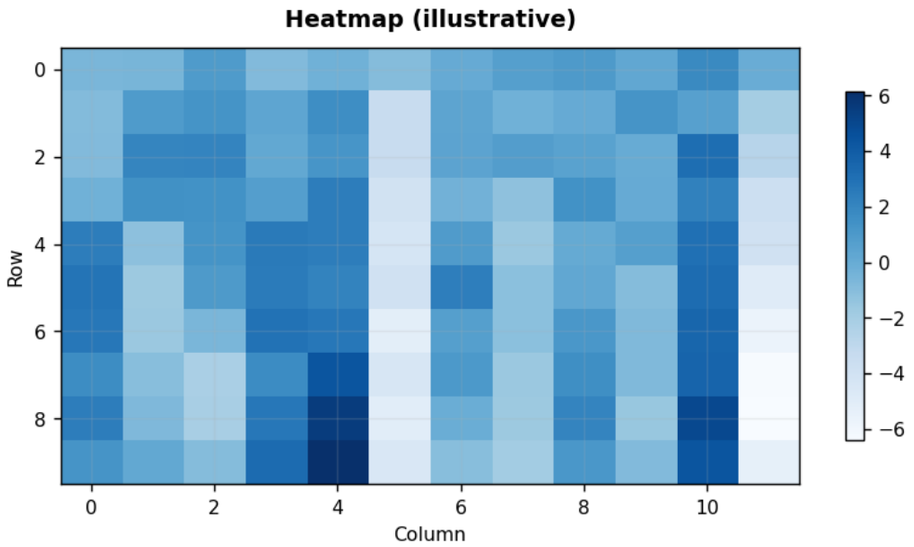


Figure 61: Output from code example 71. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_tile.

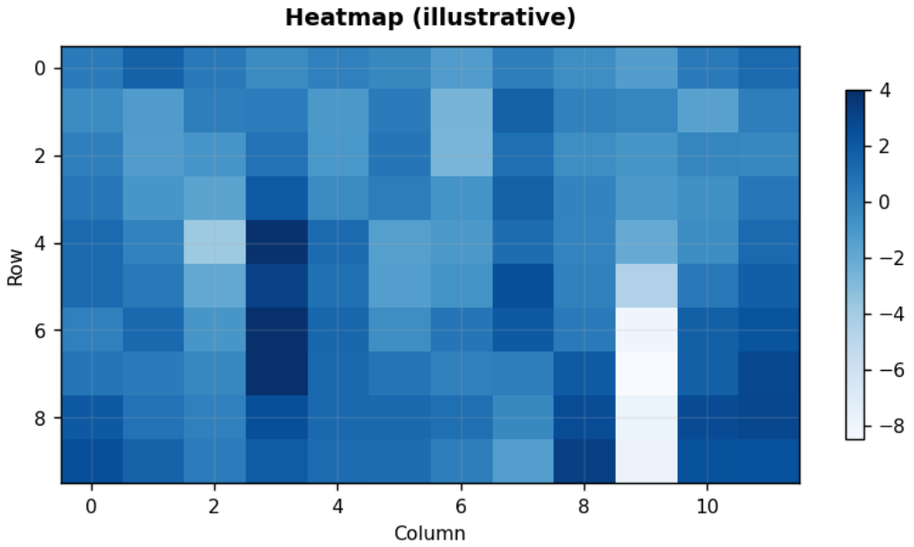


Figure 62: Output from code example 72. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: plotting / simulation functions.

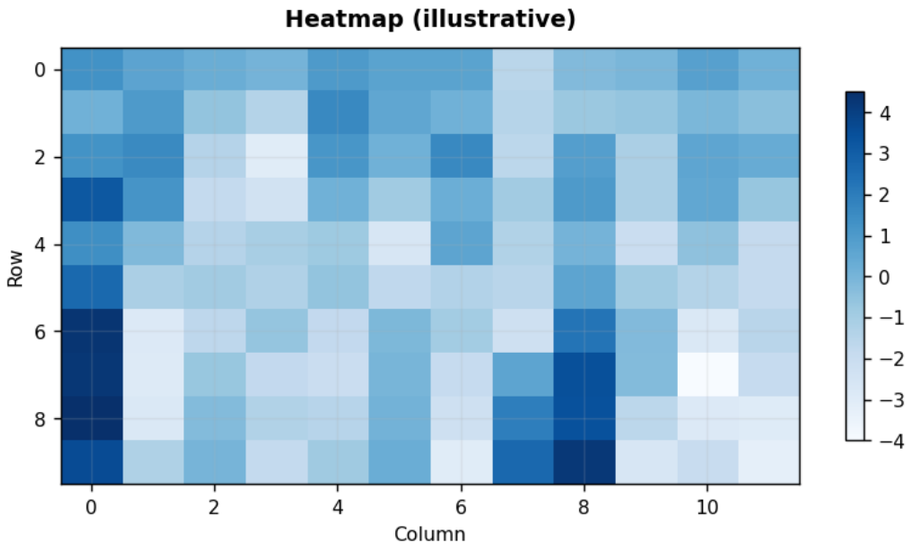


Figure 63: Output from code example 73. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: geom_text.

16. Congratulations on completing the above task, but you may have noticed that the x-axis Sample.name are not in the correct order. In fact, a good idea is to visualize different categories of data, we will learn this function next lab. But if you have some time, you may have a try by looking the help information of `facet_grid`.

This approach, called "faceting" requires one additional layer to the plotting command called facet_grid. With `facet_grid` we indicate which column contains the categories we want to use for the plot. But before we do that, we need add `Depth` back to the data frame. For Sample.name %in% c("1-S", "2-S", "3-S"), add Depth=0.5 in a new column named `Depth`; For Sample.name ends with M, add Depth=3.5; and 25 for those end with B.

17. Faceting the plot by `Depth`, set `scales` and `space` as `free` and add `x` to `switch`.

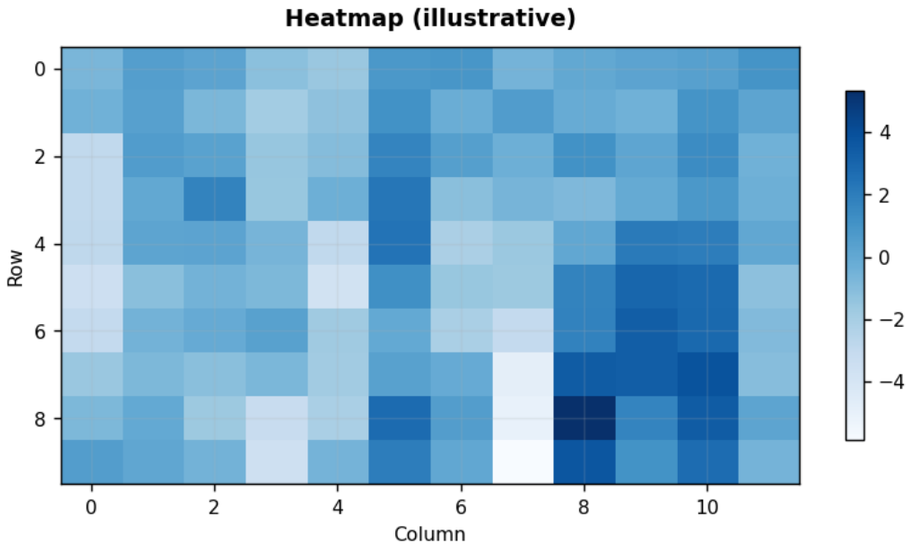


Figure 64: Output from code example 75. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_tile, ggtitle, geom_text.

- Faceting (make many panels of graphics where each panel represents the same relationship be-

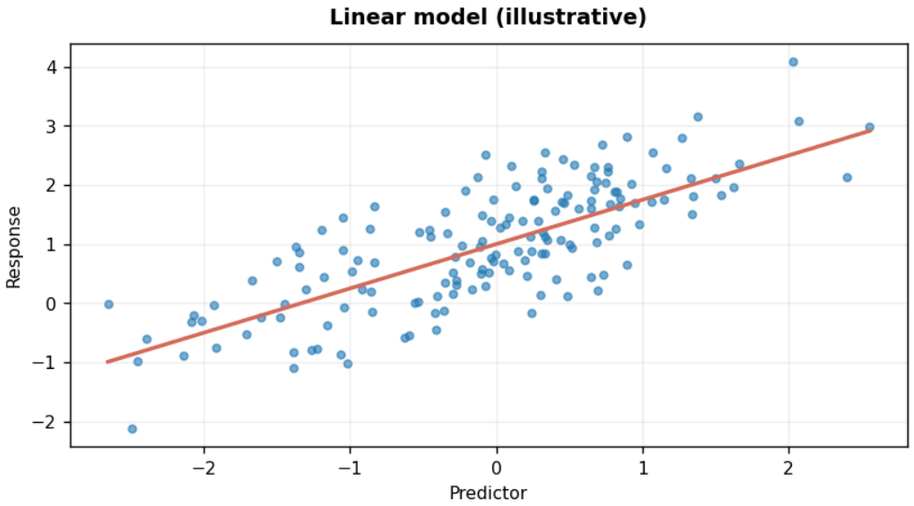


Figure 65: Output from code example 76. Illustrative reconstruction based on nearby code; not the exact original output. R packages: palmerpenguins; functions: ggplot, geom_point, geom_smooth.

tween variables, but something changes between each pane)

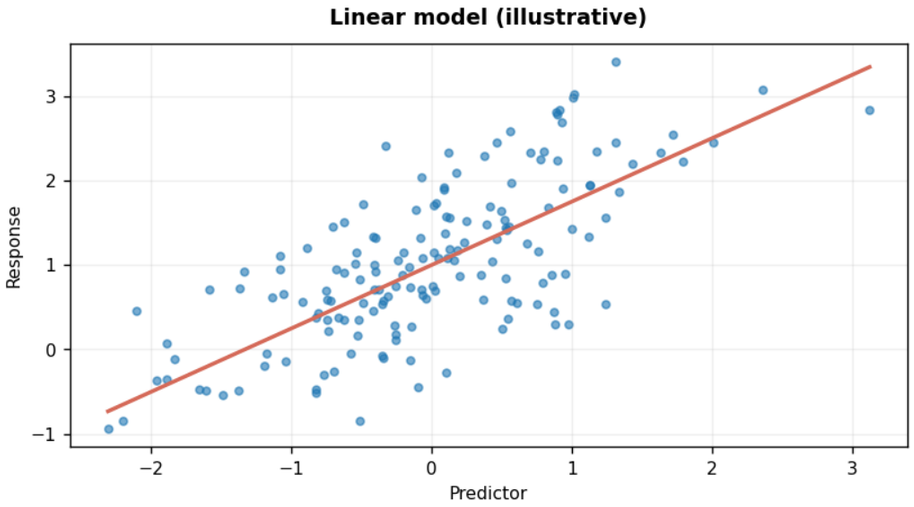


Figure 66: Output from code example 77. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_point.

- Another example
- Multi-plot

The R output arranges several exploratory plots, including grouped summaries and a regression display, for comparison.

- data manipulation

A tibble, or tbl_df, is a modern reimagining of the data.frame, keeping what time has proven to be effective, and throwing out what is not. Tibbles are data.frames that are lazy and surly: they do less (i.e. they don t change variable names or types, and don t do partial matching) and complain more (e.g. when a variable does not exist). This forces you to confront problems earlier, typically leading to cleaner, more expressive code. Tibbles also have an enhanced print() method which makes them easier to use with large datasets containing complex objects.

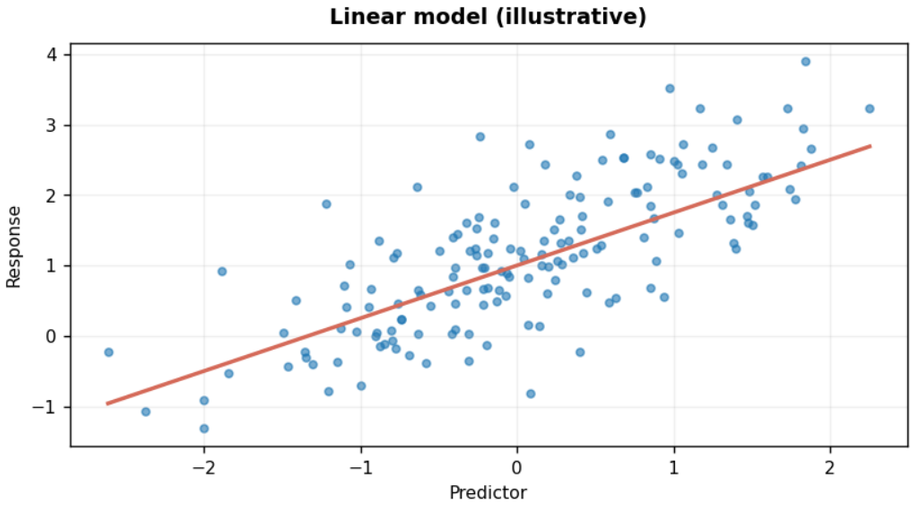


Figure 67: Output from code example 78. Illustrative reconstruction based on nearby code; not the exact original output. R packages: reshape; functions: ggplot, geom_point.

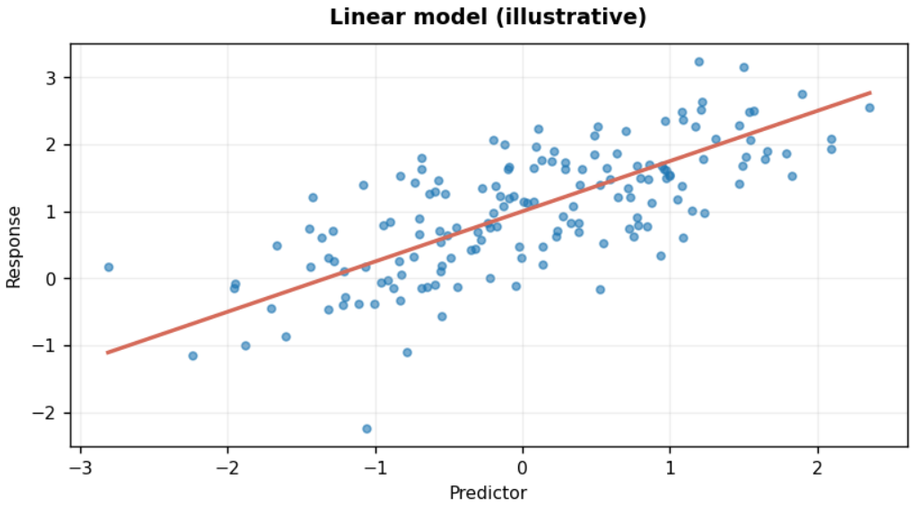


Figure 68: Output from code example 78. Illustrative reconstruction based on nearby code; not the exact original output. R packages: reshape; functions: ggplot, geom_point.

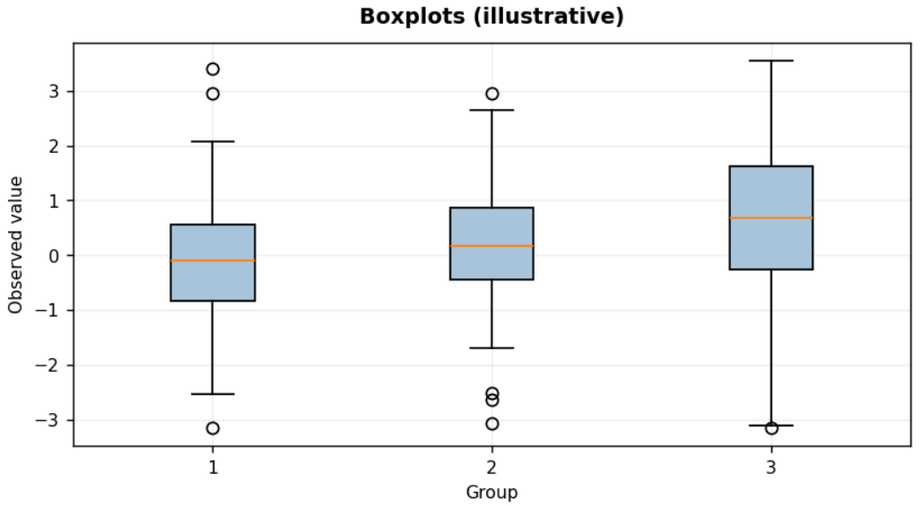


Figure 69: Output from code example 79. Illustrative reconstruction based on nearby code; not the exact original output. R packages: Rmisc, ggpubr; functions: ggplot, geom_line, ggtitle, geom_point, geom_smooth, geom_density, plot.

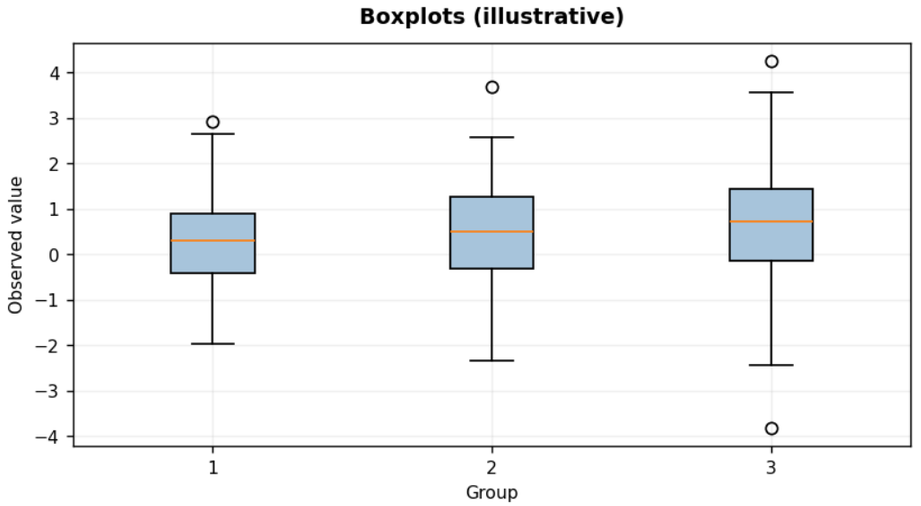


Figure 70: Output from code example 79. Illustrative reconstruction based on nearby code; not the exact original output. R packages: Rmisc, ggpubr; functions: ggplot, geom_line, ggtitle, geom_point, geom_smooth, geom_density, plot.

- %>%

For example, if we wanted to start with x, and first apply function f(), then g(), and then h(), the usual R command would be h(g(f(x))) which is hard to read because you have to start reading at the innermost set of parentheses. Using the pipe command %>%, this sequence of operations becomes x %>% f() %>% g() %>% h().

- select
- filter
- slice

filter rows based on row number:

- arrange
- mutate
- summarise

This function is used to create a summary table. It reduces the data frame to a single row containing summary statistics.

- group_by
- the comparison between whether to use %>% or not

13. Find the flight with the longest departure delay among flights from the same origin and destination (use `filter()`). Relocate the origin, destination, and departure delay to the first three columns and sort by origin and dest.
14. Find the flight with the longest departure delay among flights from the same origin and destination (use `top_n()` or `slice_max()`). Relocate the origin, destination, and departure delay to the first three columns and sort by origin and dest.
15. How do departure delays vary at different times of the day? Summarize the averaged departure delays by hours and create an new column named as `dep_delay_level` which `cut()` the averaged departure delays into three levels (low, median, and high).
16. How do departure delays vary at different times of the day? Illustrate your answer with a `geom_smooth()` plot.

how many data points are in the data set

- for loop
- bootstrap estimate of a sampling distribution

Ellipses

copy of x

LAB 10

Part 1

1. Load the `bloodpress.txt`
2. Use `pairs.panels()` function from `psych` pacakge to draw scatterplots, histograms, and calculate correlations between variables.
3. Fit a simple linear regression model of BP vs Stress. Is Stress significant?
4. Fit a simple linear regression model of BP vs Weight.
5. Fit a simple linear regression model of BP vs BSA.
6. Fit a multiple linear regression model of BP vs Weight + BSA. Is BSA still significant? Why?
7. Predict BP for Weight=92 and BSA=2 for the two simple linear regression models and the multiple linear regression model, by hand and by `predict()` function.

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

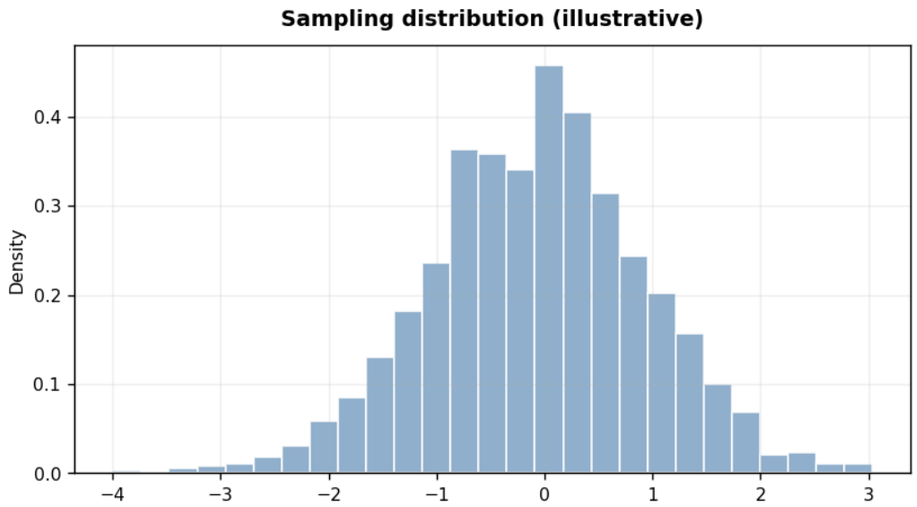


Figure 71: Output from code example 86. Illustrative reconstruction based on nearby code; not the exact original output. R packages: dplyr, ggplot2; functions: ggplot, geom_histogram.

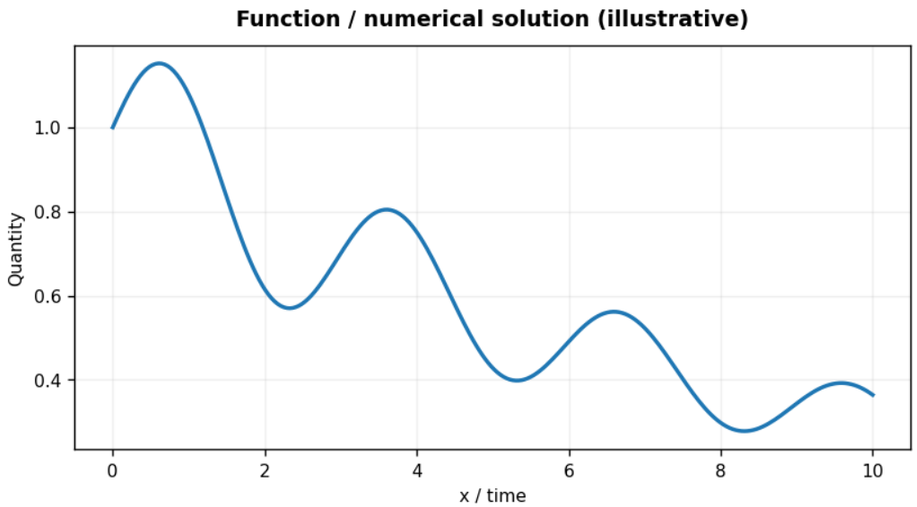


Figure 72: Output from code example 90. Illustrative reconstruction based on nearby code; not the exact original output. R packages: psych; functions: plotting / simulation functions.

8. Fit a multiple linear regression model of BP vs Age + Weight. Set argument `x` and `y` as TRUE. Save the output of `lm()` as `model.5`. How do we interpret each estimated coefficients?
9. Use the `plot_ly` function in the `plotly` package to create a 3D scatterplot of the data with the fitted plane for a multiple linear regression model of BP vs Age + Weight.
10. Extract the matrix `x` and `y` of `model.5` and assign it to a new object `X` and `y`. Remember, if you save the output of `lm()` as an object, this object contains many elements. After we set `x=TRUE` and `y=TRUE` in question 8, we can find `x` and `y` in this list.
11. Calculate X^TX, X^Ty, $(X^TX)^{-1}$, and $(X^TX)^{-1}X^Ty$. Use `t()` for transpose, `%*%` for matrix multiplication, and `solve()` for inverse of matrix. For the last one, is your result same as the estimated values you obtained in question 7? –Of course!
12. Use the anova function to display the ANOVA table with sequential (type I) sums of squares for the `model.5`.

$$SS_{\text{Variable}} = \sum_{i=1}^{n} (\hat{y}_{\text{Variable},i} - \bar{y})^2$$

$\hat{y}_{\text{Variable},i}$ is the model including only varible i.

$$F = \frac{\text{Mean Square for the Variable (MS_Variable)}}{\text{Mean Square for Residuals (MS_Residuals)}}$$

If (F $\approx$ 1) : It indicates that the sizes of MS_Variable and MS_Residuals are approximately the same, suggesting that the explanatory power of the independent variable for the dependent variable is comparable to the random error, and the null hypothesis ((H_0)) cannot be rejected.

If (F $\gg$ 1) : It indicates that MS_Variable is significantly greater than MS_Residuals, suggesting that the independent variable has a significant influence on the dependent variable, and the null hypothesis ((H_0)) can be rejected.

$SS_{Total} = SS_{Age} + SS_{Weight} + SS_{Residuals} $

12. Use the `residuals` element in fitted model or `residuals()` function to extract the fitted residuals. Calculate the sum of square of these residual values. Extract the `df.residual` element in fitted model and use the above elements to calculate the MSE. Is your result same as the `anova()` output?
13. Fit a multiple linear regression model of BP vs Age + Weight + Pulse. Save the output of `lm()` as `model.6`.
14. Use `anova()` function to obtain the ANOVA table for `model.6`. We may consider `model.6` as full model, and `model.5` as reduced model in this question. Based on the obtained ANOVA table and the output of question 11, calculate the F-statistic for testing the reduced model by hand. You may use the Residuals Sum sq and the corresponding Residuals Df from both tables. Then, calculate the p-value using $pf()$ function, don't forget about the `lower.tail`.
15. Use `anova()` function to do the F-test on `model.5` and `model.6`. Compare the output with your answers of question 14. What is the conclustion of the F-test?

RSS $= \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$

Sum of Sq $=$ RSS(Model 1) $-$ RSS(Model 2) (It represents the variation in the interpretation of the dependent variable by the newly added variable Pulse.)

$$F = \frac{\text{Sum of Sq/Df}}{\text{RSS(Model 2)/Res.Df(Model 2)}} F = \frac{0.49557}{0.270525} \approx 1.8319$$

16. Plot the qqPlot for residuals of `model.5`. What is the x-axis and y-axis of the qqPlot? What can we say about the qqPlot?
17. Plot the residual vs fitted plot of `model.5`. You may extract `fitted.values` from `model.5` and use it as `x` in `plot()`.
18. Directly use `plot()` function on `model.5`.

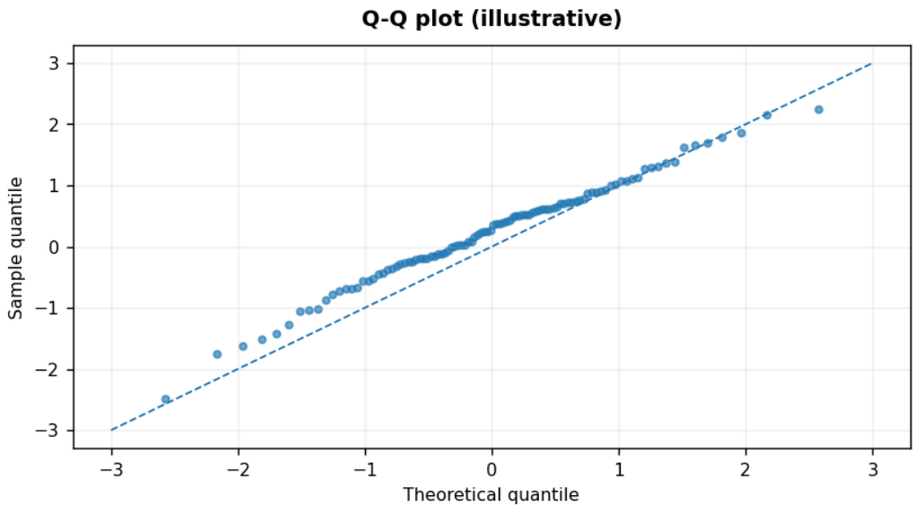


Figure 73: Output from code example 105. Illustrative reconstruction based on nearby code; not the exact original output. R packages: car; functions: plotting / simulation functions.

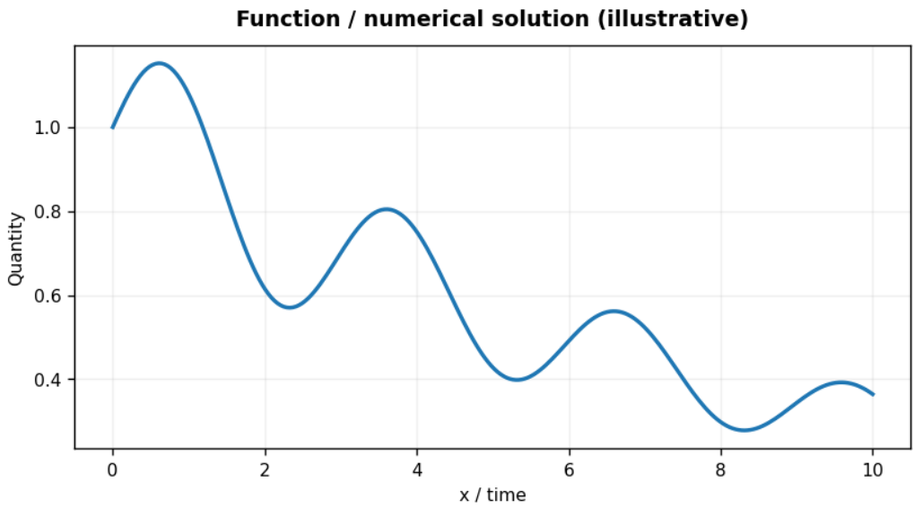


Figure 74: Output from code example 106. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: plot.

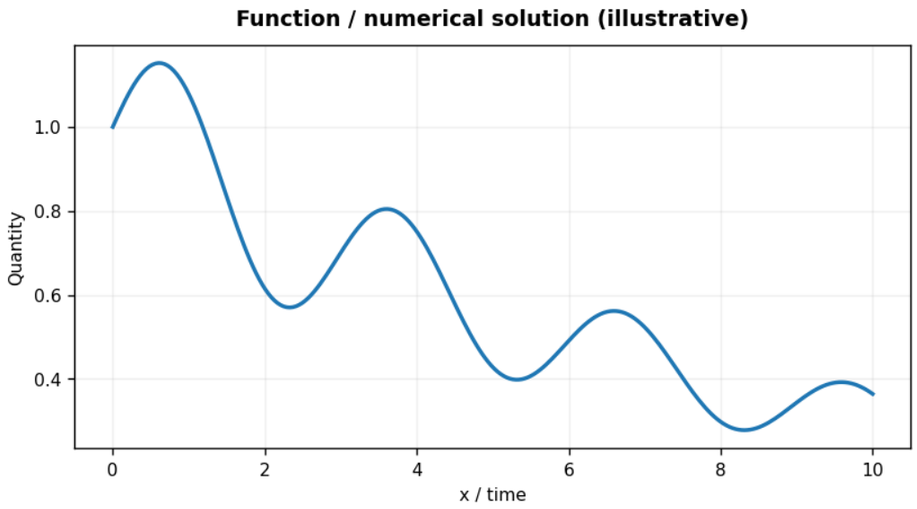


Figure 75: Output from code example 107. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: plot.

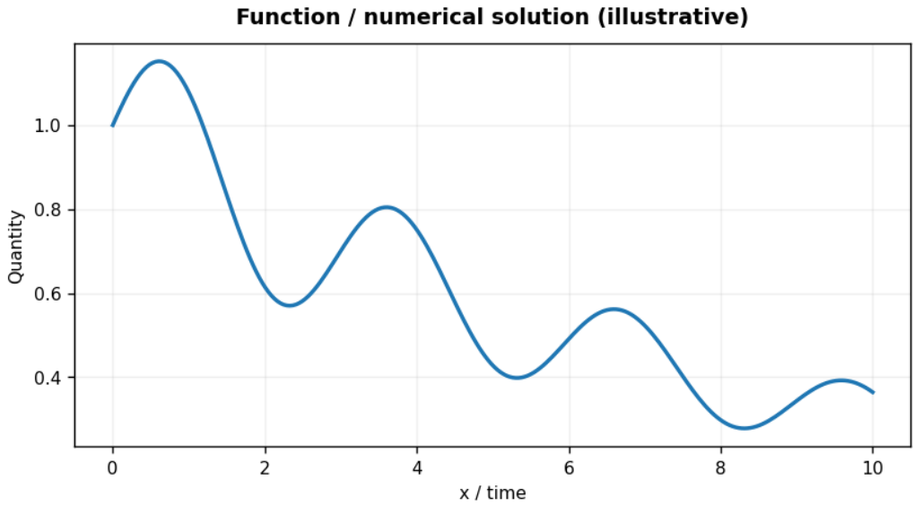


Figure 76: Output from code example 107. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: plot.

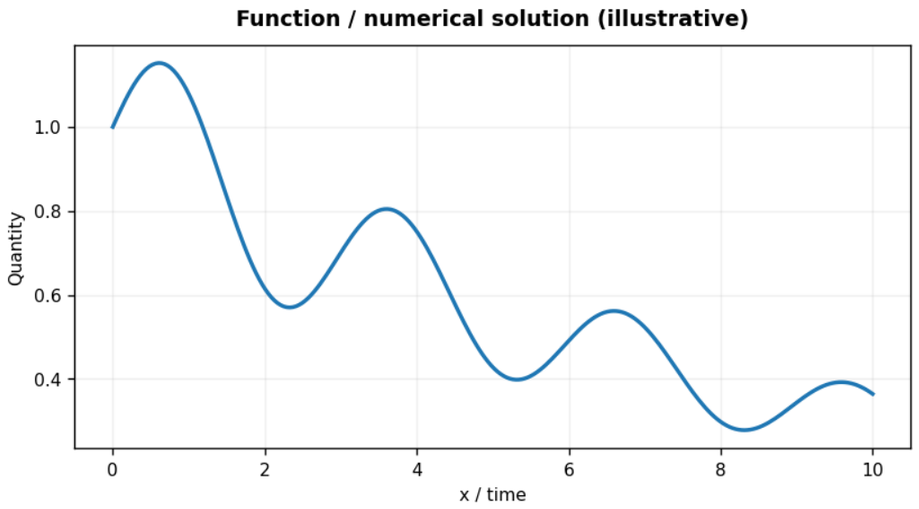


Figure 77: Output from code example 107. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: plot.

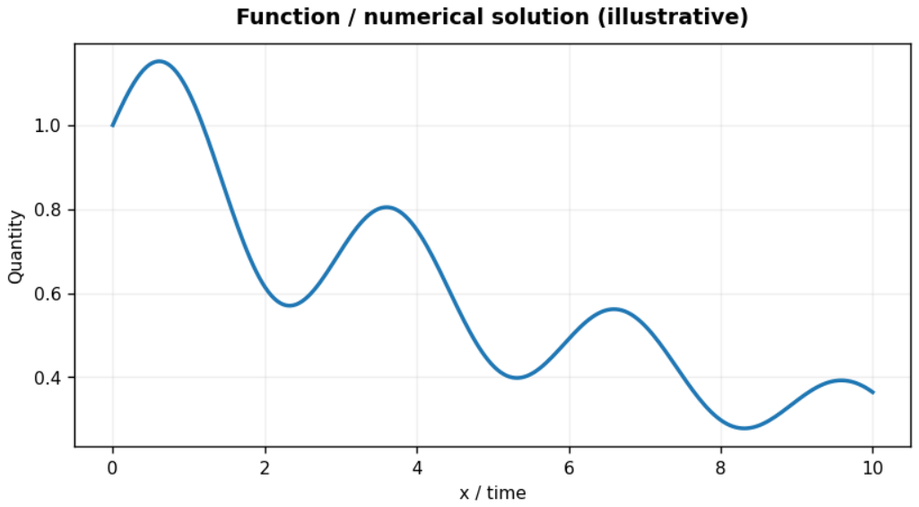


Figure 78: Output from code example 107. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: plot.

Part 2

19. Load the `hospital_infct.txt` data and select observations with Stay <= 14.
20. Create new dummy/indicator columns (`i1`, `i2`, `i3`, `i4`) for regions using `ifelse()` function. For example, i1 = 1 when Region = 1 and i1 = 0 when Region is not equal to 1; i2 = 1 when Region = 2 and i2 = 0 when Region is not equal to 2; ...
21. Fit a multiple linear regression model of InfctRsk on Stay + Xray + i2 + i3 + i4.
22. Can we include i1 + i2 + i3 + i4 in this multiple linear regression? Why?

No. In the context of using dummy variables for categorical data in regression analysis, it's essential to designate a reference category. This reference category is represented by a coefficient of zero, while the coefficients for the other categories are interpreted as deviations from this reference point.

24. Conduct an F-test (use `anova()` function) to see if at least one of i2, i3, and i4 are useful.

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Confusion Matrix

Total population $= \mathrm{P} + \mathrm{N}$	Predicted Positive (PP)	Predicted Negative (PN)	Informedness, bookmaker informedness (BM) $= \mathrm{TPR} + \mathrm{TNR} - 1$	Prevalence threshold (PT) $= \frac{\sqrt{\mathrm{TPR} \times \mathrm{FPR}} - \mathrm{FPR}}{\mathrm{TPR} - \mathrm{FPR}}$
Positive (P) [a]	True positive (TP), hit [b]	False negative (FN), miss, underestimation	True positive rate (TPR), recall, sensitivity (SEN), probability of detection, hit rate, power $= \frac{\mathrm{TP}}{\mathrm{P}} = 1 - \mathrm{FNR}$	False negative rate (FNR), miss rate type ll error [c] $= \frac{\mathrm{FN}}{\mathrm{P}} = 1 - \mathrm{TPR}$
Negative (N) [d]	False positive (FP), false alarm, overestimation	True negative (TN), correct rejection [e]	False positive rate (FPR), probability of false alarm, fall-out type I error [7] $= \frac{\mathrm{FP}}{\mathrm{N}} = 1 - \mathrm{TNR}$	True negative rate (TNR), specificity (SPC), selectivity $= \frac{\mathrm{TN}}{\mathrm{N}} = 1 - \mathrm{FPR}$

⟵ **Proof idea**
Identify the exact statement before proving it.

example

Lung Cancer Classification (https://www.kaggle.com/datasets/mysarahmadbhat/lung-cancer?select=survey+lung+cancer.csv)

The effectiveness of cancer prediction system helps the people to know their cancer risk with low cost and it also helps the people to take the appropriate decision based on their cancer risk status. The data is collected from the website online lung cancer prediction system.

Total no. of attributes: 16 No. of instances: 284

Attribute information:

Gender: M(male), F(female) Age: Age of the patient Smoking: YES=2, NO=1. Yellow fingers: YES=2, NO=1. Anxiety: YES=2, NO=1. Peer_pressure: YES=2, NO=1. Chronic Disease: YES=2, NO=1. Fatigue: YES=2, NO=1. Allergy: YES=2, NO=1. Wheezing: YES=2, NO=1. Alcohol: YES=2, NO=1. Coughing: YES=2, NO=1. Shortness of Breath: YES=2, NO=1. Swallowing Difficulty: YES=2, NO=1. Chest pain: YES=2, NO=1. Lung Cancer: YES, NO.

Goal: It is your job to classify Lung Cancer using other variables

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

Example Code

Below is the further remark

margin of error

For a given level of confidence c, The greatest possible distance between the point estimate and the value of the parameter it is estimating.

Sometimes called the maximum error of estimate or error tolerance.

⟵ **Motivation**
Decomposing a complicated signal into simpler modes makes dynamics easier to understand.

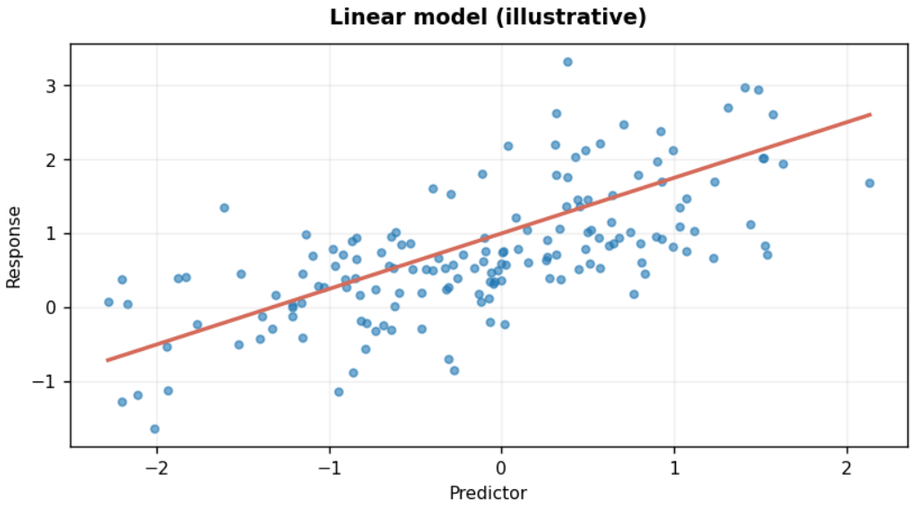


Figure 79: Output from code example 116. Illustrative reconstruction based on nearby code; not the exact original output. R packages: pROC; functions: plot.

Why n-1 in sampling variance

?

- degree of freedom
- ubiased estimation. (是指在已知 sample 时，这个无偏估计是所有估计中的平均值，这样我们才能称其为 “无偏”。)

Estimating the population variance from a sample uses n minus one degrees of freedom after estimating the sample mean.

- sampling proportion is unbiased

E($\hat{P}$) = E(X/n)=npn (binomial distribution) =p Var—also 如果 E 满足 variance 不满足呢

Simple random samlple

We know the total population but do not know p of our interests.

Based on hypergeometric probability calculation we can estimate p as $\hat{p} =$ intrests$/N = X_1 + ... + X_N/N$ where X_i~ Bernoulli(p) and $\hat{p}$ is a random variable. Therefore, $E[\hat{p}] = p$ (EX_i=p based on Bernouli distribution).

⟵ **Proof idea** Condition explicitly; it often simplifies the calculation.

- If X_i are independent then E($X_1 + X_2$) = 2E(X_1)

Here, eg. E$\hat{p}^2$= 1/n EX_1^2+n-1/nEX_1X_2

Since $X_1 = 1$ $X_1^2 = X_1$

and EX_1X_2 = PX_1, X_2

Finally, we can look at the distribution of $\hat{p}$, suppose we know the true p is 0.54, we can use **simulation** to randomly sample $X_1, ...X_n$ from Np people who support 1 and ……who support

- God’s perspective

doing this 500 times,

based on the central limit theory

Statistical inference and Probability distribution

- estimation
- condifidence intervals
- hypothesis testing

Sample Space and Probability Measure

σ algebra F is a collection of satisfying :

full and null set

completerment

countably union

measurble we can find a function that takes the elements of F and output a real number

A probability measure is a mapping P:F–> R satisfying the following 3 axioms:

countably for mutually exclusive events $A_1, A_2, ... \in F$

⟵ **My thought**
From a probabilistic model to a usable numerical quantity: what is observed, what is random?

interpret the probability

- frequentist view
- Bayesian view

Borel set is a combination of open set in some spaces

Moment Generating Functions

$M_x(t) = E(e^{tx})$

case 1

M_x may not exist. When it exists in a neighborhood of 0, using talor

$$e^{tx} = 1 + tX + (tX)^2/2 + ...$$

$$M_x(t) = 1 + t\mu + t^2\mu/2 + ...$$

$\mu_j = E(X^i)$ is the j-th moment of X. Therefore,

$$E(X^i) = M^{(j)}(0)$$

eg. Then we could also get the variance by take the 2nd order derivatives

case 2

$\int$

eg. normal

- X~N(0,1)

idea: try to write an integral of a certain distribution's pdf and we get the result of this integral as 1. here, we get the pdf of the N(t,1) and finally we get $e^{t^2/2}$

- X~N(μ, σ^2)

Then X=$\mu + \sigma Z$ where Z~N(0,1)

$$M_x(t) = E[e^{tx}] = e^{\mu t}E[e^{\sigma t Z}] = e^{\mu t}M_Z(\sigma t) = e^{\mu t + \sigma^2 t^2/2}$$

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Gamma disteibution

the family of gamma distributions generalizes the family of exponential distributions. The gamma distribution with shape r and rate λ

addition rule r +

λ stays

why MGF is useful? – to determine if two random variables have the identical CDF / to prove the addition property of distributions

MGF includes all characteristics of a distribution, from whom we could get pdf, cdf, expectation, variance

- Thm If X and Y are random variables with the same MGF, which is finite on [-t, t] for some t >0 then X and Y have the same distribution

A gamma distribution with shape r =1 is an exponential distribution

A more general function than MGF is the characteristic function.

$\phi_X(t) = E(e^{itX})$

MGF and Fourier Transform

From $f(t)$ to $\sum_{n=0}^{\infty} a_n \sin(nt)$ is called demodulation （解调）

This also relevant to the base vector

传播

⟵ **Proof idea**
Check the initial and boundary conditions after finding a general solution.

Normal -> Chi-squared distribution

MGF

the MGF of the Gamma(1/2, 1/2) distribution is the chi-squared

standard normal random variables' sum of their each own squers follow a χ^2 distribution

the addition rule lies on the sum of the dgree of freedom

$(n-1)S^2/\sigma^2$

consider

Preface

The first section here is the knowledge that attracts most of my interests on this module, followed by the lecture notes I tapped when I am on the journey of this module.

- Motivation

The history of the development of statistics has helped me gain a deeper understanding of this fascinating and practical field of knowledge. Each advancement represents a leap from specific facts to broader, generalized conclusions. What truly captivates me is the remarkable alignment between natural phenomena and statistical principles; there always seems to be a coincidence where reality and theory intersect.

Initially, I only grasped the surface of this knowledge. Then, as I delved deeper, I began to understand the underlying principles through observable phenomena. Eventually, almost miraculously, I realized how incredibly useful these theories are and how they coincide with natural occurrences. This process of moving closer and closer to generalization has profoundly illustrated to me the truth in the saying: "Mathematics is the art of giving the same name to different things."

Theorems come, theorems go. Only examples are lying forever. (Practice to use statistics to interpret real world examples)

把公式推导单独放一栏（后）

error?

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

- Independent = Disjoint? NO!
  Independent vs Disjoint: If P(E) > 0 and P(F) > 0, then E and F can NOT be both independent and disjoint.

Subsets are dependent. If E ⊂ F and neither P(E) = 0 nor P(F) = 1, then E and F are dependent.
Complements are dependent. If neither P(E) = 0 nor P(E) = 1, then E and E C are dependent.

My reading during this course:

十堂极简概率课中信出版 diaconis

心理统计

The lady tasting tea

概率论与数理统计（第三版）崩诗松等老师编著

Probability....

bivariate data

linear regression

Paired quantitative observations can be plotted to reveal the direction and strength of their association.

A fitted regression line summarizes the average linear pattern while individual points retain their variation.

background

- Regression to the mean (Galton's thinking)

Regression to the mean describes how unusually high or low measurements tend to be less extreme on a later measurement.

Predicting far outside the observed range is unstable because the fitted pattern has not been checked there.

it is not stable to predict the data outside our data sample'

Residual plot should have no pattern

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

across the whole range, it could not be showing a certain trend or a specific shape.

positive and negative points sperate averagely .

A residual plot without a systematic pattern is more consistent with a linear mean model.

A curved or widening residual pattern signals that the linear model or constant-variance assumption may be inadequate.

skew

Data skewed to the left(negatively skewed) have a longer left tail, and the mean and media are to the left of the mode.

Data skewed to the right(positively skewed) have a longer right tail, and the mean and media are to the right of the mode.

mean is much sensitive than the median to the extreame value(not a resistant measure of center) so the outlier drags the mean to the skew..... A long distribution tail pulls the mean farther than the median, so an outlier can change the mean substantially.

Compare $Q_2 - Q_1$ and $Q_3 - Q_2$ to decide left/right skew or symmetry.

ps: the **trimmed mean** is more resistant.

Quartile spacing helps distinguish left skew, right skew, and a roughly symmetric distribution.

Intervals measured in standard deviations summarize the spread around a mean.

The distribution sketches compare how much mass lies near or far from the mean.

As long as we know the mean and standard deviation of the data, we can determine how far a certain proportion of the data falls, without knowing the specific distribution shape of the data.

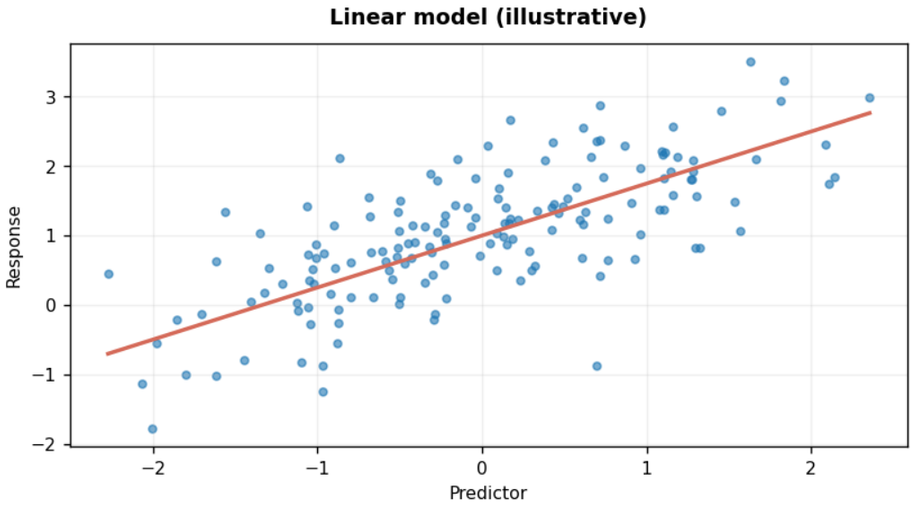


Figure 80: Output from code example 118. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: plot, abline, lm.

outlier (eg of normal distribution)

⟵ **Proof idea**
Condition explicitly; it often simplifies the calculation.

a mild outlier if it lies more than 1.5(iqr) away from the nearest quartile (the nearest end of the box);

an extreme outlier if it lies more than 3(iqr) away from the nearest quartile.

(These definitions and distances are based on the hypothetical Normal distribution (bell shaped, symmetric, normal tails). When there is a reason to suspect that the distribution is skewed, the bounds should be changed.)

quartile

Q_1(First quartile): at least 25% of the sorted values are less than or equal to Q_1 and at least 75% of the values are greater than or equal to Q_1

Q_3(Third quartile): ..

modified boxplot

A modified boxplot is a box plot where the whiskers only extend to the largest (or smallest) observation that is not an outlier and the outliers are plotted using a full circle (mild) or empty circle (extreme).

If there are no outliers, then the the whiskers end at the maximum (or minimum)

Measures of relative standing

z Scores

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

eg. height(among women or men (come from different populations)instead of just comparing the height itself)

tutorial

dotplot 没有纵轴 in r(vs scallar plot)

price of a textbook is discrete

zip code is categorical

dotplot and scallar plot

do not manipulate data—experimental

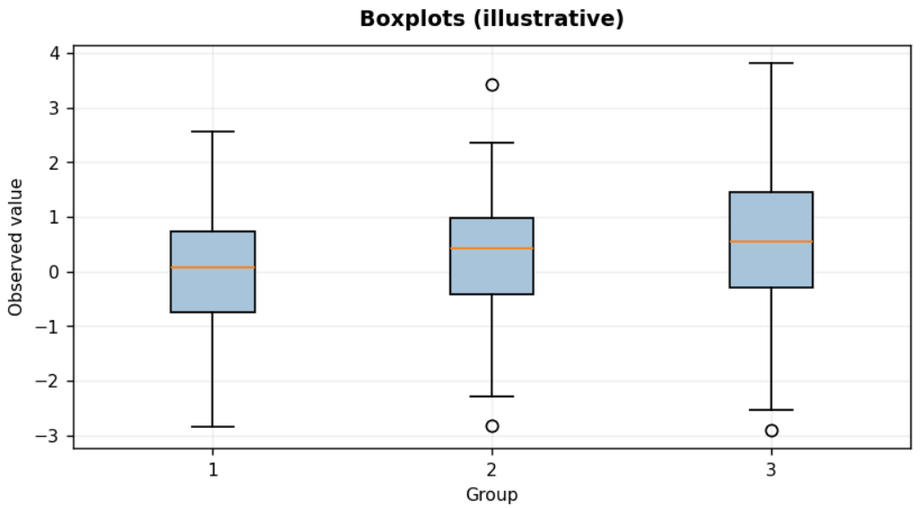


Figure 81: Output from code example 119. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: boxplot.

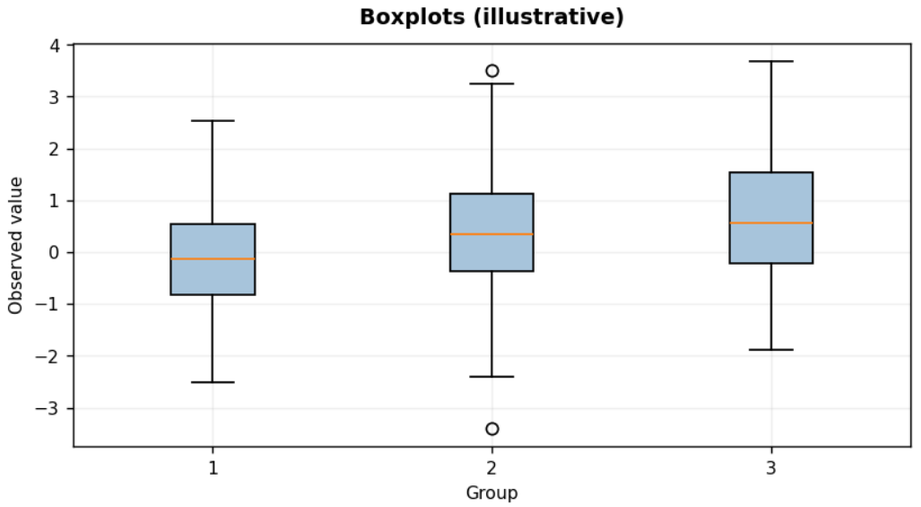


Figure 82: Output from code example 120. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: ggplot, geom_boxplot, geom_point.

table 2.1

11

huge data for 顺序的 shuffle is ok

but ..

the likelihood is totally different

The role of statistics and the Data Analysis Process

Intro

stat is a large field in math involving the collection, organization, analysis,interpretation, and presentation of data(a collection of observations on one or more variables(A characteristic whose value may change from one observation to another))

Statistics is the scientific discipline that provides methods to help us make sense of data.

It is important to be able to:

1 Extract information from tables, charts, and graphs.

2 Follow numerical arguments.

3 Understand the basics of how data should be gathered, summarized, and analysed to draw statistical conclusions.

The Data Analysis Process

1 Understanding the nature of the research problem or goals.

2 Deciding what to measure and how.

3 Collecting data.

4 Data summarization and preliminary analysis.

5 Formal Data Analysis (Statistical Methods).

6 Interpretation of the results.

populations and samples

⟵ **Motivation** Why does this definition matter later? Try an example and a counterexample.

population: The entire collection of individuals or objects about which information is desired

sample: A sample is a subset of the population, selected for study.

then select the sample

then we could summarize it using 2 branches of stat.— Decriptive stat.(methods for organizing and summarizing data.) or inferential stat.(generalizing from a sample(incomplete information) to the population from which the sample was selected and assessing the reliability of such generalizations.So we run the risk(An important aspect of statistics and making statistics inferences involves quantifying the chance of making an incorrect conclusions.))

descriptive stat

inferential stat

sample

Types of data

uni data set and bivariate and multivariate

categorical and numerical(discrete and continuous) with plot using excel (data analysis) or rstudio plot (ggplot2)

⟵ **Proof idea** Begin with epsilon-delta or a sequence characterization.

for categorial data we could use a bar chart which is a graph of a frequency distribution for categorical data.

for a small numerical data we could use dotplot

- discrete

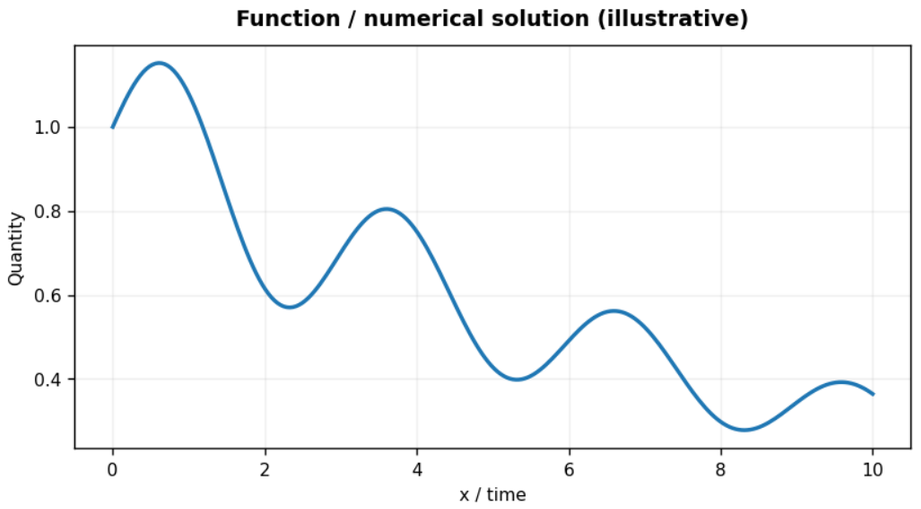


Figure 83: Output from code example 121. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: ggplot, geom_dotplot, ggtitle.

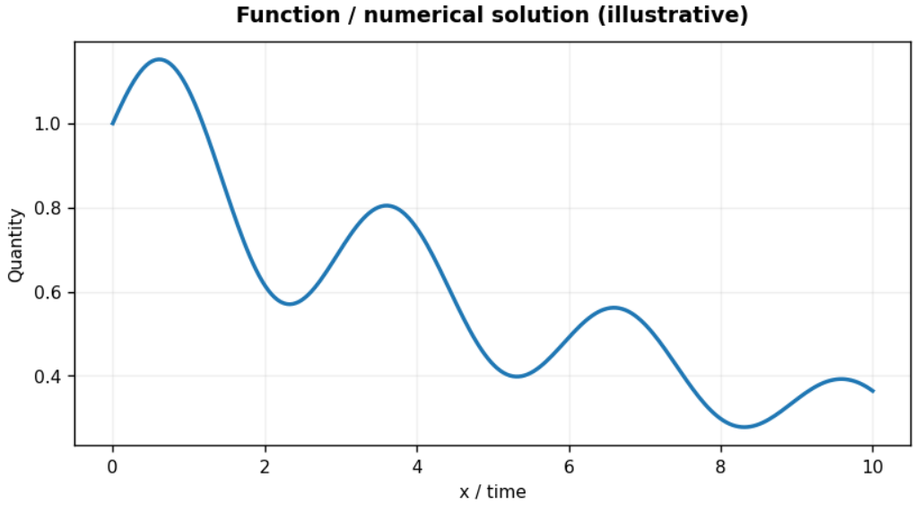


Figure 84: Output from code example 122. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: abline.

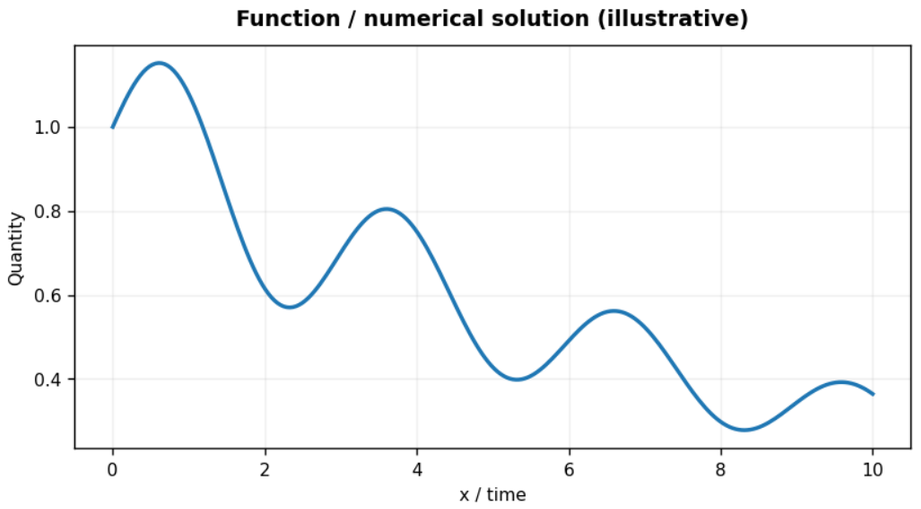


Figure 85: Output from code example 123. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: ggplot, geom_dotplot, ggtitle.

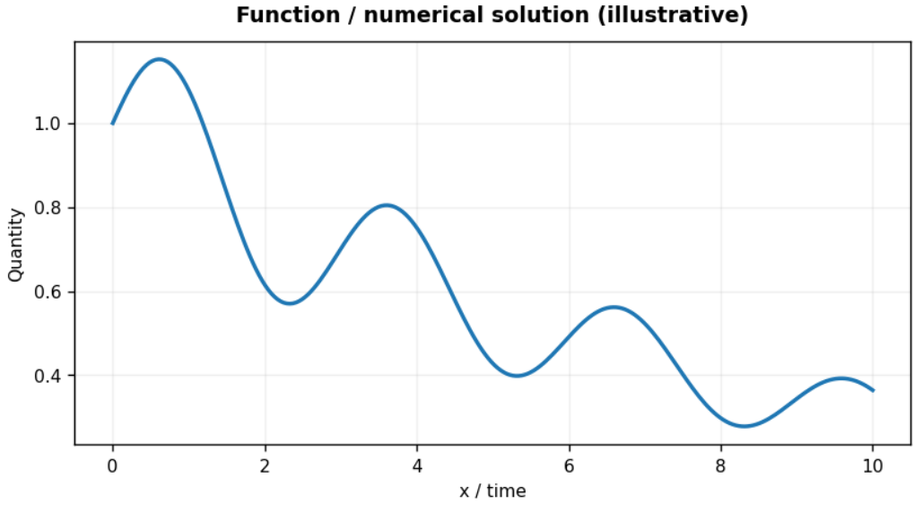


Figure 86: Output from code example 124. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: ggplot, geom_dotplot, ggtitle.

- continuous

The dot plot compares graduation rates for all athletes with basketball players.

collect data sensibly

chapter 2

Two types of studies: Observational studies and Experiments.

Observational

A study in which the investigator observes characteristics of a sample selected from one or more existing populations. The goal is to draw conclusions about the corresponding population or about differences between two or more populations.

In an observational study, it is impossible to draw clear cause-and-effect conclusions

Experiments

A study in which the investigator observes how a response variable behaves when one or more explanatory variables, also called factors, are manipulated.

A well-designed experiment can result in data that provide evidence for a cause-and-effect relationship.

A designed experiment assigns treatments and compares responses to support causal questions.

- Experimental conditions: Any particular combination of values for the explanatory variables, which are also called treatments.

comparison

- Both observational studies and experiments can be used to compare groups, but in an experiment the researcher controls who is in which group, whereas this is not the case in an observational study.
- In an observational study, it is impossible to draw clear cause-andeffect conclusions

confounding vars

A variable that is related to both how the experimental groups were formed and the response variable of interest.

- Two methods for data collection: Sampling and Experimentation.
- distinguish between selection bias, measurement or response bias, and non-response bias.
- select a simple random sample from a given population.
- distinguish between simple random sampling, stratified random sampling, cluster sampling, systematic sampling, and convenience sampling

variable

response variable–y

The response variable is the focus of a question in a study or experiment.

explanotory variable–x

An explanatory variable is one that explains for changes in the response variable.

experiments and obeservational study

bias

selection bias: When the way the sample is selected systematically excludes some part of the population of interest.

measurement or response bias

eg: survey question/scale(The scale or a machine used for measurements is not calibrated properly)

Non-response Bias:When responses are not obtained from all individuals selected for inclusion in the sample.

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

⟵ **Proof idea**
Identify the exact statement before proving it.

non-response bias can distort results if those who respond differ in important ways from those who do not respond (e.g. laziness a confounding variable).

random sampling

⟵ **My thought** From a probabilistic model to a usable numerical quantity: what is observed, what is random?

def: A sample that is selected from a population in a way that ensures that every different possible sample of size n has the same chance of being selected.

the same chance to be selected

counter eg:

Consider 100 students in a classroom, 60 females and 40 males. If we randomly sample 6 females, and 4 males, then each female has a 6/60 = 0.1 chance of being selected. Same for males, 4/40=0.1. However, not every group of 10 students is equally likely to be selected. This is not simple random sampling

The random selection process allows us to be confident that the sample adequately reflects the population, even when the sample consists of only a small fraction of the population.

eg.Voting Sample Size in a country

stratified and cluster

- stratified random sampling:In stratified random sampling, separate simple random samples are independently selected from each subgroup. Each subgroup is called a strata.

In general, it is much easier to produce relatively accurate estimates of characteristics of a homogeneous group than of a heterogeneous group.

stratified: according to certain characteristic

eg.Even with a small sample, it is possible to obtain an accurate estimate of the average grade point average (GPA) of students graduating with high honours from a university (Similar high grades, homogenous, thus only sample a few students). On the other hand, producing a reasonably accurate estimate of the average GPA of all seniors at the university, a much more diverse group of GPAs, is a more difficult task. **Not only does this ensure that students at each GPA level are represented, it also allows for a more accurate estimate of the overall average GPA.**

- cluster reflect general characteristic about the whole entire population

cluster: randomly groups

Cluster sampling involves dividing the population of interest into non-overlapping subgroups, called clusters. Clusters are then selected at random, and then all individuals in the selected clusters are included in the sample.

systematic sampling

⟵ **Motivation** From a probabilistic model to a usable numerical quantity: what is observed, what is random?

A value k is specified (e.g. k = 50 or k = 200). Then one of the first k individuals is selected at random, after which every k-th individual in the sequence is included in the sample. A sample selected in this way is called a 1 in k systematic sample.

In the case of large samples, it can ensure that the sample is evenly distributed in the population.

Random Variable

Random Variable (R.V.)

A numerical variable whose value depends on the outcome of a chance experiment. A random variable associates a numerical value with each outcome of a chance experiment. (Think of it as a rule that translates each result of a chance event into a number.)

In shorta: random variables convert random events into numbers.

A real-valued random variable X is a function: $X : S \to \mathbb{R}$, where S is the sample space of a chance experiment.

Continuous random variable X: S–> R is continuous if its set of possible values includes an entire interval on the number line(measurement), which could not be count.

Discrete random variable: X: S–> R if its set of possible value is a collection of isolated points along the number line.(counting)

eg

Examples: Coin Tossing (Discrete Random Variable) If we flip a coin 5 times, let X be the number of heads we get. Possible values of X {0,1,2,3,4,5} (where 0 means no heads, and 5 means all heads). Here, X turns each outcome of multiple coin tosses into a count of heads.

Departure Time (Continuous Random Variable) Imagine tracking when people leave a subway station between 10 PM and 11 PM. Let Y represent the time (in hours) someone leaves, so Y can be any number from 10 to 11. Here, Y assigns each departure time to a point in the range [10,11].

⟵ **Proof idea** Identify the exact statement before proving it.

sample space of multivariables

G:gender; F: year—corresponding to a certain student:S

(G, H) : S → $\mathbb{R}^2$, S={1,2,3,4}

...

Probability Mass Function and Cumulative Distribution Function for Discrete Random Variables

Probability Mass Function (PMF): $p_x(x) := P(X = x), \forall x$

Cumulative Distribution Function (CDF): $F_x(x) := P(X \leq x), \forall x$

- Note that the domain of a cdf is $(-\infty, \infty)$

pmf $\Longrightarrow$ cdf

$$F_x(x) = \sum_{y \leq x} p_x(y)$$

cdf $\Longrightarrow$ pmf Suppose X takes ordered values $x_1, x_2, x_3, \cdots$, then

$$\begin{aligned} p_X(x_i) &= P(X = x_i) = P(x_{i-1} < X \leq x_i) \\ &= P(X \leq x_i) - P(X \leq x_{i-1}) \\ &= F(x_i) - F(x_{i-1}) \end{aligned}$$

- remark: The probability of a discrete distribution varies depending on the inclusion and exclusion of the boundary values.

Expectation and Variance for Discrete Random Variables

⟵ **My thought** From a probabilistic model to a usable numerical quantity: what is observed, what is random?

$$\frac{0 \cdot f_0 + 1 \cdot f_1 + 2 \cdot f_2 + \cdots + n \cdot f_n}{N} = \frac{1}{N} \sum_{i=0}^{n} i \cdot f_i$$

Note that in $\frac{1}{N} \sum_{i=0}^{n} i \cdot f_i$,

$$\lim_{N \to \infty} \frac{f_i}{N} = P(X = i)$$

So the average number will be

$$\sum_{i=0}^{n} i \cdot P(X = i)$$

- Definition: Expectation Given a discrete random variable X, the expectation of X is

$$E[X] = \sum_{x} x \cdot p_X(x)$$

- Properties of Expectation
- If c is a constant, then $E[c] = c$.
- If $X \geq 0$ then $E[X] \geq 0$.
- If $a \leq X \leq b$ then $a \leq E[X] \leq b$.

- Proof of 3: First show $E[X] \geq a$, then show $E[X] \leq b$,

$$E[X] = \sum_x x p_X(x) \geq \sum_x a p_X(x),$$
$$= a \sum_x p_x(x) = a.$$

Similarly, $E[X] \leq b$.

- Suppose X is a discrete random variable and $Y = g(X)$, then

$$E[Y] = \sum_y y p_Y(y) = \sum_y y P(Y = y)$$
$$= \sum_y y \sum_{\{x:g(x)=y\}} P(X = x)$$
$$= \sum_y \sum_{\{x:g(x)=y\}} y P(X = x)$$
$$= \sum_y \sum_{\{x:g(x)=y\}} g(x) P(X = x)$$
$$= \sum_x g(x) P(X = x)$$

- First moment of X (mean):

$$E[X] = \sum_x x p_X(x).$$

- Second moment of X :

$$E\left[X^2\right] = \sum_x x^2 p_x(x).$$

- In general, $E[g(X)] \neq g(E[X])$. For example, let $g(x) = x^2$, and consider X such that

$$p_X(x) = \begin{cases} 0.5, & \text{for } x = -1 \\ 0.5, & \text{for } x = 1 \end{cases}$$

Then clearly $E\left[X^2\right] = 1 \neq 0 = (E[X])^2$.

- There are exceptions (e.g. when g is linear)!
- Linearity of Expectation

E[aX + b] = aE[X] + b

proof:

Suppose $g(x) = ax + b$. Then

$$E[g(X)] = \sum_x g(x) p_X(x),$$
$$= \sum_x (ax + b) p_x(x),$$
$$= \sum_x ax p_x(x) + \sum_x b p_x(x),$$
$$= a \sum_x x p_x(x) + b \sum_x p_x(x),$$
$$= aE[X] + b = g(E[X]),$$

this implies

$$E[aX + b] = aE[X] + b$$

- Remark: Apart from this case, always assume $E[g(X)] \neq g(E[X])$.

joint distribution of X and Y

The joint probability mass function (i.e., joint pmf) of X and Y for discrete random variables is defined as $p_{X,Y}(x, y) := P(X = x \text{ and } Y = y)$

P[(X,Y) ∈ A] =$\sum_{(x,y)\in A} p_{X,Y}(x, y)$,where A belongs to a subset of the $\mathbb{R}^2$ where X and Y taking values.

⟵ **Motivation**
From a probabilistic model to a usable numerical quantity: what is observed, what is random?

Marginal Probability Mass Function

Let X and Y have the joint probability mass function $p_{X,Y}(x,y)$ with space $\mathcal{S}$. The probability mass function of X (or Y) alone, is called the marginal probability mass function of X (or Y) and defined by:

$$p_X(x) = P(X = x) = \sum_y p_{X,Y}(x,y)$$
$$p_Y(y) = P(Y = y) = \sum_x p_{X,Y}(x,y)$$

For the marginal of X, we sum over all values of y.

For the marginal of Y, we sum over all values of x.

eg.–helping understand both sample space, event and Marginal probability mass function

A joint probability table sums across the other variable to obtain a marginal distribution.

Functions of multiple random variables

⟵ **Proof idea**
Condition explicitly; it often simplifies the calculation.

it has the same philosophy as one random variable things, such as expectation:

$$E[g(X,Y)] = \sum_x \sum_y g(x,y) p_{X,Y}(x,y)$$

, also the linearity of expectation

eg. Z = X + 2Y.

For a function of two random variables, expectation weights each possible input pair by its joint probability.

independence

- P($A \cap B$)=P(A)P(B) or P(A|B)=P(A)
- Two discrete random variables X and Y are independent if P(X = x and Y = y) = P(X = x)P(Y = y):

$p_{X,Y}(x,y) = p_X(x)p_Y(y), \forall x, y$

$p_{X|Y}(x|y) = p_X(x), \forall x, y$

it is easy to detect the dependent as long as we find one example, eg: $P_{X|Y}(1|1)$ is not equal to $P(X = 1)$

independence, expectations(mean) and variance

Independence and Expectations If X and Y are independent, then

$$E[XY] = E[X]E[Y].$$

Proof: We use $E[g(X,Y)]$ where $g(x,y) = xy$.

$$\begin{aligned} E[XY] &= \sum_x \sum_y xy p_{X,Y}(x,y) \\ &= \sum_x \sum_y xy p_X(x) p_Y(y), \quad (\text{ by inde} \\ &= \left(\sum_x x p_X(x) \right) \left(\sum_y y p_Y(y) \right) = E[X]E[Y]. \end{aligned}$$

(by independence)

Similarly, if X and Y are independent, then

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

- Independence and Variances

It is always true that

$$\text{Var}(aX) = a^2\,\text{Var}(X), \quad \text{and} \quad \text{Var}(X+a) = \text{Var}(X)$$

In general, when we have a sum of random variables X and Y

$$\text{Var}(X+Y) \neq \text{Var}(X) + \text{Var}(Y).$$

It is only true if X and Y are independent

- Sum of Variance for independent R.V. If two random variables X and Y are independent then

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

- Sum of Variance for independent R.V.

If two random variables X and Y are independent then

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y).$$

Proof: Independence implies $E[XY] = E[X]E[Y]$. Thus

$$\begin{aligned}
&\text{Var}(X+Y) = E\left[(X+Y-(E[X]+E[Y]))^2\right], \\
&= E\left[(X-E[X])^2 + 2(X-E[X])(Y-E[Y]) + (Y-E[Y])^2\right], \\
&= \text{Var}(X) + \text{Var}(Y) + 2E[(X-E[X])(Y-E[Y])]
\end{aligned}$$

As X is indep to Y, then $X-\mu_X$ is indep to $Y-\mu_Y$ so

$$E[(X-E[X])(Y-E[Y])] = E[X-E[X]]E[Y-E[Y]] = 0$$

Example: Assume independence, $\text{Var}(3X-5Y) =$

$$\text{Var}(3X) + \text{Var}(-5Y) = 9\,\text{Var}(X) + 25\,\text{Var}(Y)$$

- Example. Independence, mean and variance

Let Y be the random variable denoting the total number of heads by tossing a coin n times. Find the mean and variance of Y.

Let

$$Y_i = \begin{cases} 1, & \text{if the } i^{\text{th}} \text{ toss gets a head} \\ 0, & \text{otherwise.} \end{cases}$$

Then $Y = Y_1 + Y_2 + \cdots + Y_n$ where $Y_1, Y_2, \cdots, Y_n$ are independent. For any $i = 1, 2, \cdots, n$, we have

y	0	1
$P_{Y_i}(y)$	$1/2$	$1/2$

As $E[Y_i] = \frac{1}{2}$ and $\text{Var}(Y_i) = \frac{1}{4}$ for $i = 1, 2, \cdots, n$

$$E[Y] = E[Y_1] + E[Y_2] + \cdots + E[Y_n] = \frac{n}{2}$$
$$\text{Var}(Y) = \text{Var}(Y_1) + \text{Var}(Y_2) + \cdots + \text{Var}(Y_n) = \frac{n}{4}$$

(for not independent cases we have the same result as E but not Var because Var is not linear)

Continuous Random Variables

We have to consider intervals instead of one when referring to continuous random variables or the probability will equals to 0, which has no meaning.

eg. $P(a \leq Y \leq b) = P(a < Y < b)$ if Y is a continuous variable.

⟵ **My thought**
What does the definition really control? Try to draw it before proving it.

Definition. Probability Density Function (pdf)

For a continuous random variable X, the Probability Density Function (PDF) of X is f(x) where P(X = x) = 0 for all x and for any a ≤ b

discrete and continuous

For a continuous variable, probabilities are areas under a density curve rather than heights at single points.

Density function's value could be greater than 1 because the integral of it in a very tiny interval could be very small, which represents the probability(which could not be greater than 1 and since the interval could be very tiny this condition is satisfied!).(This reminder notices that the value of pdf is not the probability since the integration is the probability which is totally 1)

Covariance

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Covariance records whether two variables tend to move in the same or opposite directions.

wait for reviewing.....

Positive, negative, or near-zero covariance corresponds to different patterns of joint variation.

Descrete distributions

A probability distribution is a graph, table of formula that gives the probability for each value of the random variable.

Bernoulli is composed by binomial

Hypergeometric distribution could be approximated by Binomial when samples are large

Poisson distribution as the limit of Binomial distribution when the number of trials is large and the probabiliy of success of each trial is inverse-proportional to the number of trials

- The Poisson distribution is a discrete probability distribution that applies to occurrences of some event over a specified interval. The random variable x is the number of occurrences of the event in an interval.The probability of the event occurring x times over an interval is given by

$$P(x) = \frac{u^x \cdot e^{-\mu}}{x!}$$

  where the random variable x is the number of occurences of an event over some interval and the ovvurrences must be random and independent of each other.

中心极限定理：当泊松分布的参数 λ 较大时，泊松分布的形状会接近正态分布。这是因为中心极限定理指出，大量独立随机变量的和趋向于正态分布，而泊松分布可以看作是大量伯努利试验成功次数的分布，当试验次数足够多时，其和可以用正态分布来近似

???The occurrences must be uniformly distributed over the interval being used

⟵ **Proof idea**
Identify the exact statement before proving it.

The mean is μ

The standard deviation is $\sigma = \sqrt{\mu}$

- eg of Poisson distribution: (describing the behavior of rare events(with small probabilities). radioactive decay, arrivals of people in a line, eagles nesting in a region, patients arriving at an emergency room(the local hospital experiences a mean of 2.3 patients arriving at the emergency room during 10-11 P.M. on Fri. is known, we can find the probability that for a randomly selected Fri. between 10-11 P.M., exactly four patients arrive), Internet users logging onto a Web site)
- Comparison between Binomial:

Binomal distribution is affected by yhe smaple size n and the probability p, whereas the Poisson distribution is affected only by mean μ

A binomial distribution has a limit of possible values but a Poisson distribution has a possible values x without upper bound.

Suppose the $X_n, n \geq 1$ is a sequence of random variables such that X_n~Bin(n, p_n), where p_n~λ/n as $n- > \infty$ $lim_{n->\infty}(np_0) = \lambda$, **intuitively we can observe that** λ is the mean of

given the well-known limit $\lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^n = e^{-\lambda}$, and

$$\frac{n}{n}\frac{n-1}{n}\cdots\frac{n-k+1}{n} = \prod_{i=0}^{k-1}\left(1-\frac{i}{n}\right)$$

$$\lim_{n\to\infty}\prod_{i=0}^{k-1}\left(1-\frac{i}{n}\right) = 1$$

$$\begin{aligned}
P\left(X_n = k\right) &= \binom{n}{k} p_n^k \left(1-p_n\right)^{n-k} \\
&= \frac{n\cdots(n-k+1)}{k!} p_n^k \left(1-p_n\right)^n \left(1-p_n\right)^{-k} \\
&= \frac{1}{k!}(\underbrace{np_n}_{\to\lambda})^k \underbrace{\frac{n}{n}\frac{n-1}{n}\cdots\frac{n-k+1}{n}}_{\to 1} \underbrace{\left(1-p_n\right)^n}_{\to \mathrm{e}^{-\lambda}} \underbrace{\left(1-p_n\right)^{-k}}_{\to 1} \\
&\to \frac{\lambda^k}{k!}\mathrm{e}^{-\lambda} \quad \text{as } \underset{\smile}{n\to\infty}.
\end{aligned}$$

Check that $p_x(k), k = 0, 1, 2, \cdots$ defines a probability mass function: given the Taylor Series of e^λ around $\lambda = 0$ is given by $f(x) = \sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n$ and $e^\lambda = \sum_{n=0}^{\infty}\frac{\lambda^n}{n!}$. Hence,

$$\begin{aligned}
\sum_{k=0}^{\infty} p_X(k) &= \sum_{k=0}^{\infty}\frac{\lambda^k}{k!}\mathrm{e}^{-\lambda} \\
&= \mathrm{e}^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} \\
&= \mathrm{e}^{-\lambda}\cdot\mathrm{e}^{\lambda} \\
&= 1.
\end{aligned}$$

Geometric distribution (and Geometric series–powerful!fantastic series)

- The probability of the first happening

$$\begin{aligned}
\sum_{k=1}^{\infty} p_X(k) &= \sum_{k=1}^{\infty}(1-p)^{k-1}p \\
&= p\sum_{k=0}^{\infty}(1-p)^k \\
&= p\cdot\frac{1}{1-(1-p)} \\
&= 1.
\end{aligned}$$

- well-defined
- Tail probability of the Geometric distribution

Let $X \sim \text{Geom}(p)$ then

$$\begin{aligned}
P(X > n) &= P(X = n+1) + P(X = n+2) + P(X = n+3) + \cdots \\
&= (1-p)^n p + (1-p)^{n+1}p + (1-p)^{n+2}p + \cdots \\
&= (1-p)^n p\left(1 + (1-p) + (1-p)^2 + \cdots\right) \\
&= (1-p)^n p\frac{1}{1-(1-p)} \\
&= (1-p)^n
\end{aligned}$$

for any $n = 0, 1, 2, \ldots$

- Memoryless property of Geometric distribution

Suppose that $X \sim \text{Geom}(p)$ and $n \in \{1, 2, 3, \cdots\}$. Then

$$P(X - n = k \mid X > n) = P(X = k) \quad , k = 1, 2, 3, \cdots$$

That is, the distribution of $X - n$ under the probability function $P(\cdot \mid X > n)$ is the same as the distribution of X

the memoryless property is saying that given the first n trials are unsuccessful, the number of trials until success after the first n trials has the same distribution as the unconditional number of trials until success (independent)

- Expectation

Suppose that $X \sim \text{Geom}(p)$. Then

$$\begin{aligned} E(X) = \sum_{k=1}^{\infty} kP(X=k) &= \sum_{k=1}^{\infty} k(1-p)^{k-1}p \\ &= p\sum_{k=1}^{\infty}\left[-\frac{\mathrm{d}}{\mathrm{d}p}(1-p)^k\right] \\ &= -p\frac{\mathrm{d}}{\mathrm{d}p}\left[\sum_{k=0}^{\infty}(1-p)^k\right] \\ &= -p\frac{\mathrm{d}}{\mathrm{d}p}\left[\frac{1}{1-(1-p)}\right] \\ &= -p\frac{\mathrm{d}}{\mathrm{d}p}\left[\frac{1}{p}\right] \\ &= \frac{1}{p} \end{aligned}$$

— variance

Likewise,

$$\begin{aligned} E(X(X-1)) &= \sum_{k=1}^{\infty} k(k-1)P(X=k) \\ &= \sum_{k=2}^{\infty} k(k-1)(1-p)^{k-1}p \\ &= p(1-p)\sum_{k=2}^{\infty} k(k-1)(1-p)^{k-2} \\ &= p(1-p)\frac{\mathrm{d}^2}{\mathrm{d}p^2}\left[\sum_{k=0}^{\infty}(1-p)^k\right] \\ &= p(1-p)\frac{\mathrm{d}^2}{\mathrm{d}p^2}\frac{1}{p} \\ &= p(1-p)\cdot\frac{2}{p^3} \\ &= \frac{2(1-p)}{p^2}. \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E(X(X-1)) + E(X) - (E(X))^2 \\ &= \frac{2(1-p)}{p^2} + \frac{1}{p} - \frac{1}{p^2} \\ &= \frac{1-p}{p^2} \end{aligned}$$

Geometric distribution

The irrelevance of past events to the probability of future independent events

Given the first n trials are unsuccessful, the number of trials until success after the first n trials has the same distribution as the unconditional number of trials until success.

X~Geom(p) and n$∈${1,2,3,....}

P(X-n=k|X>n)=P(X=k), k=1,2,3,...

Tail probability for the probability calculation of more/higher than....

⟵ **My thought**
From a probabilistic model to a usable numerical quantity: what is observed, what is random?

Addition

Independent random variable with the same distribution allows the addition law

Normal distribution

interpretation:

- Standard normally distributed sth. z=1.58(corresponding to 0.9429):
  the probability of randomly selecting sth. with a value less than 1.58(unit) is equal to the area(probability) of 0.9429.
  (Or: 94.29% of sth, will have a value below 1.58(unit))

Sampling distributions and Estimators

⟵ **Motivation**
Inference uses data to distinguish plausible parameters; assumptions still matter.

We are beginning to embark a jourfa d dney that allows us to learn about populations by obtaining data from samples since it is rare that we know all values in an entire population.

Sampling distribution of a statistic is the probability distribution of a sample statistics (such as mean/proportion which tend to target the population mean/proportion), with all samples having the same sample size. This concept is important to understand. The behavior of a statistic can be known by understanding its distribution((The random variable in this case is the value of that sample statistics)). Under certain condition, the distribution of sampling mean/proportion approximates a normal distribution.

Though statistics does not depend on unknown parameters, the distribution of it depend on unknown parameters.(eg. Normal distribution of sample means depends on population mean(an unknow parameter) and standard deviation)

(ps: the advantage of sampling with replacement:

when selecting a relatively small sample from a large population, it makes no significant difference whether we sample with or without replacement.

Sampling with replacement results in independent events that are unaffected by previous outcomes, and independent events are easier to analyze and they result in simpler formulas.)

For a fixed sample size, the mean of all possible sample means is equal to the mean of population though sample means vary(sampling variability)

Unbiased estimators and biased estimators

⟵ **Proof idea**
Often the log-likelihood is simpler than the likelihood.

Statistics that target population parameters: Mean, variance, proportion

Statistics that do target population parameters: Median, Range, Standard Deviation

(the bias is relatively small then sampling standard deviation in large samples so s is ofent used to estimate σ)

The Central Limit Thm

It is the foundation for estimation population parameters and hypothesis testing

(Recall: A random variable is a variable that has a single numerical value(eg. x=1,x=2), determined by chance, for each outcome of a procedure 随机变量是一种变量，对于一个过程的每个结果都有一个随机确定的单一数值 (可以理解为，现象作用其上

A probability distribution is a graph, table or formula that gives the probability for each value of a random variable

The sampling distribution of the mean is the probability fistribution of sample means, with all sample having the same sample size n)

As the sample size increases, the corresponding sample means tend to vary less. The central limit thm tells us that if the sample size is large engough, the distribution of sample means can be approximated by normal distribution

Conclusion of CLT: (it is so important since it allows us to use the basic normal distribution methods in a wide variety of different circumstances)

The distribution of sample means will, as the sample size increases, approach a normal distribution

The mean of all sample means is the population mean μ..

$$\mu_{\bar{x}} = \mu$$

The standard deviation of all sample means is $\sigma/\sqrt{n}$ (i,e, the normal distribution from conclusion"The distribution of sample means will, as the sample size increases, approach a normal distribution" has standard deviation $\sigma/\sqrt{n}$)

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

$\sigma_{\bar{x}}$ is often called the standard error of the mean.

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Notice: Be careful to look at if it is from a normally ditributed population or a mean for some sample/group.

If given the population mean is a and a sample(c subjects) mean is b which has so small probbility under the sampling distribution with this populatio mean, if the mean is really a, then there is an extremely small probability of getting a sample mean of b or lower when c subjects are randomly selected. we can interpret it in such 2 ways:

1) population mean is true and their sample represents a chance event that is extreamely rare
2) population mean is not true and sample is typical.

Because the probability is too low, it seems more reasonable to conclude that the population mean is lower than a—hypothesis testing's thinking

Optional: Correction for a finite population

When sampling without replacement and the sample size n is greater than 5% of the finite populaiton size B(i.e. n>0.05N), ...

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

Using the Normal distribution as an approximation to the binomial distribution

requirement: np, n(1-p) ≥ 5

Be careful: adjust x for continuity by + or - 0.5(eg. at least 99, choose 98.5)

to hypothesis testing

a question (eg):

In a test of a gender-selection technique assume that 100 couples using a particular treatment give birth to 52 girls (and 48 boys). If the technique has no effect, then the probability of a girl is approximately 0.5. If the probability of a girl is 0.5, find the probability that among 100 newborn babies, exactly 52 are girls. Based on the result, is there strong evidence supporting a claim that the gender-selection technique increases the likelihood that a baby is a girl?

it is a binomial distribution with np=nq=100*0.5=50≥ 5

so we use the normal distribution with mean of 50 and $\sigma = \sqrt{npq} = 5$as an approximation to the binomial distribution

to answer"is there strong evidence supporting a claim that the gender-selection technique increases the likelihood that a baby is a girl?", we need to calculate more than 52(x successes among n trials is an unusually high number of successes if P($x \geq a$) is very small)

原因是因为如果只看 52 这个数字肯定概率很小，因为任何单个数字发生的可能性概率都很小

So if the answer of P($x \geq 52$) is small we could conclude that the gender selection is useful

(总结：如果是 0。5 概率来看的话 52 以上本来就不是难事，as indicated by the such large probability of P($x \geq 52$) 所以我们没有充分证据拒绝 "not effective" 这个假设前提)

此处还没有引入假设检验所以都是用 using probability to determine when results are unusual 这个思想来思考问题的：

- Unusually low: x successes among n trails is an unually low number of successes if P(x or fewer) is very small

A very small binomial tail probability marks an observed number of successes as unusual under the assumed model.

Interpretation for another example of gender selection(using only unusual results to explain): Because the probability of more than 13 girls, which is 0.001 is so low, we conclude that it is unusual to get 13 girls among 14 babies (using binomial to calculate it). This suggests that the technique of gender selection appears to be effective since it is highly unlikely that the result of 13 girls among 14 births happened by chance.

Normal distribution

If a variable is the superposition result of a large number of small independent random factors, then the variable must obey the normal distribution of variables

e.g. measure error

Assessing Normality

In general, quantile plots can be used to assess any probability distribution.

For a normal quantile plot(or normal probability plot), it is a graph of points(x,y) where each x vaue is from the original set of sample data and each y value is the cooresponding z score that is a quantile value expected from the standard normal distribution

Procedures

- Histogram(not helpful for small data set)
- outliers: reject normality if there is more than 1 outlier present(not helpful for small data set)
- normal quantile plot:

sort data from lowest to highest

- Sampling distribution of pˆ when population is infinite

A normal quantile plot compares ordered observations with expected normal quantiles to check the distributional shape.

inferences from 2 samples introduces the differences between two populaton means using matched pairs but correlaition and regression analyze the association between the 2 variables and if such an association exists we wnat to describe it with an equation that can be used for predictions

Paired observations support studying correlation and regression, unlike a comparison of two unrelated sample means.

paired sampled data(or called bivariate data)

- a correlation exists between two variables when one of them is related to the other in some way.
- the linear correlation coefficient r measures the strength of the linear association between the paired x- and y-quantitative values in a sample. Its value is computed by using the formula(Pearson(1857-1937) product moment correlation coefficient)

.....otherwise there is not sufficient evidence to support the conclusion of a significant linear equation

!!!: interpreting r: explained variation: the value of r^2 is the proportion of the variation in y that is explained by the linear association between x and y.(and the other percentage is explained by factors other thanx such as characteristics not included in the study)

- correlation does not imply causality,just the association
- average suppress individual variation and may inflate the correlation coefficient(the linear correlation coefficient became higher when reginal averages were used)

point estimation and confident interval

- Tomorrow is the exam Good luck! 12.29
- 12.30 Everything on MTH113 exam was fine

All in all, the exam did not decide anything. I think there is a long way to go and MTH113 just include not many about statistics, more than a half has been teached during high school.....I will keep exploring the charismatic statistics world since we could not only understand it using the very basic life common sense but also prove them in a rigorous way, while during the proof, such as regression, we also could enjoy its explanation by linear algebra which I also enjoy.....

- I love you MTH113!!!

The closing course image accompanies the reflection on learning statistics beyond an exam.

⟵ **Motivation**
From a probabilistic model to a usable numerical quantity: what is observed, what is random?

⟵ **Proof idea**
Often the log-likelihood is simpler than the likelihood.

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Model selection: AIC, AICc, and BIC

AIC (Akaike Information Criterion)

By Akaike (1974), although the true distribution $P(x)$ is unknown, we can use the maximum likelihood estimate (MLE) to approximate it. The AIC is derived from an estimate of the KL divergence between the true model and a candidate model.

AIC is defined as:

$$AIC = -2\log(L) + 2k$$

AICc

AICc is a corrected version of AIC that adjusts for small sample sizes. It is defined as:

$$AICc = AIC + \frac{2k(k+1)}{n-k-1}$$

when n goes to infinity, AICc converges to AIC.

BIC (Bayesian Information Criterion)

- Bayesian Thm

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

A goes to θ, B goes to x

$$\hat{\theta} = \arg\max_{\theta} P(\theta|x) = \arg\max_{\theta} \frac{P(x|\theta)P(\theta)}{P(x)} = \arg\max_{\theta} P(x|\theta)P(\theta)$$

- BIC formula

$$BIC = -2\log(L) + k\log(n)$$

loglikelihood penalized by the number of parameters and the sample size

loglikelihood could be replaced by the loss function

BIC is more popular used in high-dimensional data analysis. If not in high-dimensional data analysis, AIC and BIC values are similar.

⟵ **My thought**
Decomposing a complicated signal into simpler modes makes dynamics easier to understand.